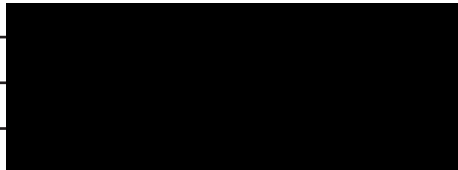


Weight and Balance Manual

FOR
ENTERTAINMENT
ONLY

MODEL	
CL-600-2B16 (CL-650)	



CHALLENGER 650

INTRODUCTION

1. Scope

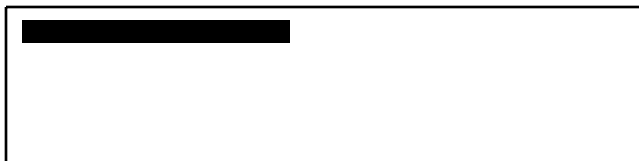
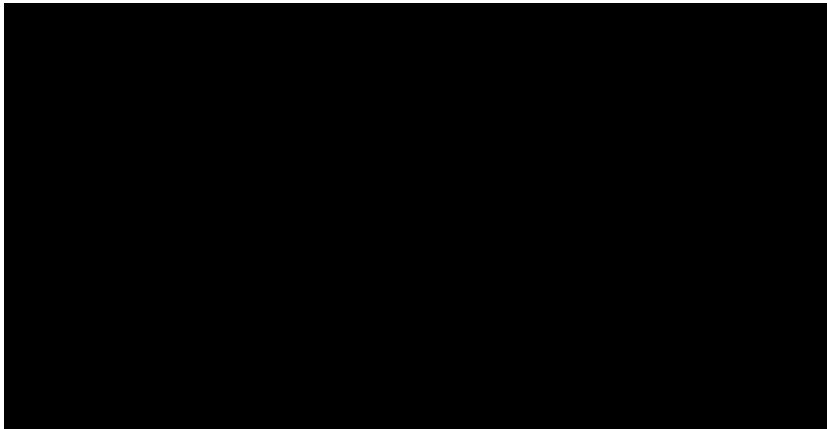
This manual applies to the Challenger 650. The Challenger 650 is a marketing designation that is applicable to the Model CL-600-2B16, Aircraft serial No. 6050 and subs.

The Weight and Balance Manual (WBM) prepared by Bombardier Business Aircraft agrees with specification No. 100 of the Air Transport Association of America (ATA-100), Revision 32, dated January 1994. The WBM contains data on the aircraft loads and aircraft operations.

The weight and balance procedure is a necessary pre-flight check to make sure of the safe and satisfactory operation of the aircraft. Personnel must obey the procedures given in the WBM for the calculations of weight and balance. Only fully qualified and licensed personnel can do these procedures.

Only data about basic aircraft configuration are given in this manual. Data for the standard and optional changes and Service Bulletins approved by Bombardier Business Aircraft, Customer Support will be included as soon as possible after the change or Service Bulletin is issued.

Upon appropriate agreement, changes or installations done by the operator or completion center and approved by Bombardier Business Aircraft, Customer Support may be included in future revisions of the manual. The Completion Center provide the data related to the completed aircraft, and this data must be inserted in Chapter 2 of this manual.



3. How to Use the Manual

The manual is divided into two chapters (1 – CONTROL, 2 – AIRCRAFT REPORTS).

The ATA-100 has a three-element number system to identify the different subjects contained in the WBM. The first element identifies the chapter, the second element identifies a section in the chapter and the third element identifies a subject in the section. The example that follows shows and briefly describes a typical three-element number:

“01-00-30”

The first element identifies Chapter 01 for CONTROL,

The second element identifies Section 00 for GENERAL,

The third element identifies Subject 30 for AIRCRAFT STATIONS.

4. Instructions for Continued Airworthiness (ICA) Manuals

The instructions for continued airworthiness (ICA) include data necessary to the continued airworthiness of the aircraft. The ICA's also include all data applicable to the components that have an interface with the aircraft systems, and their continued airworthiness.

The instructions for continuing airworthiness (ICAs) for the first 24 months/1800 flight hours/900 landings will be available by Aircraft Entry-into-Service.

The remaining scheduled maintenance tasks for more than 24 months/1800 flight hours/900 landings will be published one year prior to their in-service requirements.

ICAs for the aircraft obey the Code of Federal Regulations, CFR 25.1529, and appendix H to part 25. The technical manuals that follow are ICA manuals:

- Aircraft Maintenance Manual (AMM)
- System Description Section (SDS)
- Aircraft Illustrated Parts Catalog (AIPC)
- Structural Repair Manual (SRM)
- Time Limits/Maintenance Checks (TLMC)
- Non-Destructive Testing Manual (NDTM)
- Weight and Balance Manual (WBM)
- Wiring Diagram Manual (WDM)
- Component Maintenance Manual (CMM)

NOTE: Troubleshooting procedures will be covered through a web based application called SmartFix.

5. Acronyms

Refer to the Subjects that follow for the technical definition of the acronyms found in the Weight and Balance Manual:

- DEFINITIONS (Subject 01–00–20)
- AIRCRAFT STATIONS (Subject 01–00–30)
- REFERENCE POINTS (Subject 01–00–40).

In the Weight and Balance Manual, terms are spelled out the first time they are used and the acronym is given in parentheses. Then only the acronym is given.

6. Revisions and Temporary Revisions

Bombardier supplies revisions to the WBM when required. Bombardier also supplies temporary revisions as necessary. Temporary revisions are included at the subsequent revision to the applicable subject.

7. Warnings, Cautions and Notes

Warnings, cautions and notes added to the text are there to show important parts of the text when necessary. Their definitions are as follows:

A. WARNING

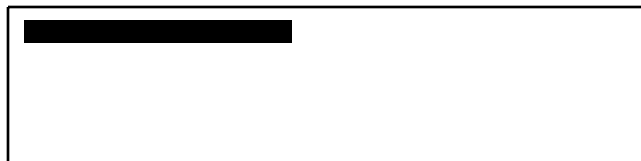
A WARNING TELLS THE USER OF MATERIALS, PROCEDURES, OR LIMITS THAT MUST BE USED CAREFULLY TO PREVENT INJURY TO PERSONS. THE WARNING ALWAYS COMES BEFORE THE PROCEDURE TO WHICH IT APPLIES AND IS WRITTEN IN UPPER CASE LETTERS.

B. CAUTION

A CAUTION TELLS THE USER OF PROCEDURES THAT MUST BE FOLLOWED TO PREVENT DAMAGE TO EQUIPMENT. THE CAUTION ALWAYS COMES BEFORE THE PROCEDURE TO WHICH IT APPLIES AND IS WRITTEN IN UPPER CASE LETTERS.

C. Note

A NOTE tells the user of procedures that must be followed to make the work easier. The NOTE always follows the procedure to which it applies.



8. Front Matter

The WBM includes the front matter that follows:

- Title Page
- Record of Revisions
- Record of Temporary Revisions
- List of Effective Pages
- Table of Contents

A. Title Page

The Title Page is found at the front of each volume of the WBM. It contains the logo, the aircraft model and effectivity, the title of the manual, and the volume number. It also contains the Bombardier Inc. address and copyright statement.

B. Record of Revisions

The Record of Revisions is found after the Title Page of this Volume of the WBM. The user must enter the data in the Record of Revisions when he/she receives a revision, and put the revision into the WBM.

C. Record of Temporary Revisions

The Record of Temporary Revisions is found after the Record of Revisions. The user must update the Record of Temporary Revisions when he/she puts a Temporary Revision into the WBM, or removes a Temporary Revision from the WBM.

D. List of Effective Pages

The List of Effective Pages is found at the front of each Chapter. It identifies each chapter/section/subject and page and also gives the effectivity and date of each page.

E. Table of Contents

The Table of Contents is found after the List of Effective Pages for each chapter. The table of contents gives an index for the data in each chapter. It gives the titles, ATA numbers, effectivity and the page number of each system, subsystem or component.

CHAPTER 01 – CONTROL**TABLE OF CONTENTS**

Subject	Ch–Se–Su	Effectivity	Page
GENERAL	01–00–00	ALL	
Description			1
DIMENSIONS AND AREAS	01–00–10	ALL	
Dimensions			1
DEFINITIONS	01–00–20	ALL	
General			1
Acronyms			
Definitions			3
AIRCRAFT STATIONS	01–00–30	ALL	
General			1
Buttock Line (BL)			
Fuselage Station (FS)			
Horizontal Stabilizer Station (HSS)			
Nacelle Station (XN)			
Vertical Stabilizer Station (VSS)			2
Water Line (WL)			
Wing Station (WS)			
Winglet Station (WGLTS)			
REFERENCE POINTS	01–00–40	ALL	
General			1
Leading–Edge Mean–Aerodynamic–Chord (LEMAC)			
Main–Landing–Gear (MLG) Axle Center–Line			
Mean Aerodynamic Chord (MAC)			
Nose–Landing–Gear (NLG) Axle Center–Line			
Weighing Datum (WD)			
Zero Reference Datum (ZRD)			
Forward Jacking Point Center–Line			
Aft Jacking Point Center–Line			2
DESIGN WEIGHT LIMITS	01–00–50	ALL	
General			1
CENTER OF GRAVITY LIMITS	01–00–60	ALL	
General			1
MAC Conversion			
Longitudinal CG Envelope			4

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Subject	Ch–Se–Su	Effectivity	Page
TAKE-OFF TRIM SETTINGS General	01–00–70	ALL	1
BALANCE EFFECT CAUSED BY MAJOR OPERATION General	01–00–80	ALL	1
FUEL General	01–20–00	ALL	1
FUEL TANK CONFIGURATION AND CAPACITIES Fuel Tank Configuration Description Fuel Tank Capacities Unusable Fuel Usable Fuel	01–20–10	ALL	1 2 3
FUEL MANAGEMENT Fuel Transfer System Fuel Replenishment Pressure Refuel Procedure Gravity Refuel Procedure Fuel Burn Sequence EICAS Messages	01–20–20	ALL	1 5
FLUIDS General	01–30–00	ALL	1
SYSTEM FLUIDS System Fluid Capacities	01–30–10	ALL	1
SERVICE FLUIDS Service Fluid Capacities	01–30–20	ALL	1
PERSONNEL General	01–40–00	ALL	1
FLIGHT COMPARTMENT General Crew Arrangement Crew Weight and Balance Arms	01–40–10	ALL	1
PASSENGER COMPARTMENT General	01–40–20	ALL	1

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL**

Subject	Ch-Se-Su	Effectivity	Page
BALANCE CHANGES DUE TO MOVEMENT	01-40-40	ALL	
General			1
Aircraft on the Ground			
Aircraft in Flight			
CARGO	01-60-00	ALL	
General			1
CARGO COMPARTMENTS	01-60-10	ALL	
Limitations			1
General			
PASSENGER COMPARTMENT STORAGE	01-60-20	ALL	
Limitations			1
General			
GROUND OPERATIONS	01-80-00	ALL	
General			1
STATIC LOADS	01-80-20	ALL	
Maximum Permitted Jacking Weight			1
Jacking on Structure			
Jacking on Landing Gear			
Maximum Loading			2
WEIGHING PROCEDURES	01-80-30	ALL	
General			1
Job Set-Up Information			
Tools and Equipment			
Reference Information			2
Weighing on Tripod Jacks Procedure			4
Weighing			
Center of Gravity			12
Example			13
Weighing on Platform Scales and Jack Procedure			14
Weighing			15
Center of Gravity			25
Example			
Close Out			29
BALANCE LIMITS	01-80-60	ALL	
Static Stability			1
Towing			2
Nose Landing Gear Load			

CHALLENGER⁶⁵⁰
CL-650 WEIGHT AND BALANCE MANUAL

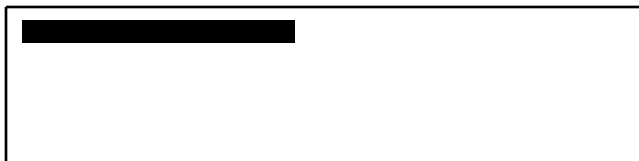
Subject	Ch-Se-Su	Effectivity	Page
LOADING PROCEDURES AND EXAMPLES	01-90-00	ALL	
General			1
OPERATING WEIGHT EMPTY	01-90-10	ALL	
General			1
Procedure			
Example			4
WEIGHT BUILD-UP	01-90-20	ALL	
General			1
Procedure			
ZFW Calculation			6
RW Calculation			9
TOW Calculation			12
FCBP Calculation			15
LW Calculation			17
Payload Item Calculation			19
Weight Build-Up Envelope			
CG SLIDE METHOD	01-90-50	ALL	
Procedure			1
Example			2

GENERAL

1. Description

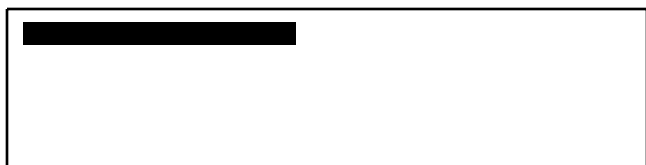
This section gives general data applicable to the aircraft as follows:

- DIMENSIONS AND AREAS (Subject 01-00-10)
- DEFINITIONS (Subject 01-00-20)
- AIRCRAFT STATIONS (Subject 01-00-30)
- REFERENCE POINTS (Subject 01-00-40)
- DESIGN WEIGHT LIMITS (Subject 01-00-50)
- CENTER OF GRAVITY LIMITS (Subject 01-00-60)
- TAKE-OFF TRIM SETTINGS (Subject 01-00-70)
- BALANCE EFFECT CAUSED BY MAJOR OPERATION (Subject 01-00-80).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



01-00-00

Page 2
Oct 27/2015

DIMENSIONS AND AREAS

1. Dimensions

Refer to Figure 1.

This part of the manual shows the general external and primary dimensions of the aircraft. The dimensions are in imperial units. The equivalent values in SI units are also given.

Dimensions are measured in planes parallel or perpendicular to the fuselage reference plane. Chord lines are measured as projections on the wing reference plane. The wing reference plane passes through WL53.25 at the center line of the aircraft and has a dihedral angle of 2.33 with respect to the fuselage reference plane.

The Table 1 that follows shows the dimensions and areas of the primary aircraft assemblies.

NOTE: The values found in Table 1 are for general information only. Do not use them to do inspections or to reject assemblies.

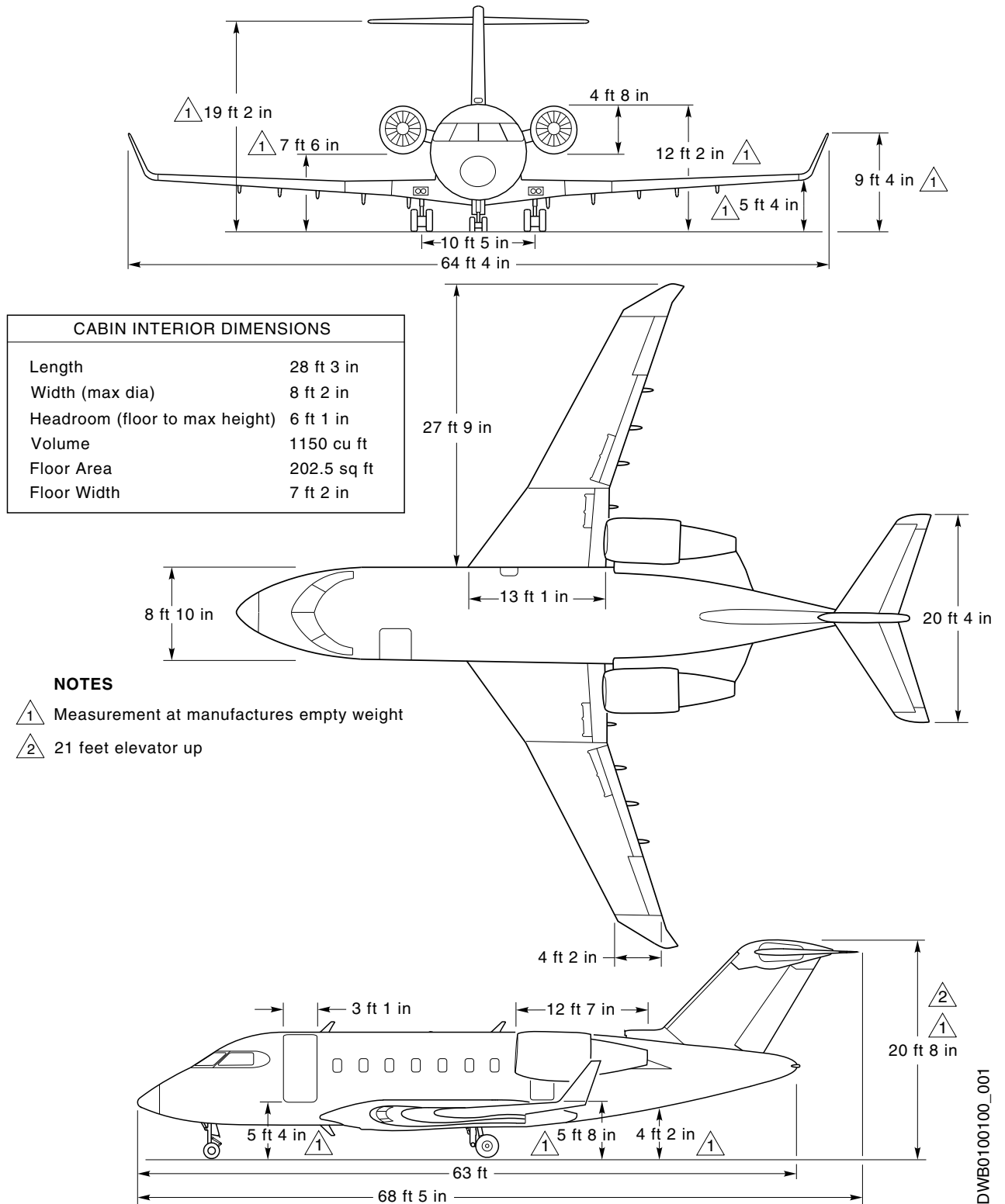
Table 1 – General Dimensions and Areas of the Aircraft	
Dimensions/Areas	Imperial (SI) Units
Aircraft	
Total Height	20 ft 8 in (6.30 m)
Total Height (elevator up)	21 ft (6.40 m)
Total Length	68 ft 5 in (20.85 m)
Wing	
Leading Edge MAC (LEMAC)	488.025 in (12.40 m)
Mean Aerodynamic Chord (MAC)	92.64 in (2.35 m)
Span	64 ft 4 in (19.61 m)
Horizontal Stabilizer	
Span	20 ft 4 in (6.20 m)
Fuselage	
Diameter (external)	8 ft 10 in (2.69 m)
Length	63 ft (19.2 m)
Static Ground Angle (nominal)	0 Degrees

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

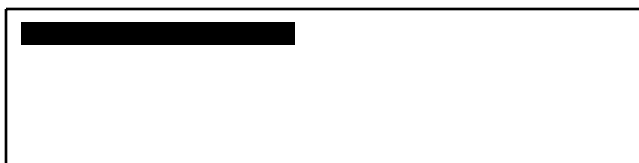
Table 1 – General Dimensions and Areas of the Aircraft	
Dimensions/Areas	Imperial (SI) Units
Area	
Aileron Area (total, left and right)	15.0 ft ² (1.39 m ²)
Elevator Area	23.1 ft ² (2.15 m ²)
Flap Area (total, left and right)	84.0 ft ² (7.80 m ²)
Gross Wing Area [1]	520.0 ft ² (48.31 m ²)
Rudder Area	21.9 ft ² (2.03 m ²)
Total Upper–Exposed Horizontal–Stabilizer	92.5 ft ² (8.59 m ²)
Total Upper–Exposed Vertical–Stabilizer	120.7 ft ² (11.21 m ²)
Total Upper–Exposed Wing [1]	450.0 ft ² (41.81 m ²)
Note: [1] Wing Area includes ailerons, flaps, spoilers and fuselage area.	
Landing Gear	
Main–Landing–Gear Tire Dimension	H27 x 8.5–16 PR [2]
Main–Landing–Gear Vertical Travel of Axle	14.0 in (0.36 m)
Main–Landing–Gear Wheel Dimension	14.0 in (0.36 m)
Main–Landing–Gear Wheel Spacing	17.6 in (0.45 m)
Minimum Turning Circle Diameter	102 ft (31.09 m)
Nose–Landing–Gear Tire Dimension	18 x 4.4–12 PR [2]
Nose–Landing–Gear Vertical Travel of Axle	14.0 in (0.36 m)
Nose–Landing–Gear Wheel Dimension	10.0 in (0.25 m)
Nose–Landing–Gear Wheel Spacing	11.5 in (0.29 m)
Track	125.0 in (3.18 m)
Wheel Base	314.0 in (7.98 m)
NOTE: [2] PR = Ply Rating	

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

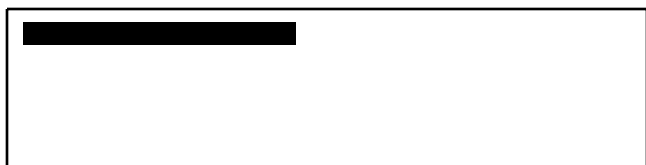


Aircraft Basic Dimensions and Ground Clearances
Figure 1



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



DEFINITIONS

1. General

This section contains a list of acronyms and the definitions of terms found in the Weight and Balance Manual (WBM). The list of acronyms also includes abbreviations.

2. Acronyms

Acronym	Term
ACM	Additional Crew Member
A/C	Aircraft
BL	Buttock Line
BWE	Basic Weight Empty
CG	Center of Gravity
CPLT	Copilot
DWE	Delivered Weight Empty
FCBP	Fuel Curve Boundary–Points
FLT#	Flight Number
FOB	Fuel On Board
FS	Fuselage Station
F/A	Flight Attendant
HSS	Horizontal Stabilizer Station
I/B	Inboard
LEMAC	Leading–Edge Mean–Aerodynamic–Chord
MAC	Mean Aerodynamic Chord
MFW	Minimum Flight Weight
MLG	Main Landing Gear
MLW	Maximum Landing Weight
MRW	Maximum Ramp Weight
MTOW	Maximum Take–Off Weight
MWE	Manufacturer’s Weight Empty

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

Acronym	Term
MZFW	Maximum Zero–Fuel Weight
NLG	Nose Landing Gear
OWE	Operating Weight Empty
O/B	Outboard
PLT	Pilot
RW	Ramp Weight
SB	Service Bulletin
STGR	Stringer
TEW	True Empty Weight
TOW	Take–Off Weight
VSS	Vertical Stabilizer Station
WD	Weighing Datum
WGLTS	Winglet Station
WL	Water Line
WRP	Wing Reference Plane
WS	Wing Station
ZFW	Zero Fuel Weight
ZRD	Zero Reference Datum
3D	Three Dimensions

3. Definitions

Basic Weight Empty (BWE)

The BWE is defined as follows:

TEW + Temporary fixed equipment necessary for particular flights.

NOTE: Examples of temporary equipment are temporary ballast, flight test equipment or ferry kit.

Delivered Weight Empty (DWE)

The DWE includes the MWE, all fixed interior equipment (standard and optional) and the customer options.

Fuel Curve Boundary–Points (FCBP)

A group of points defined by the fuel curve and the longitudinal CG envelope. These FCBP define the area to use for the check of the aircraft longitudinal CG position, given a particular loading.

Manufacturer's Weight Empty (MWE)

The MWE includes the weight of the structure, power plant, systems and interior provisions. The MWE does not include engine oil and the unusable fuel.

Maximum Landing Weight (MLW)

The MLW is the maximum aircraft weight permitted during a landing at the nominal descent velocity.

Maximum Ramp Weight (MRW)

The MRW is the maximum possible aircraft weight. An aircraft is permitted to have the MRW only during ground handling and taxiing.

Maximum Take–Off Weight (MTOW)

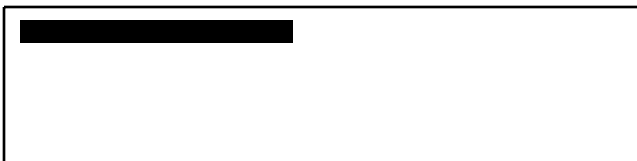
The MTOW is the maximum weight for the take–off operation.

Maximum Zero–Fuel Weight (MZFW)

The MZFW is the maximum permitted aircraft weight before the load of the usable fuel is added to the aircraft.

Mean Aerodynamic Chord (MAC)

The MAC is the chord of an imaginary wing with a constant chord that has the same aerodynamic characteristics as the aircraft wing.



Minimum Flight Weight (MFW)

The MFW is the minimum permitted aircraft weight for flight operations.

Operating Items

The operating items include the engine oil, unusable fuel, crew, crew equipment and passenger service items. Operating items also include other items necessary for the operation of the aircraft during usual flights.

Operating Weight Empty (OWE)

The OWE includes the TEW and operating items. The operating items do not include the engine oil and unusable fuel already included in the TEW.

Payload

The payload includes the passengers, baggage and cargo.

Ramp Weight (RW)

The RW includes the ZFW and usable fuel.

Take-Off Weight (TOW)

The TOW is the weight at the start of the take-off operation.

True Empty Weight (TEW)

The TEW is the weight of the aircraft in a condition where only the crew and fuel are necessary for the flight. The condition can change as follows:

1. $TEW = MWE + \text{Engine Oil} + \text{Unusable Fuel}$.
2. $TEW = DWE + \text{Engine Oil} + \text{Unusable Fuel}$.

Useful Load

The useful load is the difference between the RW and the OWE.

Xarm

The Xarm is the distance along the longitudinal axis of the aircraft from the Zero Reference Datum (ZRD) up to the CG of the item.

Zero Fuel Weight (ZFW)

The ZFW includes the OWE and payload.

AIRCRAFT STATIONS

1. General

To calculate the arm of a given component on the aircraft, aircraft coordinates are used. This section shows the six primary components of the aircraft in relation to their reference planes. The six primary components are:

- Fuselage
- Wing
- Winglet
- Vertical Stabilizer
- Horizontal Stabilizer
- Power Plant.

A. Buttock Line (BL)

The BL are vertical lines perpendicular to the Yarm axis. They are used to measure the distance in inches along the Yarm axis. The plane of symmetry of the aircraft at BL0.00 gives the zero reference datum (ZRD).

B. Fuselage Station (FS)

The FS are vertical lines perpendicular to the Xarm axis. They are used to measure the distance in inches along the Xarm axis. The fuselage station values start at the ZRD and increase thru the rear end of the aircraft, as shown. The ZRD is at 144 inches (3.658 m) forward of the aircraft nose. Fuselage stations, therefore, are numbered from FS144.00 at the tip of the aircraft nose to FS899.82 at the rear extremity of the aircraft fuselage.

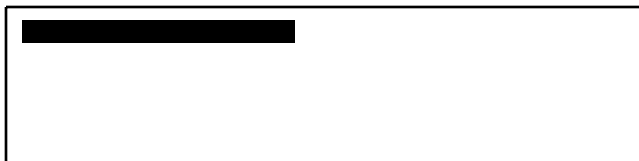
C. Horizontal Stabilizer Station (HSS)

The HSS are vertical lines parallel to the BL. They are measured from the symmetrical center line of the aircraft. The HSS are numbered from HSS0.00 (center line) to HSS122.00 at each stabilizer tip.

D. Nacelle Station (XN)

Nacelle Stations (XN) are numbered from XN100.72 to XN247.90. The XN200.0 aligns with FS674.737.

The engine flanges are identified as A4, A5, A7 and A8.



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

E. Vertical Stabilizer Station (VSS)

The VSS are lines related to the vertical–stabilizer reference axis. The ZRD point for the VSS is at the intersection of WL102.5 and FS794.7. The VSS are perpendicular to the vertical stabilizer 67.50% spar.

F. Water Line (WL)

The WL are horizontal lines perpendicular to the Zarm axis and they are parallel to the fuselage center line. They are used to measure the distance in inches along the Zarm axis.

G. Wing Station (WS)

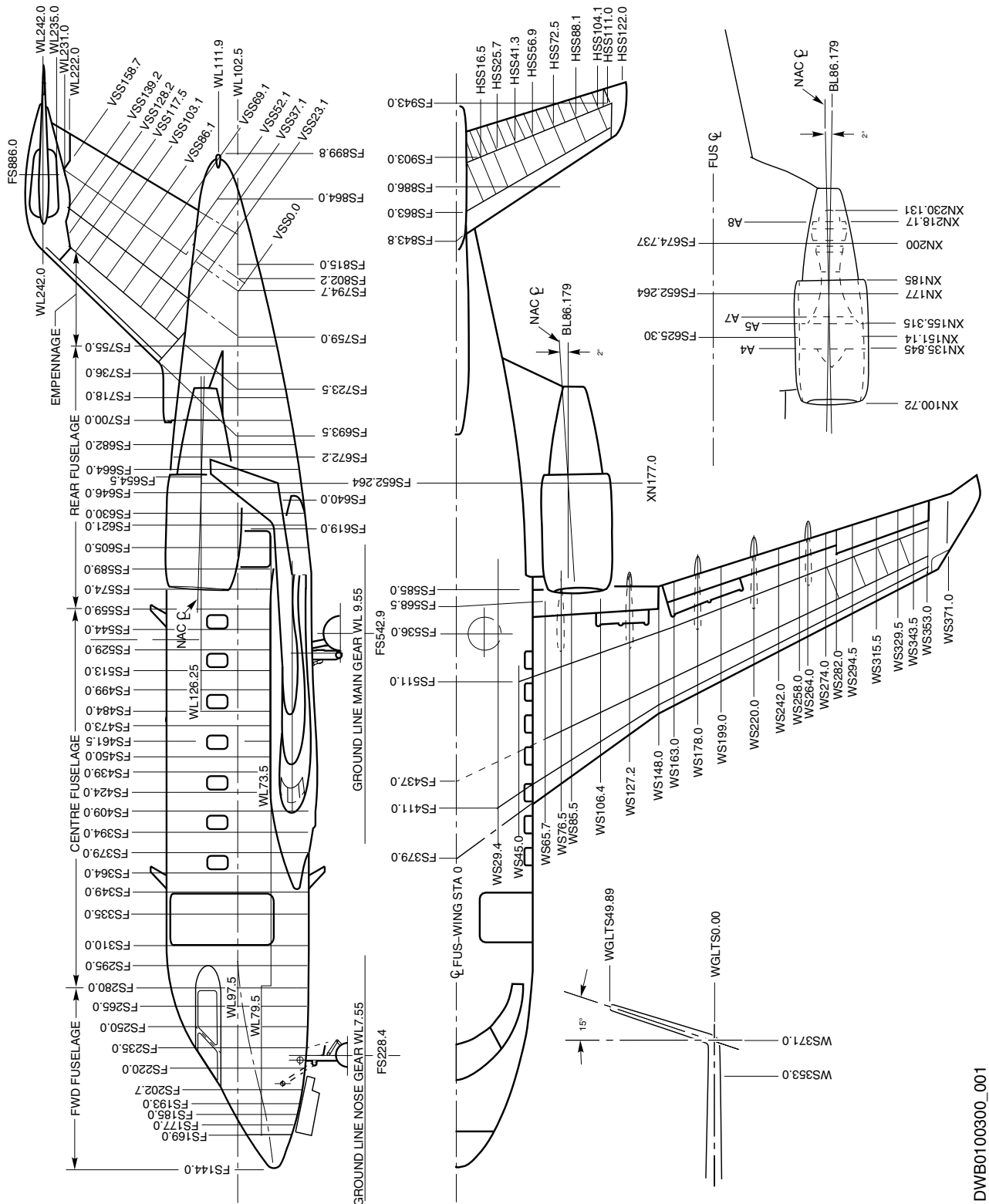
The WS are vertical lines perpendicular to the wing reference plane (WRP). The WS are numbered from WS0.00 (center line) to WS371.08 at each winglet base.

H. Winglet Station (WGLTS)

The WGLTS are horizontal lines parallel to the wing reference plane (WRP). The winglets are numbered from WGLTS 0.00 to WGLTS 49.89 at each wing tip.

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

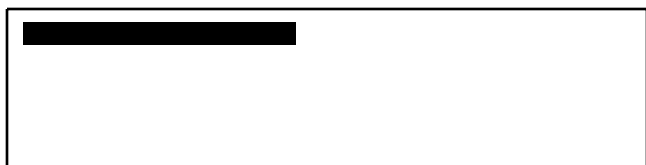


Aircraft Stations
Figure 1

DWB0100300_001

***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



REFERENCE POINTS

1. General

This section gives data related to the reference points.

A. Leading-Edge Mean-Aerodynamic-Chord (LEMAC)

LEMAC is the distance along the Xarm axis from the Zero Reference Datum (ZRD) to the leading edge, at the location where the wing chord is equal to the MAC.

B. Main-Landing-Gear (MLG) Axle Center-Line

Refer to Figure 2 for the MLG axle center-line between fully-extended and fully-compressed positions. It is necessary to measure the center-line when gear reaction is used for weighing.

C. Mean Aerodynamic Chord (MAC)

The MAC is parallel to the fuselage center line. The location of the MAC is between the leading edge and trailing edge of the wing. Refer to Figure 1.

D. Nose-Landing-Gear (NLG) Axle Center-Line

The NLG axle center-line is approximately 228.4 inches (5.80 m) aft of the ZRD. Refer to Figure 1. It is necessary to measure the center-line when gear reaction is used for weighing.

E. Weighing Datum (WD)

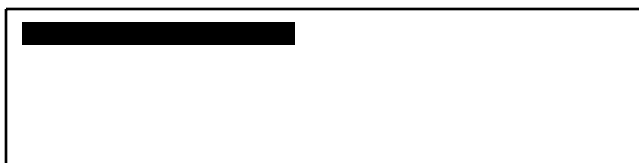
The WD is a reference point that is on the bottom of the aircraft fuselage. The point is forward of the wing fairing on the center-line at FS375.00.

F. Zero Reference Datum (ZRD)

The ZRD is a point that is in the plane of symmetry of the aircraft. The location of this point is at fuselage station FS0.00, approximately 144 inches (3.658 m) forward of the nose radome at buttock line BL0.00, and at water line WL0.00. Refer to Figure 1.

G. Forward Jacking Point Center-Line

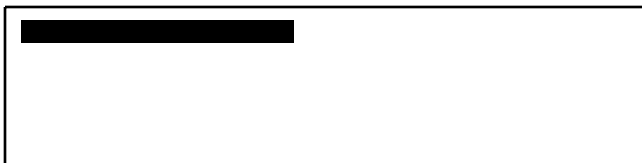
The center line of the forward jacking point is 278 inches (7.06 m) aft of the ZRD. Refer to Figure 1.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

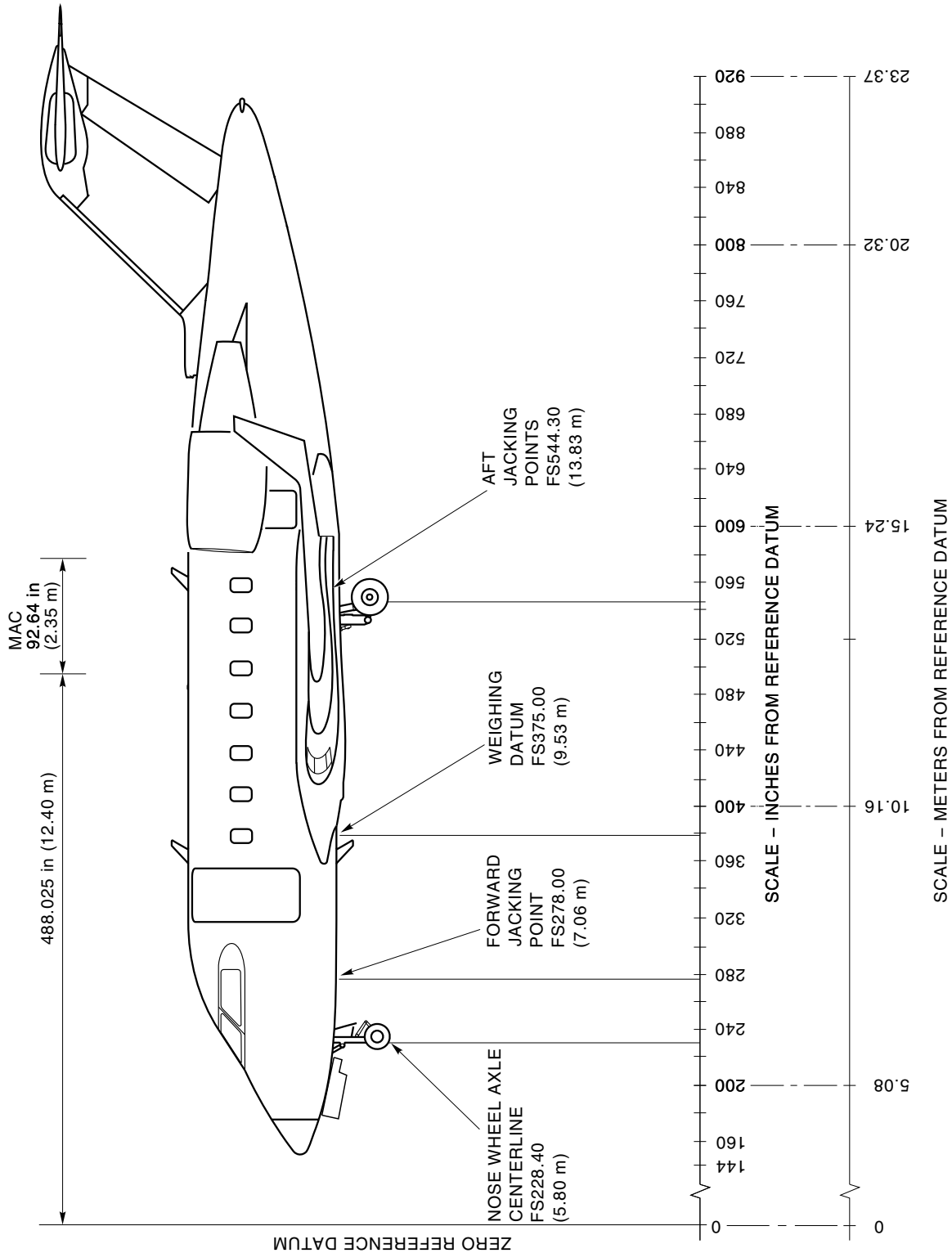
H. Aft Jacking Point Center-Line

The center line of the aft jacking points is 544.3 inches (13.83 m) aft of the ZRD. Refer to Figure 1.



CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

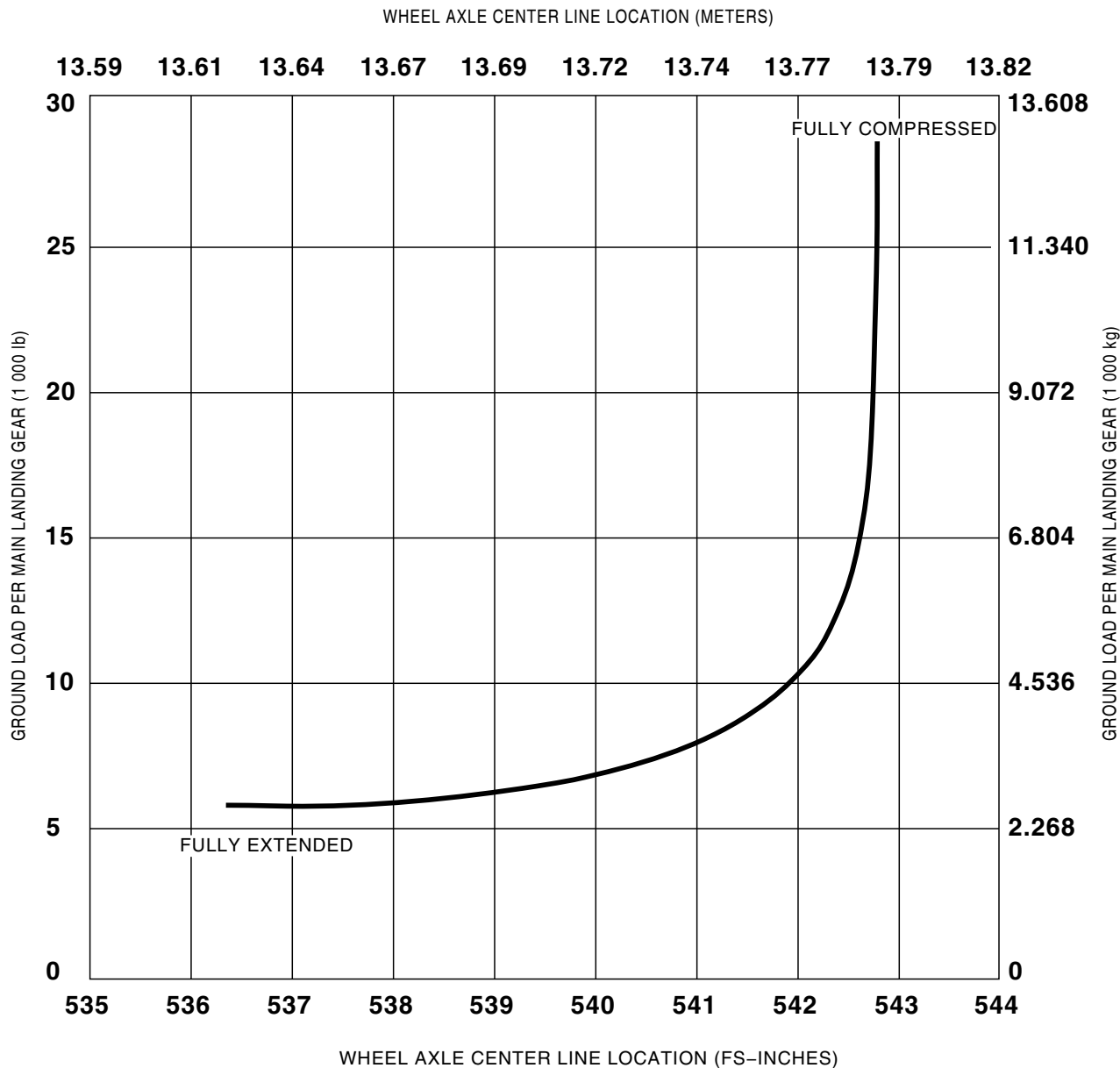


Reference Datum Points
Figure 1

DWB0100400_001

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

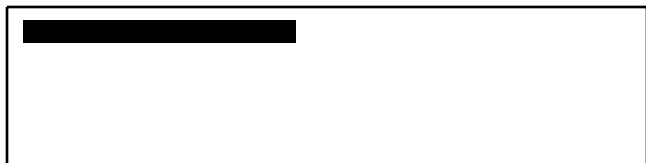


NOTES

1. INFLATION PRESSURE FOR MAIN LANDING GEAR SHOCK ABSORBER IS 470 PSI (3 240.56 kPa) (AT FULLY EXTENDED POSITION).
2. FOR ACCURATE WHEEL AXLE CENTER LINE MEASUREMENT, USE GSE REF. NO. 08-00-05. AN INACCURACY IN THE MEASUREMENT OF THE WHEEL AXLE CENTER LINE WILL AFFECT THE CG CALCULATION.

DWB0100400_002

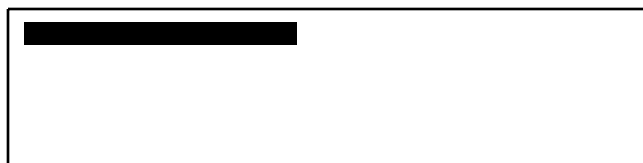
Main Landing Gear – Distance Aft of Reference Datum
Figure 2



DESIGN WEIGHT LIMITS**1. General**

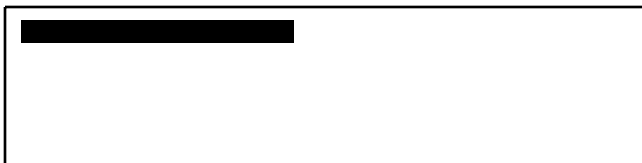
The Table 1 that follows gives the maximum and minimum design weight limits of the aircraft:

Table 1 – Design Weight Limits		
Description	Weight	
	lb	kg
Maximum Ramp Weight (MRW)	48 300	21 909
Maximum Take-Off Weight (MTOW)	48 200	21 863
Maximum Landing Weight (MLW)	38 000	17 237
Maximum Zero-Fuel Weight (MZFW)	32 000	14 515
Minimum Flight Weight (MFW)	26 000	11 794



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



CENTER OF GRAVITY LIMITS**1. General**

This section contains the tables and chart related to the longitudinal CG envelope. The longitudinal CG envelope shows the weight and CG limits for the take-off, flight and landing conditions.

The balance limits are different for ground operations (Subject 01-80-60).

2. MAC Conversion

Use the Table 1A (imperial units) or Table 1B (SI units) that follows to quickly find the relation between the Xarm and the MAC (%MAC). This will make it easier to use the longitudinal CG envelope.

The formula to convert from Xarm to %MAC is:

$$\%MAC = [(Xarm - LEMAC) / MAC] \times 100$$

NOTE: The Weight and Balance Manual gives the definitions of LEMAC and MAC (Subject 01-00-40) . The values of LEMAC and MAC are in Table 1 (Subject 01-00-10) .

Table 1A – Xarm to %MAC (Imperial Units) [1]										
Xarm (in.)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
502	15.1	15.2	15.3	15.4	15.5	15.6	15.7	15.8	15.9	16.1
503	16.2	16.3	16.4	16.5	16.6	16.7	16.8	16.9	17.0	17.1
504	17.2	17.4	17.5	17.6	17.7	17.8	17.9	18.0	18.1	18.2
505	18.3	18.4	18.5	18.6	18.8	18.9 [2]	19.0	19.1	19.2	19.3
506	19.4	19.5	19.6	19.7	19.8	19.9	20.1	20.2	20.3	20.4
507	20.5	20.6	20.7	20.8	20.9	21.0	21.1	21.2	21.3	21.4
508	21.6	21.7	21.8	21.9	22.0	22.1	22.2	22.3	22.4	22.5
509	22.6	22.7	22.9	23.0	23.1	23.2	23.4	23.4	23.5	23.6
510	23.7	23.8	23.9	24.0	24.2	24.3	24.4	24.5	24.6	24.7
511	24.8	24.9	25.0	25.1	25.2	25.3	25.4	25.6	25.7	25.8

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 1A – Xarm to %MAC (Imperial Units) [1]**

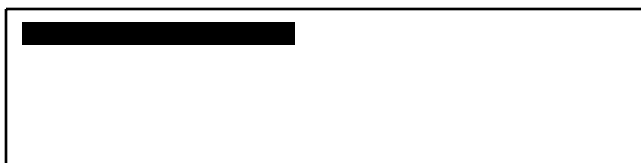
Xarm (in.)	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
512	25.9	26.0	26.1	26.2	26.3	26.4	26.5	26.6	26.7	26.9
513	27.0	27.1	27.2	27.3	27.4	27.5	27.6	27.7	27.8	27.9
514	28.0	28.1	28.3	28.4	28.5	28.6	28.7	28.8	28.9	29.0
515	29.1	29.2	29.3	29.4	29.5	29.7	29.8	29.9	30.0	30.1
516	30.2	30.3	30.4	30.5	30.6	30.7	30.8	31.0	31.1	31.2
517	31.3	31.4	31.5	31.6	31.7	31.8	31.9	32.0	32.1	32.2
518	32.4	32.5	32.6	32.7	32.8	32.9	33.0	33.1	33.2	33.3
519	33.4	33.5	33.7	33.8	33.9	34.0	34.1	34.2	34.3	34.4
520	34.5	34.6	34.7	34.8	34.9	35.1	35.2	35.3	35.4	35.5
521	35.6	35.7	35.8	35.9	36.0	36.1	36.2	36.4	36.5	36.6
522	36.7	36.8	36.9	37.0	37.1	37.2	37.3	37.4	37.5	37.6
523	37.8	37.9	38.0	38.1	38.2	38.3	38.4	38.5	38.6	38.7
524	38.8	38.9	39.0	39.2	39.3	39.4	39.5	39.6	39.7	39.8

[1] The Xarm is equivalent to the FS. The Xarm is the distance in inches aft of the ZRD.

[2] Example: FS505.5 = 18.9 %MAC

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL**

Table 1B – Xarm to %MAC (SI Units) [1]										
Xarm (m)	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
12.7	12.9	13.3	13.8	14.2	14.6	15.0	15.5	15.9	16.3	16.7
12.8	17.2	17.6	18.0	18.4	18.9 [2]	19.3	19.7	20.1	20.6	21.0
12.9	21.4	21.8	22.3	22.7	23.1	23.5	24.0	24.4	24.8	25.2
13.0	25.7	26.1	26.5	26.9	27.4	27.8	28.2	28.6	29.1	29.5
13.1	29.9	30.3	30.8	31.2	31.6	32.1	32.5	32.9	33.3	33.7
13.2	34.2	34.6	35.0	35.4	35.9	36.3	36.7	37.1	37.6	38.0
13.3	38.4	38.8	39.3	39.7	40.1	40.5	41.0	41.4	41.8	42.2
[1] The Xarm is equivalent to the FS. The Xarm is the distance in meters aft of the ZRD. [2] Example: FS505.5 = 12.84 m = 18.9 %MAC										



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

3. Longitudinal CG Envelope

Refer to Figure 1.

The Table 2A (imperial units) and the Table 2B (SI units) that follow give the center of gravity limits:

Table 2A – Center of Gravity Limits (MTOW 48 200 lb) (Imperial Units)									
Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
lb	in.	lb in.		in.	lb in.		in.	lb in.	
26 000	506.55	13 170	20.0	506.55	13 170	20.0	520.45	13 532	35.0
26 200	506.55	13 272	20.0	506.55	13 272	20.0	520.45	13 636	35.0
26 400	506.55	13 373	20.0	506.55	13 373	20.0	520.45	13 740	35.0
26 600	506.55	13 474	20.0	506.55	13 474	20.0	520.45	13 844	35.0
26 800	506.55	13 576	20.0	506.55	13 576	20.0	520.45	13 948	35.0
27 000	506.55	13 677	20.0	506.55	13 677	20.0	520.45	14 052	35.0
27 200	506.55	13 778	20.0	506.55	13 778	20.0	520.45	14 156	35.0
27 400	506.55	13 880	20.0	506.55	13 880	20.0	520.45	14 260	35.0
27 600	506.55	13 981	20.0	506.55	13 981	20.0	520.45	14 364	35.0
27 800	506.55	14 082	20.0	506.55	14 082	20.0	520.45	14 468	35.0
28 000	506.55	14 183	20.0	506.55	14 183	20.0	520.45	14 573	35.0
28 200	506.55	14 285	20.0	506.55	14 285	20.0	520.45	14 677	35.0
28 400	506.55	14 386	20.0	506.55	14 386	20.0	520.45	14 781	35.0
28 600	506.55	14 487	20.0	506.55	14 487	20.0	520.45	14 885	35.0
28 800	506.55	14 589	20.0	506.55	14 589	20.0	520.45	14 989	35.0
29 000	506.55	14 690	20.0	506.55	14 690	20.0	520.45	15 093	35.0
29 200	506.55	14 791	20.0	506.55	14 791	20.0	520.45	15 197	35.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2A – Center of Gravity Limits (MTOW 48 200 lb) (Imperial Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	lb	in.		lb	in.		lb	in.	
29 400	506.55	14 893	20.0	506.55	14 893	20.0	520.45	15 301	35.0
29 600	506.55	14 994	20.0	506.55	14 994	20.0	520.45	15 405	35.0
29 800	506.55	15 095	20.0	506.55	15 095	20.0	520.45	15 509	35.0
30 000	506.55	15 197	20.0	506.55	15 197	20.0	520.45	15 613	35.0
30 200	506.55	15 298	20.0	506.55	15 298	20.0	520.45	15 718	35.0
30 400	506.55	15 399	20.0	506.55	15 399	20.0	520.45	15 822	35.0
30 600	506.55	15 501	20.0	506.55	15 501	20.0	520.45	15 926	35.0
30 800	506.55	15 602	20.0	506.55	15 602	20.0	520.45	16 030	35.0
31 000	506.55	15 703	20.0	506.55	15 703	20.0	520.45	16 134	35.0
31 200	506.55	15 804	20.0	506.55	15 804	20.0	520.45	16 238	35.0
31 400	506.55	15 906	20.0	506.55	15 906	20.0	520.45	16 342	35.0
31 600	506.55	16 007	20.0	506.55	16 007	20.0	520.45	16 446	35.0
31 800	506.55	16 108	20.0	506.55	16 108	20.0	520.45	16 550	35.0
32 000	506.55	16 210	20.0	506.55	16 210	20.0	520.45	16 654	35.0
32 200	506.55	16 311	20.0	506.55	16 311	20.0	520.45	16 758	35.0
32 400	506.55	16 412	20.0	506.55	16 412	20.0	520.45	16 863	35.0
32 600	506.55	16 514	20.0	506.55	16 514	20.0	520.45	16 967	35.0
32 800	506.55	16 615	20.0	506.55	16 615	20.0	520.45	17 071	35.0
33 000	506.55	16 716	20.0	506.55	16 716	20.0	520.45	17 175	35.0
33 200	506.55	16 818	20.0	506.55	16 818	20.0	520.45	17 279	35.0
33 400	506.55	16 919	20.0	506.55	16 919	20.0	520.45	17 383	35.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2A – Center of Gravity Limits (MTOW 48 200 lb) (Imperial Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	lb	in.		lb	in.		lb	in.	
33 600	506.55	17 020	20.0	506.55	17 020	20.0	520.45	17 487	35.0
33 800	506.55	17 121	20.0	506.55	17 121	20.0	520.45	17 591	35.0
34 000	506.55	17 223	20.0	506.55	17 223	20.0	520.45	17 695	35.0
34 200	506.55	17 324	20.0	506.55	17 324	20.0	520.45	17 799	35.0
34 400	506.55	17 425	20.0	506.55	17 425	20.0	520.45	17 903	35.0
34 600	506.55	17 527	20.0	506.55	17 527	20.0	520.45	18 008	35.0
34 800	506.55	17 628	20.0	506.55	17 628	20.0	520.45	18 112	35.0
35 000	506.55	17 729	20.0	506.55	17 729	20.0	520.45	18 216	35.0
35 200	506.55	17 831	20.0	506.55	17 831	20.0	520.45	18 320	35.0
35 400	506.55	17 932	20.0	506.55	17 932	20.0	520.45	18 424	35.0
35 600	506.55	18 033	20.0	506.55	18 033	20.0	520.45	18 528	35.0
35 800	506.55	18 135	20.0	506.55	18 135	20.0	520.45	18 632	35.0
36 000	506.55	18 236	20.0	506.55	18 236	20.0	520.45	18 736	35.0
36 200	506.55	18 337	20.0	506.55	18 337	20.0	520.45	18 840	35.0
36 400	506.55	18 439	20.0	506.55	18 439	20.0	520.45	18 944	35.0
36 600	506.55	18 540	20.0	506.55	18 540	20.0	520.45	19 048	35.0
36 800	506.55	18 641	20.0	506.55	18 641	20.0	520.45	19 153	35.0
37 000	506.55	18 742	20.0	506.55	18 742	20.0	520.45	19 257	35.0
37 200	506.55	18 844	20.0	506.55	18 844	20.0	520.45	19 361	35.0
37 400	506.55	18 945	20.0	506.55	18 945	20.0	520.45	19 465	35.0
37 600	506.55	19 046	20.0	506.55	19 046	20.0	520.45	19 569	35.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2A – Center of Gravity Limits (MTOW 48 200 lb) (Imperial Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	lb	in.		lb	in.		lb	in.	
37 800	506.55	19 148	20.0	506.55	19 148	20.0	520.45	19 673	35.0
38 000	506.55	19 249	20.0	506.55	19 249	20.0	520.45	19 777	35.0
38 200	506.06	19 331	19.5	506.06	19 331	19.5	520.56	19 885	35.1
38 400	505.56	19 414	18.9	505.56	19 414	18.9	520.67	19 994	35.2
38 600	505.07	19 496	18.4	505.07	19 496	18.4	520.78	20 102	35.4
38 800	504.58	19 578	17.9	504.58	19 578	17.9	520.89	20 211	35.5
39 000	504.08	19 659	17.3	504.08	19 659	17.3	521.00	20 319	35.6
39 200	503.59	19 741	16.8	503.59	19 741	16.8	521.12	20 428	35.7
39 400	503.09	19 822	16.3	503.09	19 822	16.3	521.23	20 536	35.8
39 500	502.85	19 862	16.0	502.85	19 862	16.0	521.28	20 591	35.9
39 600	502.85	19 913	16.0	502.98	19 918	16.2	521.34	20 645	36.0
39 800	502.85	20 013	16.0	503.25	20 030	16.4	521.45	20 754	36.1
40 000	502.85	20 114	16.0	503.53	20 141	16.7	521.56	20 862	36.2
40 200	502.85	20 214	16.0	503.80	20 253	17.0	521.67	20 971	36.3
40 400	502.85	20 315	16.0	504.07	20 364	17.3	521.78	21 080	36.4
40 600	502.85	20 416	16.0	504.34	20 476	17.6	521.89	21 189	36.6
40 800	502.85	20 516	16.0	504.61	20 588	17.9	522.01	21 298	36.7
41 000	502.85	20 617	16.0	504.88	20 700	18.2	522.12	21 407	36.8
41 200	502.85	20 717	16.0	505.15	20 812	18.5	522.23	21 516	36.9
41 400	502.85	20 818	16.0	505.42	20 925	18.8	522.34	21 625	37.0
41 600	502.85	20 918	16.0	505.69	21 037	19.1	522.45	21 734	37.2

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2A – Center of Gravity Limits (MTOW 48 200 lb) (Imperial Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	lb	in.		lb	in.		lb	in.	
41 800	502.85	21 019	16.0	505.97	21 149	19.4	522.56	21 843	37.3
42 000	502.85	21 120	16.0	506.24	21 262	19.7	522.67	21 952	37.4
42 200	502.85	21 220	16.0	506.51	21 375	20.0	522.78	22 061	37.5
42 400	502.85	21 321	16.0	506.78	21 487	20.2	522.89	22 171	37.6
42 600	502.85	21 421	16.0	507.05	21 600	20.5	523.01	22 280	37.8
42 800	502.85	21 522	16.0	507.32	21 713	20.8	523.12	22 389	37.9
43 000	502.85	21 622	16.0	507.59	21 826	21.1	523.23	22 499	38.0
43 200	502.85	21 723	16.0	507.86	21 940	21.4	523.23	22 603	38.0
43 400	502.85	21 824	16.0	508.13	22 053	21.7	523.23	22 708	38.0
43 600	502.85	21 924	16.0	508.41	22 166	22.0	523.23	22 813	38.0
43 800	502.85	22 025	16.0	508.68	22 280	22.3	523.23	22 917	38.0
44 000	502.85	22 125	16.0	508.95	22 394	22.6	523.23	23 022	38.0
44 200	502.85	22 226	16.0	509.22	22 507	22.9	523.23	23 127	38.0
44 400	502.85	22 326	16.0	509.49	22 621	23.2	523.23	23 231	38.0
44 600	502.85	22 427	16.0	509.76	22 735	23.5	523.23	23 336	38.0
44 750	502.85	22 502	16.0	509.96	22 821	23.7	523.23	23 414	38.0
44 800	502.91	22 530	16.1	510.03	22 849	23.8	523.23	23 441	38.0
45 000	503.16	22 642	16.3	510.30	22 964	24.0	523.23	23 545	38.0
45 200	503.41	22 754	16.6	510.57	23 078	24.3	523.23	23 650	38.0
45 400	503.66	22 866	16.9	510.85	23 192	24.6	523.23	23 755	38.0
45 600	503.92	22 979	17.2	511.12	23 307	24.9	523.23	23 859	38.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2A – Center of Gravity Limits (MTOW 48 200 lb) (Imperial Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	lb	in.		lb	in.		lb	in.	
45 800	504.17	23 091	17.4	511.39	23 422	25.2	523.23	23 964	38.0
46 000	504.42	23 203	17.7	511.66	23 536	25.5	523.23	24 068	38.0
46 200	504.67	23 316	18.0	511.93	23 651	25.8	523.23	24 173	38.0
46 400	504.92	23 428	18.2	512.20	23 766	26.1	523.23	24 278	38.0
46 600	505.17	23 541	18.5	512.47	23 881	26.4	523.23	24 382	38.0
46 800	505.42	23 654	18.8	512.74	23 996	26.7	523.23	24 487	38.0
47 000	505.67	23 767	19.0	513.02	24 112	27.0	523.23	24 592	38.0
47 200	505.92	23 880	19.3	513.29	24 227	27.3	523.23	24 696	38.0
47 400	506.18	23 993	19.6	513.56	24 343	27.6	523.23	24 801	38.0
47 600	506.43	24 106	19.9	513.83	24 290	27.8	523.23	24 906	38.0
47 700	506.55	24 163	20.0	513.96	24 516	28.0	523.23	24 958	38.0
47 800	506.71	24 221	20.2	514.12	24 575	28.2	523.23	25 010	38.0
47 900	506.86	24 279	20.3	514.27	24 634	28.3	523.23	25 063	38.0
48 000	507.02	24 337	20.5	514.43	24 693	28.5	523.23	25 115	38.0
48 100	507.17	24 395	20.7	514.58	24 751	28.7	523.23	25 167	38.0
48 200	507.33	24 453	20.8	514.74	24 810	28.8	523.23	25 220	38.0
48 300	507.48	24 511	21.0	514.89	24 869	29.0	523.23	25 272	38.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2B – Center of Gravity Limits (MTOW 21 863 kg) (SI Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	kg	m		kg	m		kg	m	
11 794	12.87	151.79	20.0	12.87	151.79	20.0	13.21	155.80	35.0
11 884	12.87	152.95	20.0	12.87	152.95	20.0	13.21	156.99	35.0
11 975	12.87	154.12	20.0	12.87	154.12	20.0	13.21	158.19	35.0
12 066	12.87	155.29	20.0	12.87	155.29	20.0	13.21	159.39	35.0
12 156	12.87	156.45	20.0	12.87	156.45	20.0	13.21	160.58	35.0
12 247	12.87	157.62	20.0	12.87	157.62	20.0	13.21	161.78	35.0
12 338	12.87	158.79	20.0	12.87	158.79	20.0	13.21	162.98	35.0
12 429	12.87	159.96	20.0	12.87	159.96	20.0	13.21	164.19	35.0
12 519	12.87	161.12	20.0	12.87	161.12	20.0	13.21	165.38	35.0
12 610	12.87	162.29	20.0	12.87	162.29	20.0	13.21	166.58	35.0
12 701	12.87	163.46	20.0	12.87	163.46	20.0	13.21	167.78	35.0
12 792	12.87	164.63	20.0	12.87	164.63	20.0	13.21	168.98	35.0
12 882	12.87	165.79	20.0	12.87	165.79	20.0	13.21	170.17	35.0
12 973	12.87	166.96	20.0	12.87	166.96	20.0	13.21	171.37	35.0
13 064	12.87	168.13	20.0	12.87	168.13	20.0	13.21	172.58	35.0
13 154	12.87	169.29	20.0	12.87	169.29	20.0	13.21	173.76	35.0
13 245	12.87	170.46	20.0	12.87	170.46	20.0	13.21	174.97	35.0
13 336	12.87	171.63	20.0	12.87	171.63	20.0	13.21	176.17	35.0
13 427	12.87	172.81	20.0	12.87	172.81	20.0	13.21	177.37	35.0
13 517	12.87	173.96	20.0	12.87	173.96	20.0	13.21	178.56	35.0
13 608	12.87	175.13	20.0	12.87	175.13	20.0	13.21	179.76	35.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2B – Center of Gravity Limits (MTOW 21 863 kg) (SI Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	kg	m		m	kg m		m	kg m	
13 699	12.87	176.31	20.0	12.87	176.31	20.0	13.21	180.96	35.0
13 789	12.87	177.46	20.0	12.87	177.46	20.0	13.21	182.15	35.0
13 880	12.87	178.64	20.0	12.87	178.64	20.0	13.21	183.35	35.0
13 971	12.87	179.81	20.0	12.87	179.81	20.0	13.21	184.56	35.0
14 062	12.87	180.98	20.0	12.87	180.98	20.0	13.21	185.76	35.0
14 152	12.87	182.14	20.0	12.87	182.14	20.0	13.21	186.95	35.0
14 243	12.87	183.31	20.0	12.87	183.31	20.0	13.21	188.15	35.0
14 334	12.87	184.48	20.0	12.87	184.48	20.0	13.21	189.35	35.0
14.424	12.87	185.64	20.0	12.87	185.64	20.0	13.21	190.54	35.0
14 515	12.87	186.81	20.0	12.87	186.81	20.0	13.21	191.74	35.0
14 606	12.87	187.98	20.0	12.87	187.98	20.0	13.21	192.95	35.0
14 697	12.87	189.15	20.0	12.87	189.15	20.0	13.21	194.15	35.0
14 787	12.87	190.31	20.0	12.87	190.31	20.0	13.21	195.34	35.0
14 878	12.87	191.48	20.0	12.87	191.48	20.0	13.21	196.54	35.0
14 969	12.87	192.65	20.0	12.87	192.65	20.0	13.21	197.74	35.0
15 060	12.87	193.82	20.0	12.87	193.82	20.0	13.21	198.94	35.0
15 150	12.87	194.98	20.0	12.87	194.98	20.0	13.21	200.13	35.0
15 241	12.87	196.15	20.0	12.87	196.15	20.0	13.21	201.33	35.0
15 332	12.87	197.32	20.0	12.87	197.32	20.0	13.21	202.54	35.0
15 422	12.87	198.48	20.0	12.87	198.48	20.0	13.21	203.72	35.0
15 513	12.87	199.65	20.0	12.87	199.65	20.0	13.21	204.93	35.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2B – Center of Gravity Limits (MTOW 21 863 kg) (SI Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	kg	m		kg m	m		kg m	m	
15 604	12.87	200.82	20.0	12.87	200.82	20.0	13.21	206.13	35.0
15 695	12.87	201.99	20.0	12.87	201.99	20.0	13.21	207.33	35.0
15 785	12.87	203.15	20.0	12.87	203.15	20.0	13.21	208.52	35.0
15 876	12.87	204.32	20.0	12.87	204.32	20.0	13.21	209.72	35.0
16 057	12.87	206.65	20.0	12.87	206.65	20.0	13.21	212.11	35.0
16 148	12.87	207.82	20.0	12.87	207.82	20.0	13.21	213.32	35.0
16 239	12.87	209.00	20.0	12.87	209.00	20.0	13.21	214.52	35.0
16 330	12.87	210.17	20.0	12.87	210.17	20.0	13.21	215.72	35.0
16 420	12.87	211.33	20.0	12.87	211.33	20.0	13.21	216.91	35.0
16 511	12.87	212.50	20.0	12.87	212.50	20.0	13.21	218.11	35.0
16 602	12.87	213.67	20.0	12.87	213.67	20.0	13.21	219.31	35.0
16 692	12.87	214.82	20.0	12.87	214.82	20.0	13.21	220.50	35.0
16 783	12.87	216.00	20.0	12.87	216.00	20.0	13.21	221.70	35.0
16 874	12.87	217.17	20.0	12.87	217.17	20.0	13.21	222.91	35.0
16 965	12.87	218.34	20.0	12.87	218.34	20.0	13.21	224.11	35.0
17 055	12.87	219.50	20.0	12.87	219.50	20.0	13.21	225.30	35.0
17 146	12.87	220.67	20.0	12.87	220.67	20.0	13.21	226.50	35.0
17 237	12.87	221.84	20.0	12.87	221.84	20.0	13.21	227.70	35.0
17 328	12.85	222.66	19.5	12.85	222.66	19.5	13.22	229.08	35.1
17 418	12.84	223.65	18.9	12.84	223.65	18.9	13.23	230.44	35.2
17 509	12.83	224.64	18.4	12.83	224.64	18.4	13.23	231.64	35.4

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2B – Center of Gravity Limits (MTOW 21 863 kg) (SI Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	kg	m		kg m	m		kg m	m	
17 600	12.82	225.63	17.9	12.82	225.63	17.9	13.23	232.85	35.5
17 690	12.80	226.43	17.3	12.80	226.43	17.3	13.23	234.04	35.6
17 781	12.79	227.42	16.8	12.79	227.42	16.8	13.24	235.42	35.7
17 872	12.78	228.40	16.3	12.78	228.40	16.3	13.24	236.63	35.8
17 917	12.77	228.80	16.0	12.77	228.80	16.0	13.24	237.22	35.9
17 963	12.77	229.39	16.0	12.78	229.57	16.2	13.24	237.83	36.0
18 053	12.77	230.54	16.0	12.78	230.72	16.4	13.24	239.02	36.1
18 144	12.77	231.70	16.0	12.79	232.06	16.7	13.25	240.41	36.2
18 235	12.77	232.86	16.0	12.80	233.41	17.0	13.25	241.61	36.3
18 325	12.77	234.01	16.0	12.80	234.56	17.3	13.25	242.81	36.4
18 416	12.77	235.17	16.0	12.81	235.91	17.6	13.26	244.20	36.6
18 507	12.77	236.33	16.0	12.82	237.26	17.9	13.26	245.40	36.7
18 598	12.77	237.50	16.0	12.82	238.43	18.2	13.26	246.61	36.8
18 688	12.77	238.65	16.0	12.83	239.77	18.5	13.26	247.80	36.9
18 779	12.77	239.81	16.0	12.84	241.12	18.8	13.27	249.20	37.0
18 870	12.77	240.97	16.0	12.84	242.29	19.1	13.27	250.40	37.2
18 960	12.77	242.12	16.0	12.85	243.64	19.4	13.27	251.60	37.3
19 051	12.77	243.28	16.0	12.86	245.00	19.7	13.28	253.00	37.4
19 142	12.77	244.44	16.0	12.87	246.36	20.0	13.28	254.21	37.5
19 233	12.77	245.61	16.0	12.87	247.53	20.2	13.28	255.41	37.6
19 323	12.77	246.75	16.0	12.88	248.88	20.5	13.28	256.61	37.8

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2B – Center of Gravity Limits (MTOW 21 863 kg) (SI Units)**

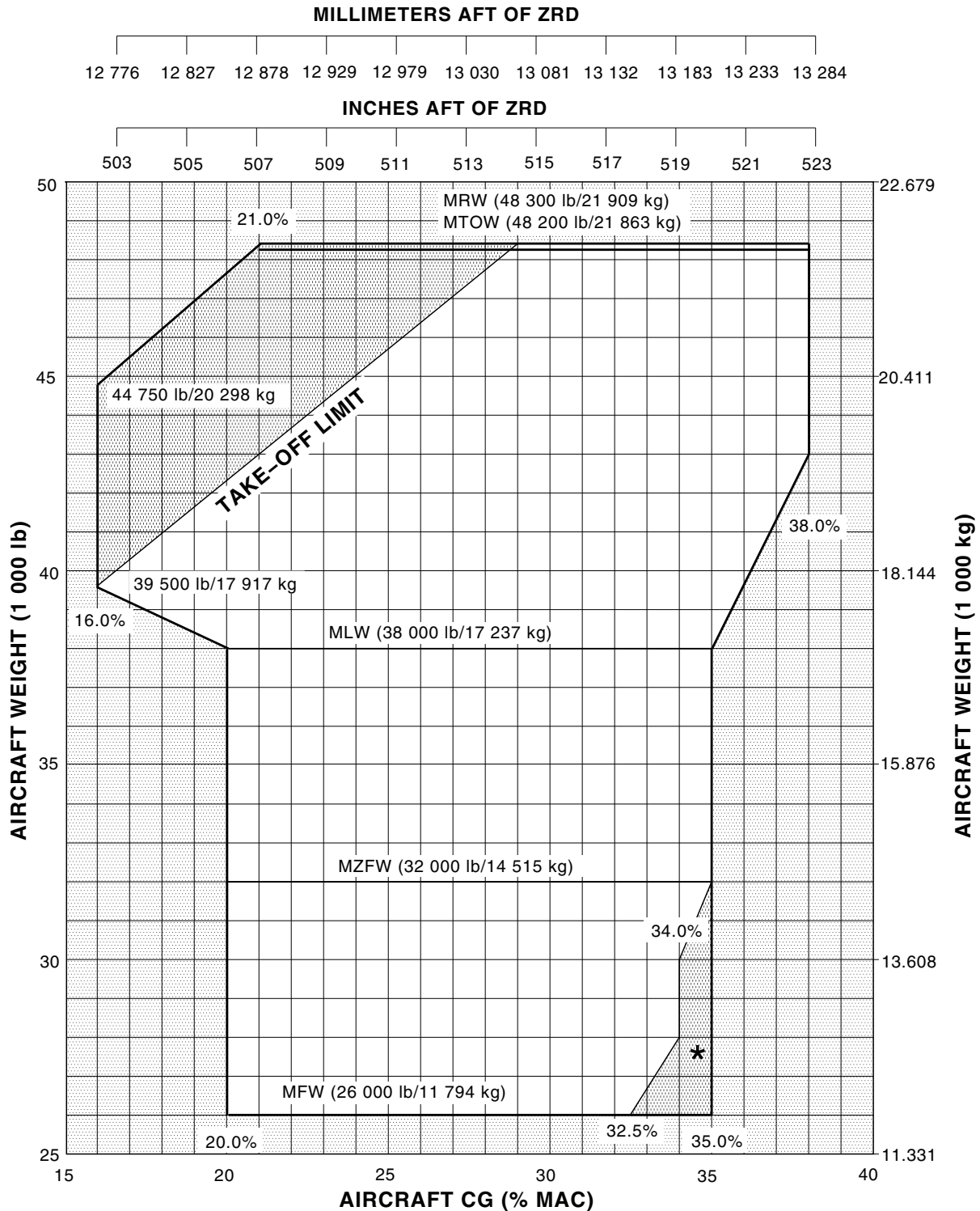
Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	kg	m		kg m	m		kg m	m	
19 414	12.77	247.92	16.0	12.89	250.25	20.8	13.29	258.01	37.9
19 505	12.77	249.08	16.0	12.89	251.42	21.1	13.29	259.22	38.0
19 596	12.77	250.24	16.0	12.90	252.79	21.4	13.29	260.43	38.0
19 686	12.77	251.39	16.0	12.91	254.15	21.7	13.29	261.63	38.0
19 777	12.77	252.55	16.0	12.91	255.32	22.0	13.29	262.84	38.0
19 868	12.77	253.71	16.0	12.92	256.70	22.3	13.29	264.05	38.0
19 958	12.77	254.87	16.0	12.93	258.00	22.6	13.29	265.24	38.0
20 049	12.77	256.03	16.0	12.93	259.32	22.9	13.29	266.45	38.0
20 140	12.77	257.19	16.0	12.94	260.63	23.2	13.29	267.66	38.0
20 231	12.77	258.34	16.0	12.95	261.95	23.5	13.29	268.87	38.0
20 298	12.77	259.21	16.0	12.95	262.93	23.7	13.29	269.77	38.0
20 321	12.77	259.58	16.1	12.95	263.25	23.8	13.29	270.07	38.0
20 412	12.78	260.87	16.3	12.96	264.57	24.0	13.29	271.28	38.0
20 503	12.79	262.16	16.6	12.97	265.89	24.3	13.29	272.48	38.0
20 593	12.79	263.45	16.9	12.98	267.21	24.6	13.29	273.68	38.0
20 684	12.80	264.75	17.2	12.98	268.53	24.9	13.29	274.89	38.0
20 775	12.81	266.04	17.4	12.99	269.85	25.2	13.29	276.10	38.0
20 866	12.81	267.34	17.7	13.00	271.18	25.5	13.29	277.31	38.0
20 956	12.82	268.63	18.0	13.00	272.49	25.8	13.29	278.51	38.0
21 047	12.82	269.93	18.2	13.01	273.82	26.1	13.29	279.71	38.0
21 138	12.83	271.23	18.5	13.02	275.15	26.4	13.29	280.92	38.0

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2B – Center of Gravity Limits (MTOW 21 863 kg) (SI Units)**

Weight	Forward Limit			Forward Take-Off Limit			Aft Limit		
	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC	Xarm	Moment/ 1 000	%MAC
	kg	m		m	kg m		m	kg m	
21 228	12.84	272.52	18.8	13.02	276.46	26.7	13.29	282.12	38.0
21 319	12.84	273.82	19.0	13.03	277.80	27.0	13.29	283.33	38.0
21 410	12.85	275.13	19.3	13.04	279.13	27.3	13.29	284.54	38.0
21 501	12.86	276.44	19.6	13.04	280.47	27.6	13.29	285.75	38.0
21 591	12.86	277.73	19.9	13.05	281.79	27.8	13.29	286.94	38.0
21 637	12.87	278.39	20.0	13.05	282.46	28.0	13.29	287.56	38.0
21 682	12.87	279.06	20.2	13.06	283.14	28.2	13.29	288.15	38.0
21 727	12.87	279.72	20.3	13.06	283.81	28.3	13.29	288.75	38.0
21 773	12.88	280.40	20.5	13.07	284.50	28.5	13.29	289.36	38.0
21 818	12.88	281.06	20.7	13.07	285.17	28.7	13.29	289.96	38.0
21 863	12.88	281.60	20.8	13.07	285.75	28.8	13.29	290.56	38.0
21 909	12.89	282.39	21.0	13.08	286.56	29.0	13.29	291.16	38.0

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL



[★ For zero fuel weight in that region, refer to Figure 3, 01-20-20, Auxiliary and Tail Tank Admissible Ratio (for ZFW CG inside lower aft zone of CG envelope)]

Center of Gravity Limits
Figure 1

DWB0100600_004

***CHALLENGER* 650**

CL-650 WEIGHT AND BALANCE MANUAL

TAKE-OFF TRIM SETTINGS

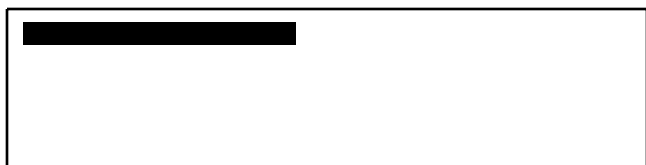
1. General

The adjustment of stabilizer settings for take-off must agree with the related CG of the aircraft. Refer to the Airplane Flight Manual – NORMAL PROCEDURES Section, FLIGHT CONTROLS – BEFORE TAKE-OFF subject to find the correct take-off trim settings.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

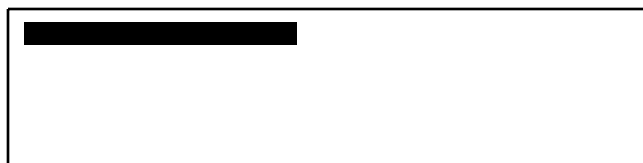
THIS PAGE INTENTIONALLY LEFT BLANK



BALANCE EFFECT CAUSED BY MAJOR OPERATION**1. General**

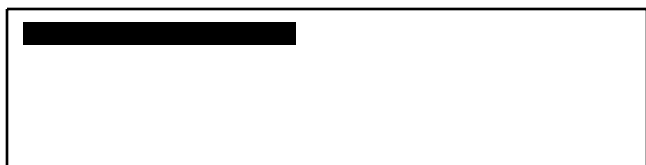
The operation of some systems changes the CG of the aircraft during flight, ground handling, take-off and landing operations. The Table 1 that follows shows the effect of these operations on the aircraft CG:

Table 1 – Balance Effect Caused by Major Operation		
Items	Aircraft Delta Moment for Items Change from Fully Deployed to Fully Stowed Position	
	Moment/1 000	
	lb in.	kg m
Gears Up Compared to Gears on Ground	–3.7	–0.043



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



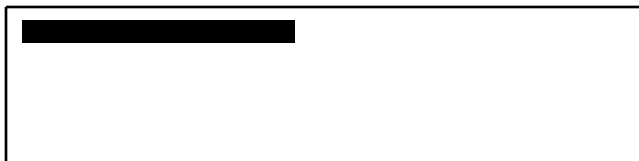
CHALLENGER⁶⁵⁰
CL-650 WEIGHT AND BALANCE MANUAL

FUEL

1. General

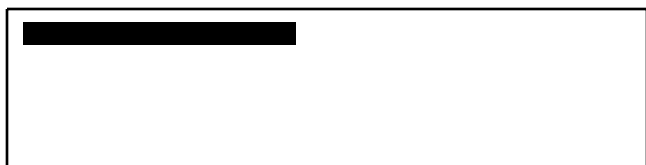
This section contains the subjects related to fuel that follow:

- Fuel Tank Configuration and Capacities (Subject 01-20-10)
- Fuel Management (Subject 01-20-20).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



01-20-00

Page 2
Oct 27/2015

FUEL TANK CONFIGURATION AND CAPACITIES**1. Fuel Tank Configuration**

Refer to Figure 1.

The configuration of the fuel tanks is as shown.

A. Description

The complete wing box structure is sealed to form three fuel tanks, the two main tanks, and the center wing auxiliary tank. Two other auxiliary tanks are located in the fuselage, fwd and aft of the center wing auxiliary tank. The tail tanks are located in the aft fuselage.

Main Tanks

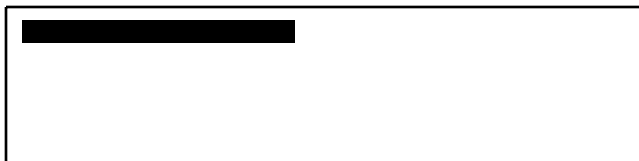
- Left main tank (L)
- Right main tank (R).

Auxiliary Tanks

- Center wing auxiliary tank (A)
- Forward auxiliary tank (FA)
- Aft auxiliary tank (AA).

Tail Tanks

- Tail cone tank (T)
- Two saddle tanks (S).



2. Fuel Tank Capacities

The Table 1 that follows shows the total volume, weight, center of gravity (CG) and moment for the fuel tanks:

Table 1 – Fuel Tank Capacities								
Description	Volume		Weight [1]		CG		Moment/1 000	
	U.S. gal	L	lb	kg	in.	m	lb in	kg m
Main Tanks (L and R) (per side)	720	2 726	4 860	2 205	506.6	12.867	2 462	28.372
Auxiliary Tanks (Total)	1 062	4 020	7 168	3 251	450.6	11.445	3 230	37.159
Tail Tanks (Total)	461	1 744	3 112	1 411	771.7	19.600	2 401	25.722
NOTE: [1] Weight is calculated with a fuel density = 6.75 lb/U.S. gal (0.809 kg/L)								

3. Unusable Fuel

The unusable fuel is the sum of the residual and trapped fuel in all the fuel system.

The residual fuel is the quantity of fuel that the engines cannot use under normal conditions in flight. It is possible to drain this residual fuel through the draining points with the aircraft in the level position.

The trapped fuel is the fuel caught in the aircraft fuel system after the residual fuel is drained.

The Table 2 that follows shows the volume, weight, CG and moment values for the unusable fuel:

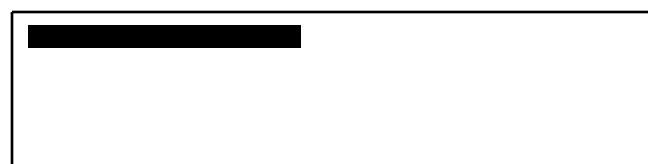


Table 2 – Unusable Fuel								
Description	Volume		Weight [1]		CG		Moment/1 000	
	U.S. gal	L	lb	kg	in.	m	lb in.	kg m
Residual Fuel	12	45	81.0	36.7	544.0	13.82	44.1	0.507
Trapped Fuel	7	27	48.0	21.8	524.0	13.31	25.2	0.290
Total Unusable Fuel	19	72	129.0	58.5	536.6	13.63	69.2	0.797
NOTE: [1] Weight is calculated with a fuel density = 6.75 lb/U.S. gal (0.809 kg/L)								

4. Usable Fuel

The usable fuel is the fuel available to operate the aircraft engines. The Table 3A (imperial units) and Table 3B (SI units) that follow show the volume, weight, CG and moment values for the usable fuel:

- The total usable fuel that the main tanks can contain is 1 440 U.S. gal (5 450.8 L)
- Total usable fuel for the auxiliary tanks is 1 062 U.S. gal (4 020.1 L)
- Total usable fuel for the tail tanks is 461 U.S. gal (1 744.8 L).

NOTE: Weight is calculated with a fuel density = 6.75 lb/U.S. gal (0.809 kg/L).

Table 3A – Usable Fuel (Imperial Units)							
Volume	Weight [1]	Main Tanks		Auxiliary Tanks		Tail Tanks	
		CG	Moment/ 1 000	CG	Moment/ 1 000	CG	Moment/ 1 000
U.S. gal	lb	in.	lb in.	in.	lb in.	in.	lb in.
50	338	473.5	160	456.0	154	745.5	252
100	675	475.8	321	458.0	309	755.3	510
150	1 013	477.6	484	460.0	466	761.0	771
200	1 350	479.0	647	460.7	622	764.5	1 032
250	1 688	480.3	811	461.1	778	766.7	1 294

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 3A – Usable Fuel (Imperial Units)**

Volume	Weight [1]	Main Tanks		Auxiliary Tanks		Tail Tanks	
		CG	Moment/ 1 000	CG	Moment/ 1 000	CG	Moment/ 1 000
U.S. gal	lb	in.	lb in.	in.	lb in.	in.	lb in.
300	2 025	481.5	975	461.1	934	768.3	1 556
350	2 363	482.6	1 140	459.3	1 085	769.5	1 818
400	2 700	483.7	1 306	457.9	1 236	770.7	2 081
450	3 038	484.8	1 473	457.7	1 390	771.5	2 343
461	3 112	–	–	–	–	771.7	2 401
500	3 375	486.0	1 640	457.5	1 544	–	–
550	3 713	487.2	1 809	457.2	1 697	–	–
600	4 050	488.3	1 978	456.8	1 850	–	–
650	4 388	489.5	2 148	456.4	2 002	–	–
700	4 725	490.6	2 318	455.9	2 154	–	–
750	5 063	491.7	2 489	455.3	2 305	–	–
800	5 400	492.8	2 661	454.7	2 455	–	–
850	5 738	493.8	2 833	454.0	2 605	–	–
900	6 075	494.9	3 007	453.3	2 754	–	–
950	6 413	495.9	3 180	452.5	2 902	–	–
1 000	6 750	496.9	3 354	451.7	3 049	–	–
1 050	7 088	497.8	3 528	450.8	3 195	–	–
1 062	7 168	–	–	450.6	3 230	–	–
1 100	7 425	498.8	3 704	–	–	–	–
1 150	7 763	499.8	3 880	–	–	–	–
1 200	8 100	500.8	4 056	–	–	–	–

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL**

Table 3A – Usable Fuel (Imperial Units)							
Volume	Weight [1]	Main Tanks		Auxiliary Tanks		Tail Tanks	
		CG	Moment/ 1 000	CG	Moment/ 1 000	CG	Moment/ 1 000
U.S. gal	lb	in.	lb in.	in.	lb in.	in.	lb in.
1 250	8 438	501.8	4 234	–	–	–	–
1 300	8 775	502.9	4 413	–	–	–	–
1 350	9 113	504.1	4 594	–	–	–	–
1 400	9 450	505.4	4 776	–	–	–	–
1 440	9 720	506.6	4 924	–	–	–	–
NOTE: [1] Weight is calculated with a fuel density = 6.75 lb/U.S. gal (0.809 kg/L)							

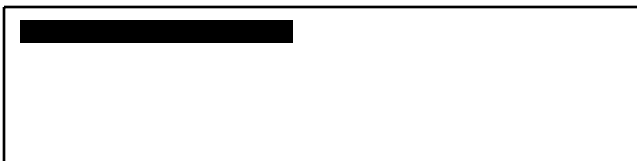
Table 3B – Usable Fuel (SI Units)							
Volume	Weight [1]	Main Tanks		Auxiliary Tanks		Tail Tanks	
		CG	Moment/ 1 000	CG	Moment/ 1 000	CG	Moment/ 1 000
L	kg	m	kg m	m	kg m	m	kg m
100	81	12.00	0.970	11.57	0.936	18.78	1.519
200	162	12.03	1.946	11.59	1.874	18.94	3.064
300	243	12.06	2.927	11.61	2.818	19.07	4.628
400	324	12.09	3.912	11.64	3.766	19.18	6.206
500	404	12.11	4.899	11.67	4.718	19.27	7.792
600	485	12.14	5.890	11.69	5.672	19.33	9.383
700	566	12.16	6.883	11.70	6.623	19.39	10.976
800	647	12.17	7.878	11.70	7.574	19.42	12.570
900	728	12.19	8.876	11.71	8.525	19.45	14.162
1 000	809	12.21	9.875	11.71	9.474	19.48	15.755

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL**

Table 3B – Usable Fuel (SI Units)							
Volume	Weight [1]	Main Tanks		Auxiliary Tanks		Tail Tanks	
		CG	Moment/ 1 000	CG	Moment/ 1 000	CG	Moment/ 1 000
L	kg	m	kg m	m	kg m	m	kg m
1 100	890	12.22	10.877	11.71	10.422	19.50	17.347
1 200	971	12.24	11.880	11.70	11.356	19.51	18.941
1 300	1 052	12.25	12.886	11.67	12.274	19.53	20.537
1 400	1 132	12.27	13.895	11.65	13.192	19.55	22.136
1 500	1 213	12.28	14.905	11.63	14.114	19.56	23.737
1 600	1 294	12.30	15.918	11.63	15.047	19.58	25.340
1 700	1 375	12.31	16.934	11.63	15.987	19.59	26.940
1 744.8	1 411	12.32	17.389	11.62	16.406	19.60	27.655
1 800	1 456	12.33	17.952	11.62	16.923	–	–
1 900	1 537	12.34	18.972	11.62	17.858	–	–
2 000	1 618	12.36	19.996	11.62	18.792	–	–
2 100	1 699	12.38	21.022	11.61	19.724	–	–
2 200	1 780	12.39	22.050	11.61	20.655	–	–
2 300	1 860	12.41	23.081	11.60	21.584	–	–
2 400	1 941	12.42	24.115	11.60	22.511	–	–
2 500	2 022	12.44	25.151	11.59	23.436	–	–
2 600	2 103	12.45	26.190	11.58	24.360	–	–
2 700	2 184	12.47	27.230	11.58	25.281	–	–
2 800	2 265	12.48	28.273	11.57	26.200	–	–
2 900	2 346	12.50	29.318	11.56	27.117	–	–
3 000	2 427	12.51	30.365	11.55	28.031	–	–

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL**

Table 3B – Usable Fuel (SI Units)							
Volume	Weight [1]	Main Tanks		Auxiliary Tanks		Tail Tanks	
		CG	Moment/ 1 000	CG	Moment/ 1 000	CG	Moment/ 1 000
L	kg	m	kg m	m	kg m	m	kg m
3 100	2 507	12.53	31.413	11.54	28.943	–	–
3 200	2 588	12.54	32.463	11.53	29.853	–	–
3 300	2 669	12.56	33.515	11.52	30.761	–	–
3 400	2 750	12.57	34.568	11.51	31.666	–	–
3 500	2 831	12.58	35.623	11.50	32.568	–	–
3 600	2 912	12.60	36.680	11.49	33.468	–	–
3 700	2 993	12.61	37.738	11.48	34.365	–	–
3 800	3 074	12.62	38.797	11.47	35.259	–	–
3 900	3 155	12.64	39.859	11.46	36.150	–	–
4 000	3 235	12.65	40.922	11.45	37.038	–	–
4 020.1	3 252	12.65	41.136	11.45	37.216	–	–
4 100	3 316	12.66	41.987	–	–	–	–
4 200	3 397	12.67	43.055	–	–	–	–
4 300	3 478	12.69	44.125	–	–	–	–
4 400	3 559	12.70	45.198	–	–	–	–
4 500	3 640	12.71	46.274	–	–	–	–
4 600	3 721	12.73	47.353	–	–	–	–
4 700	3 802	12.74	48.436	–	–	–	–
4 800	3 883	12.76	49.524	–	–	–	–
4 900	3 963	12.77	50.616	–	–	–	–
5 000	4 044	12.79	51.712	–	–	–	–

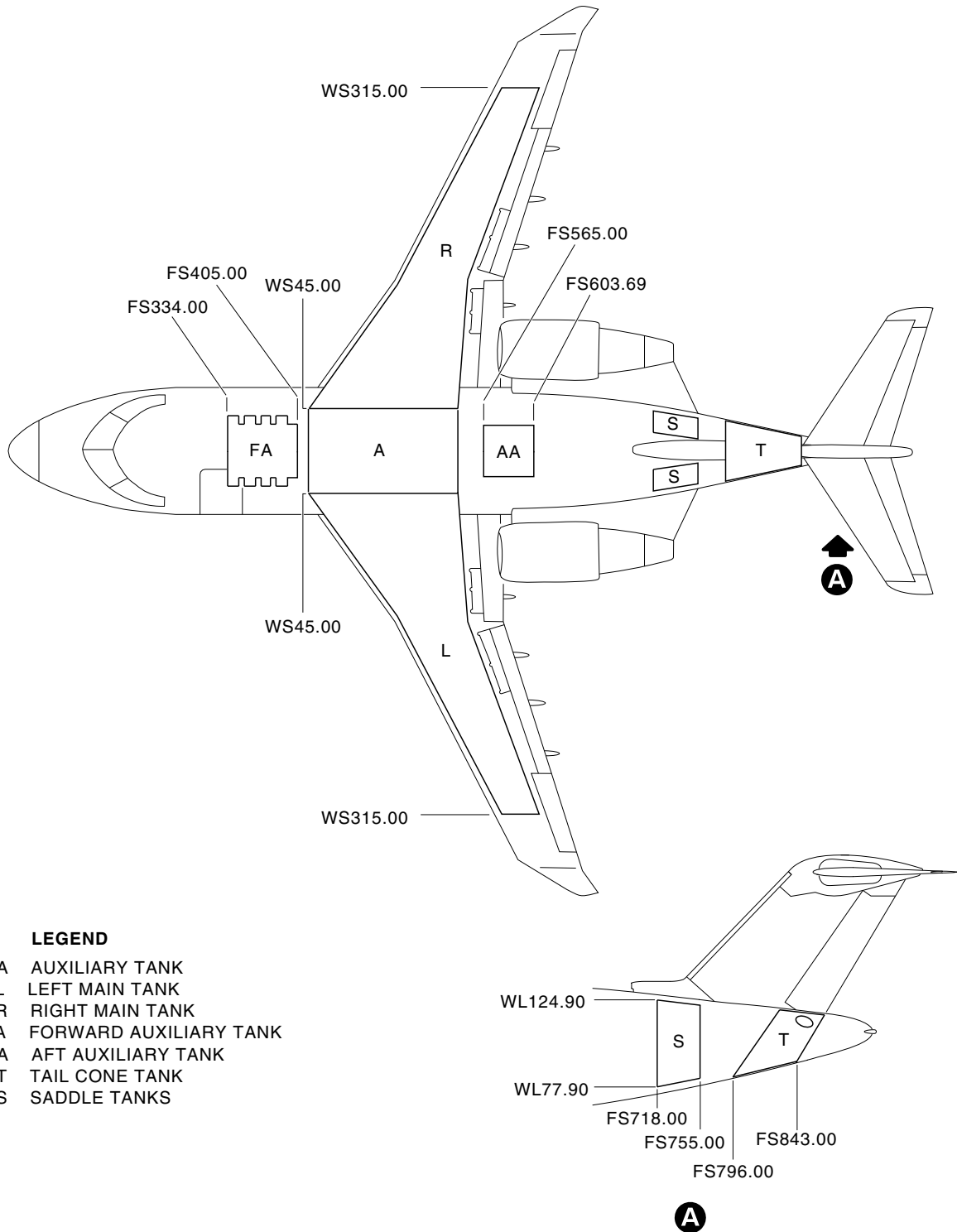


CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL**

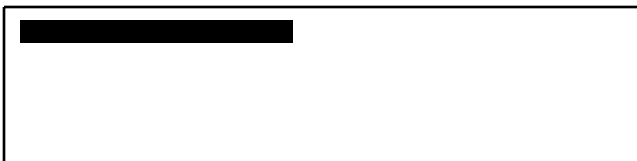
Table 3B – Usable Fuel (SI Units)							
Volume	Weight [1]	Main Tanks		Auxiliary Tanks		Tail Tanks	
		CG	Moment/ 1 000	CG	Moment/ 1 000	CG	Moment/ 1 000
L	kg	m	kg m	m	kg m	m	kg m
5 100	4 125	12.80	52.814	–	–	–	–
5 200	4 206	12.82	53.922	–	–	–	–
5 300	4 287	12.84	55.035	–	–	–	–
5 400	4 368	12.86	56.154	–	–	–	–
5 450.8	4 409	12.87	56.725	–	–	–	–
NOTE: [1] Weight is calculated with a fuel density = 0.809 kg/L							

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

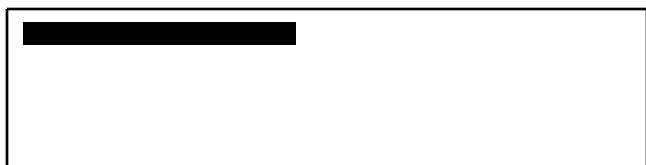


Fuel Tank Configuration
Figure 1



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



FUEL MANAGEMENT**1. Fuel Transfer System**

The fuel transfer system controls the fuel flows and fuel balance between all the fuel tanks during replenishment and fuel consumption. The fuel moves automatically from the auxiliary fuel tank to the main tanks via fuel ejectors. The fuel also moves automatically from the tail tank to the auxiliary tank via transfer pumps at a constant flow. To prevent lateral imbalance, it is also possible to do a manual transfer to make the fuel flow from wing to wing. In case of an emergency, the fuel system can dump the fuel that is in the tail tank.

2. Fuel Replenishment

To refuel the aircraft, follow the procedures given in Chapter 12 of the Aircraft Maintenance Manual (AMM TASK 12-11-00). Replenishment of the auxiliary and tail tanks is necessary only if the fuel load must be more than the full main tanks contain.

A. Pressure Refuel Procedure**(1) Normal Replenishment Mode**

The replenishment of the auxiliary tank and tail tank is done at the same time in a ratio of at least 2.65:1 (auxiliary/tail quantity), until the auxiliary tank is full. The tail tank completes its replenishment afterward.

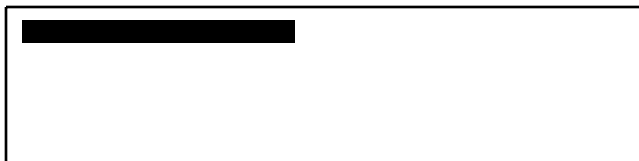
(2) Manual OVERRIDE Replenishment Mode

The replenishment of the auxiliary tank and tail tank must be done with admissible quantity ratio:

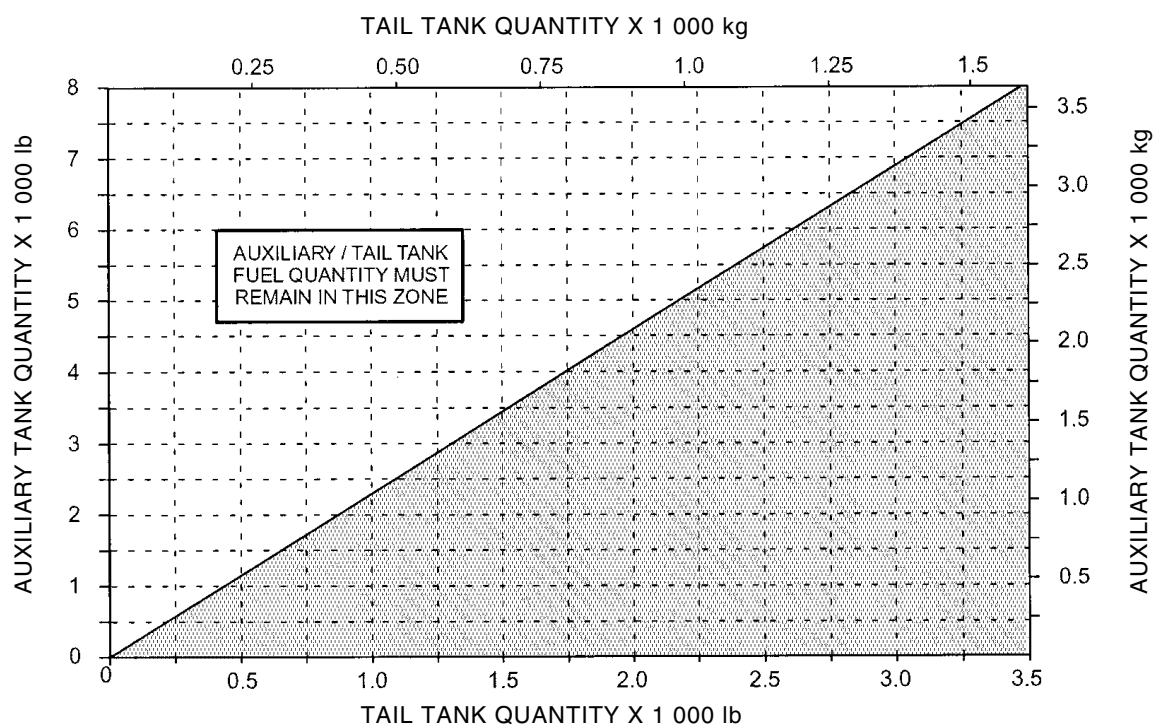
- Refer to Figure 2 if the aircraft CG at zero fuel weight (ZFW) is forward of the lower aft zone of the CG envelope (CG between 32,000 lb/35% MAC, 30,000 lb/34% MAC, 28,000 lb/34% MAC, 26,000 lb/32.5% MAC and 26,000 lb/35% MAC).
- Refer to Figure 1 or, Figure 2 if the take-off longitudinal center of gravity is forward of 34.5% MAC.
- Refer to Figure 3 if the aircraft CG at zero fuel weight (ZFW) is inside the lower aft zone of the CG envelope (CG between 32,000 lb/35% MAC, 30,000 lb/34% MAC, 28,000 lb/34% MAC, 26,000 lb/32.5% MAC and 26,000 lb/35% MAC). The replenishment of the auxiliary tank and tail tank must be completed such that the auxiliary tank is full, or at least 2.95 times the tail tank quantity.

B. Gravity Refuel Procedure

Gravity refueling is applicable for the main and auxiliary tanks only. The CG must be monitored closely during gravity refueling of the auxiliary tank.

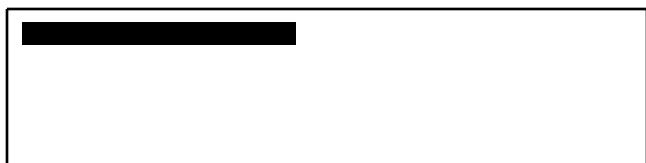


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

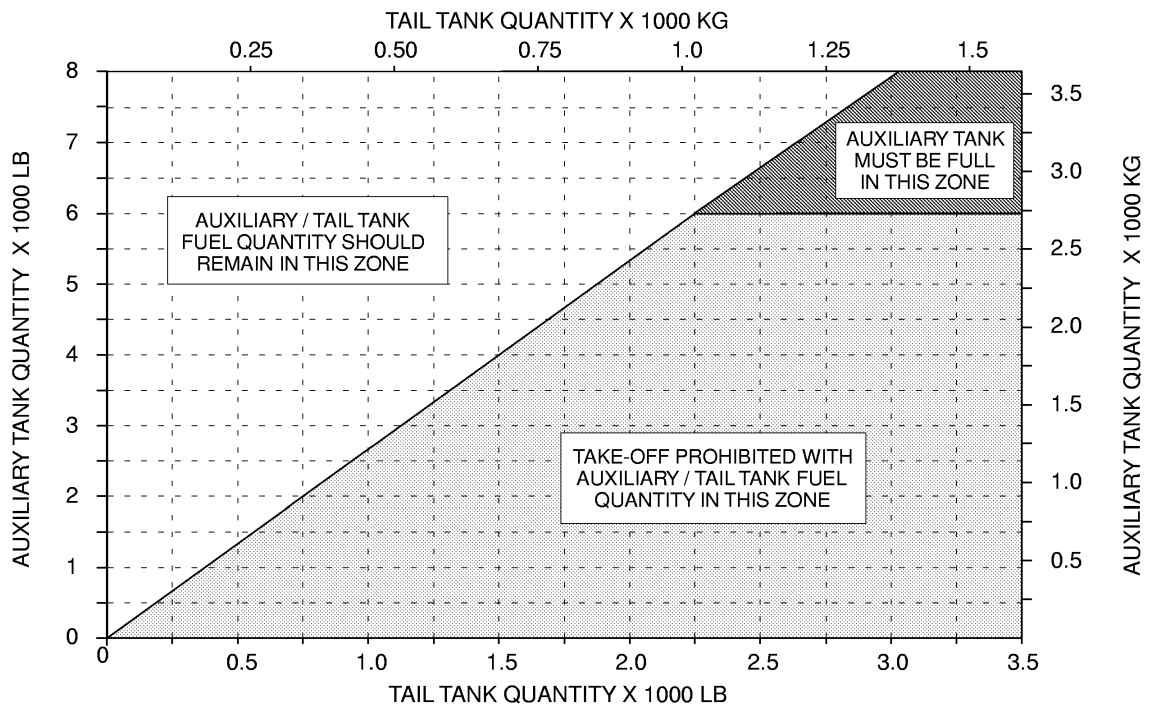


DWB0120200_002

Auxiliary and Tail Tank Quantity Admissible Ratio (Only for CG Forward of 34.5% MAC)
Figure 1



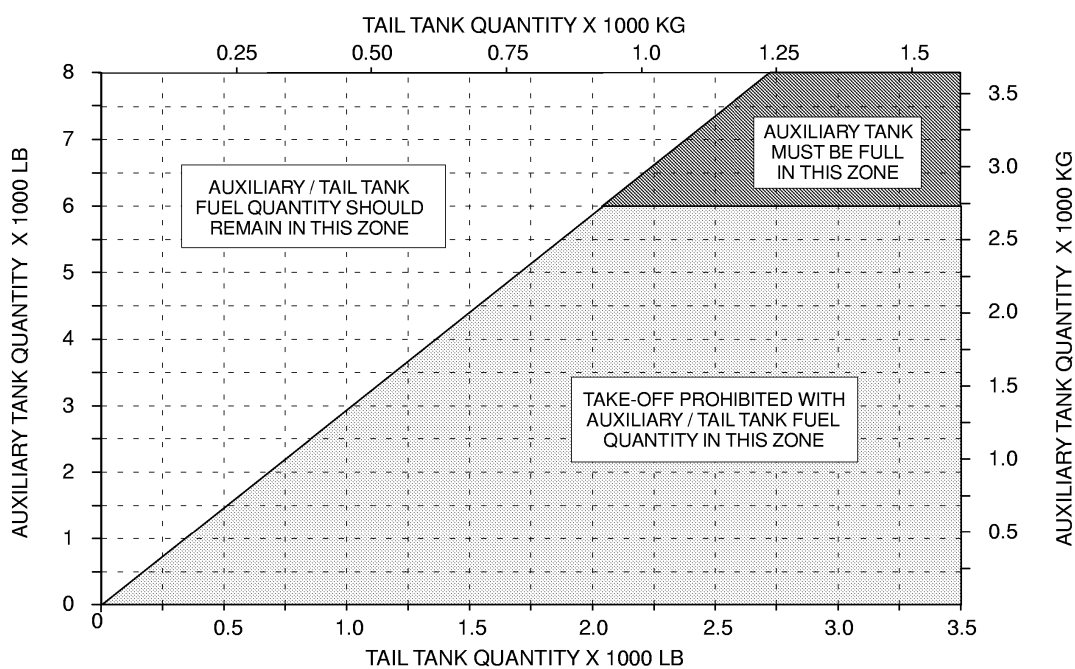
CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL



Auxiliary and Tail Tank Quantity Admissible Ratio
Figure 2

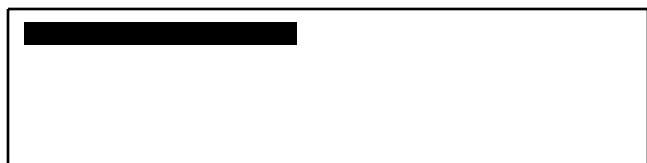
DWB0120200_003

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL



DWB0120200_004

Auxiliary and Tail Tank Quantity Admissible Ratio (for ZFW CG inside lower aft zone of CG envelope)
Figure 3



01-20-20

Page 4
Oct 27/2015

3. Fuel Burn Sequence

The fuel burn sequence is as follows:

- A. The usual fuel flow during flight is from the left main tank to the left engine and from the right main tank to the right engine. The fuel transfer from the auxiliary fuel tank starts during take-off. It can also start, when on ground, if the fuel quantity decreases and gets to less than 600 U.S. gal (approximately and in each of the main tanks).

At this point, the main tanks do not deplete but the auxiliary and tail tanks deplete in a 1:1 ratio until the auxiliary tank quantity decreases by about 70 U.S. gal. The auxiliary tank then stops to deplete while the tail tank continues to deplete.

- B. If the two conditions that follow exist:

- The fuel quantity in the tail tank gets to less than 38% of its capacity.
- The auxiliary tank stays at more than 35% of its capacity.

Then, the fuel transfer between the tail tank and the auxiliary tank stops until the auxiliary fuel tank depletes to less than 35% of its capacity.

At this point, the auxiliary tank stops again to deplete while the tail tank depletes completely. The auxiliary tank then depletes completely. In typical condition, the wing will deplete by 500.0 lb (226.8 kg) for each of the main tanks by the time the auxiliary and tail tanks get empty, and subsequently the main tanks finish to deplete.

4. EICAS Messages

The EICAS caution messages AUX TANK HVY and TAIL TANK HVY indicate a fuel system failure or fuel imbalance between the auxiliary and tail tanks (Refer to the Airplane Flight Manual (AFM) for crew responses to these messages).

The message AUX TANK HVY will show when the auxiliary and tail tank quantity proportions are as follows:

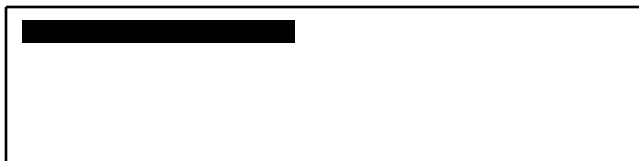
$$[\text{Auxiliary Tank Quantity (lb)}] \geq [3.983 \times \text{Tail Tank Quantity (lb)} + 3\,370.0]$$

The message TAIL TANK HVY will show when the auxiliary and tail tank quantity proportions are as follows:

$$\begin{aligned} &\text{Tail Tank Quantity (lb)} < 2\,500 \text{ lb AND Aircraft Pitch} < 5^\circ \text{ AND} \\ &[3.200 \times \text{Tail Tank Quantity (lb)}] \geq [\text{Auxiliary Tank Quantity (lb)} + 2\,000.0] \end{aligned}$$

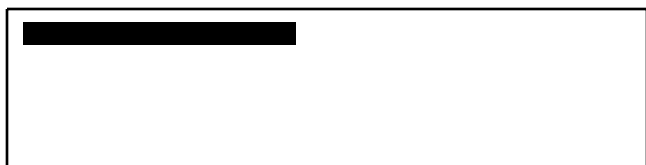
OR

$$\begin{aligned} &[\text{Tail Tank Quantity (lb)} \geq 2\,500 \text{ lb OR Aircraft Pitch} \geq 5^\circ] \text{ AND} \\ &\text{Tail Tank Quantity (lb)} > 625 \text{ lb AND Tail Tank Fuel Transfer Malfunction in Flight} \end{aligned}$$



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



01-20-20

Page 6
Oct 27/2015

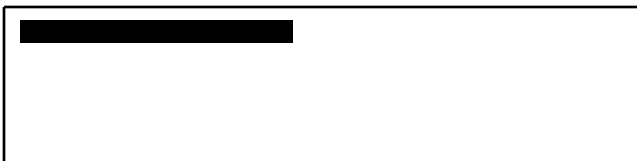
CHALLENGER⁶⁵⁰
CL-650 WEIGHT AND BALANCE MANUAL

FLUIDS

1. General

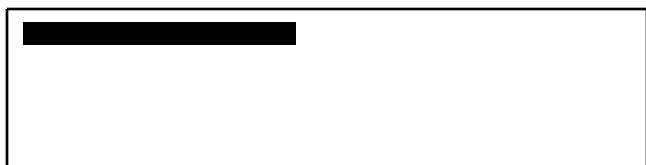
This section contains the subjects related to fluids that follow:

- System Fluids (Subject 01-30-10)
- Service Fluids (Subject 01-30-20).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



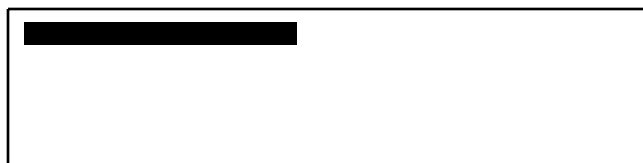
SYSTEM FLUIDS**1. System Fluid Capacities**

The Table 1 that follows gives the volume, weight, center of gravity (CG) and moment for the system fluids:

Table 1 – System Fluid Capacities								
Description	Volume		Weight		CG		Moment/1 000	
	U.S. qt	L	lb	kg	in.	m	lb in.	kg m
Engine Fluids								
Engine Oil Tanks (each of the 2)	6.8	6.5	13.7	6.21	645.4	16.39	8.8	0.102
Replenishment Oil Tank	6.0	5.7	12.1	5.49	729.0	18.52	8.8	0.102
Replenishment System Lines	3.7	3.5	7.5	3.40	687.8	17.47	5.2	0.059
Total	23.3	22.1	47.0	21.32	673.7	17.11	31.66	0.365

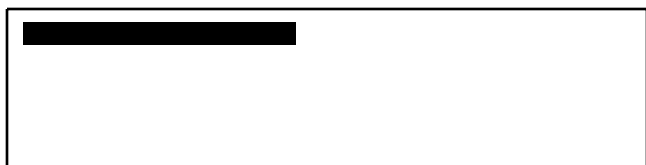
NOTE: Low level for the engine oil tank is 1.3 U.S. qt (1.23 L).

NOTE: It is possible to drain oil completely from all tanks.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK

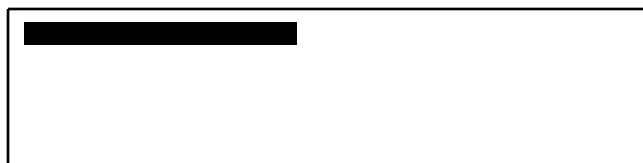


CHALLENGER⁶⁵⁰
CL-650 WEIGHT AND BALANCE MANUAL

SERVICE FLUIDS

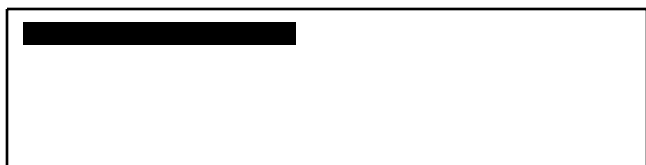
1. Service Fluid Capacities

Refer to Chapter 2, Aircraft Reports provided by the Completion Center for data related to the service fluids.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



01-30-20

Page 2
Oct 27/2015

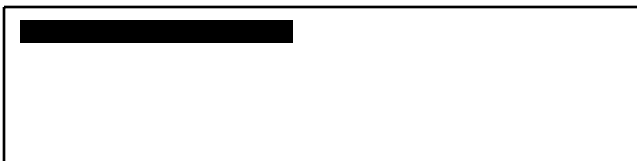
CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

PERSONNEL

1. General

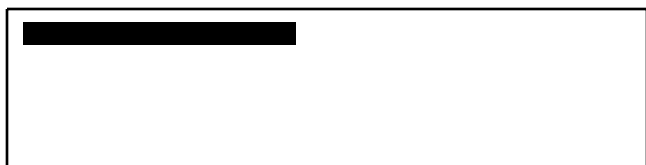
This section contains the subjects related to personnel that follow:

- Flight Compartment (Subject 01-40-10)
- Passenger Compartment (Subject 01-40-20)
- Balance Changes Due to Movement (Subject 01-40-40).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



FLIGHT COMPARTMENT**1. General**

This section contains data on the personnel in the flight compartment.

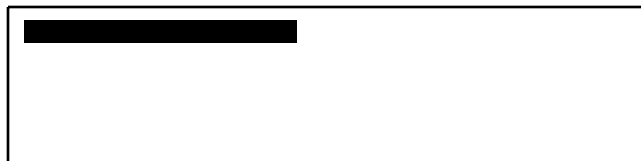
2. Crew Arrangement

The crew seats are installed side by side on seat tracks in the flight compartment. The pilot sits on the left side and the copilot sits on the right side of the flight compartment. Structural provision for a jump seat is located behind the bulkhead at FS280.00.

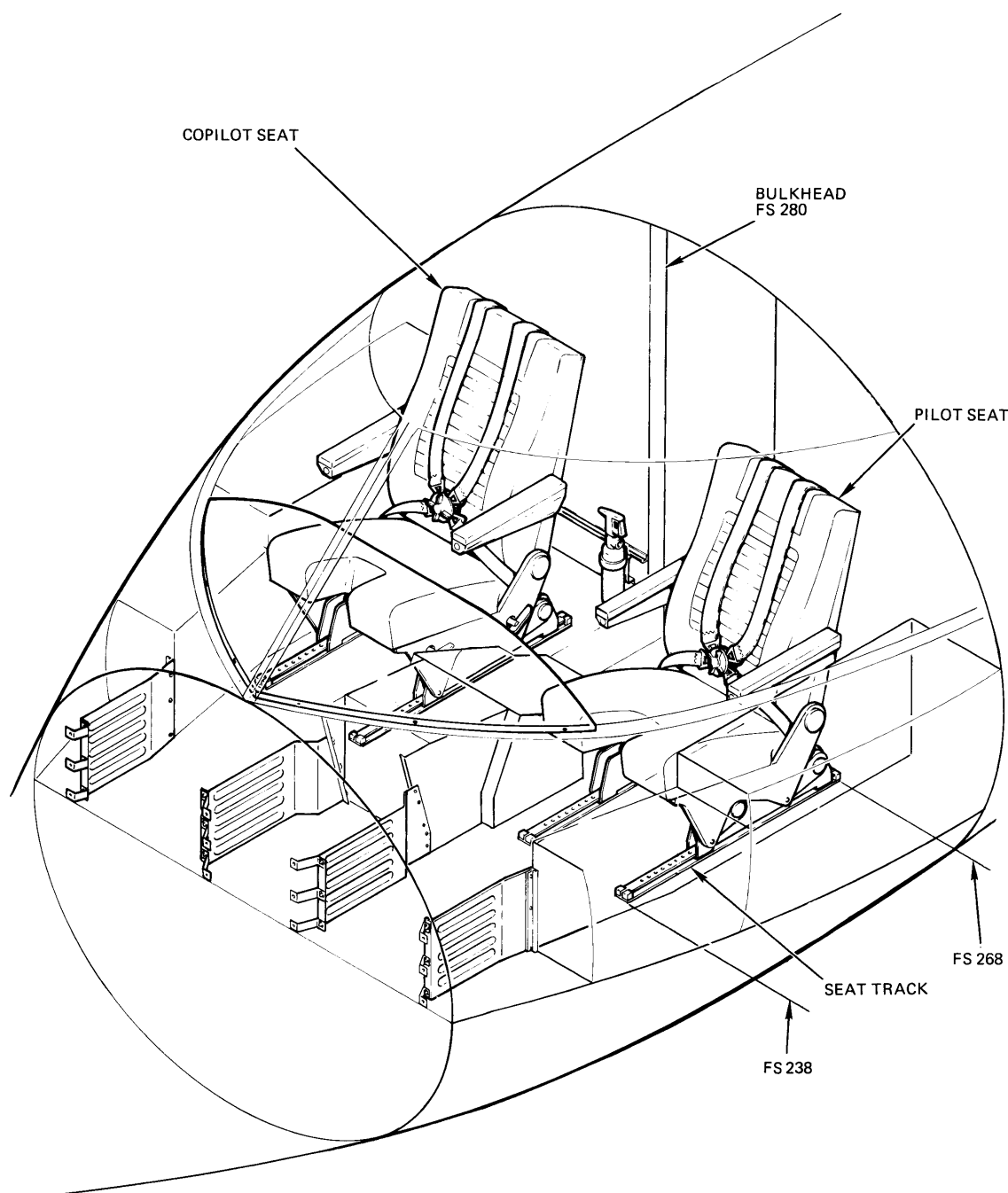
3. Crew Weight and Balance Arms

The Table 1 that follows shows the weight, center of gravity (CG) and moment for the pilot and copilot with the seats in the fully vertical position:

Table 1 – Pilot and Copilot Weight and CG and Moment						
Items	Weight		Xarm		Moment/1 000	
	lb	kg	in.	m	lb in.	kg m
Pilot	170	77.11	255.0	6.48	43.4	0.500
Copilot	170	77.11	255.0	6.48	43.4	0.500
Total	340	154.22	255.0	6.48	86.7	0.999

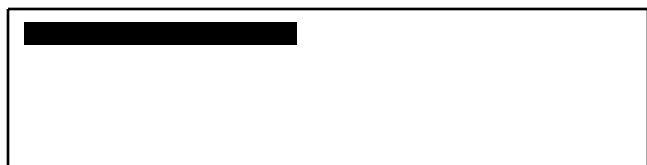


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL



DWB0140100_001

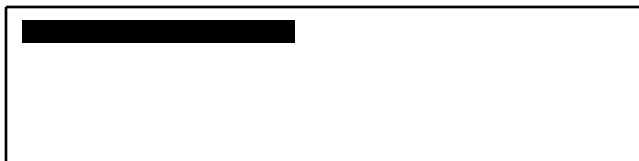
Crew Arrangement
Figure 1



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL
PASSENGER COMPARTMENT

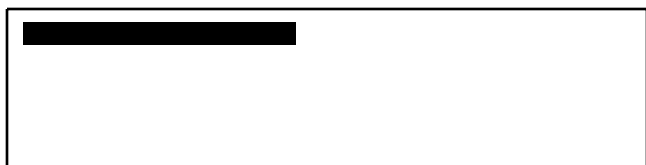
1. General

Refer to Chapter 2, Aircraft Reports provided by the Completion Center for the data related to the passenger compartment.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



BALANCE CHANGES DUE TO MOVEMENT**1. General**

The operation of some systems, passenger and crew movement can change the center of gravity (CG) during flight, ground handling, take-off and landing operations. Refer to Chapter 2, Aircraft Reports provided by the Completion Center for the data related to the effect of these movements on the aircraft CG.

2. Aircraft on the Ground

The weight and CG limits, defined by the envelope, must be met with the conditions that follow ((Subject 01-00-60), (Subject 01-80-60) and (Subject 01-90-20)):

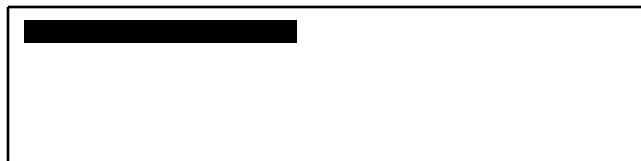
- Landing gear down
- Crew and passengers seated in normal locations.

3. Aircraft in Flight

With the weight and CG limits met with the aircraft on the ground, safe limits in flight are achieved. This is with the conditions that follow:

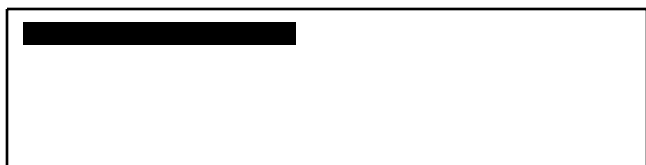
- Landing gear retracted
- Crew and passengers are allowed normal movement in flight.

NOTE: For landing, aircraft configuration must meet the conditions given for Aircraft on the Ground.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



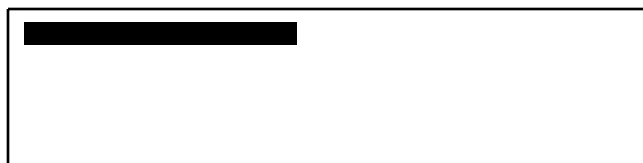
CHALLENGER⁶⁵⁰
CL-650 WEIGHT AND BALANCE MANUAL

CARGO

1. General

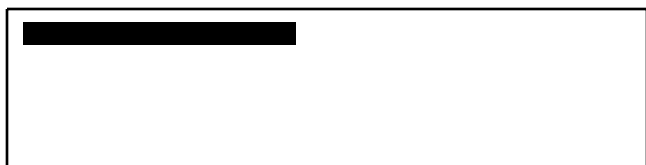
This section contains the subjects related to cargo that follow:

- Cargo Compartments (Subject 01-60-10)
- Passenger Compartment Storage (Subject 01-60-20).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



CARGO COMPARTMENTS

1. Limitations

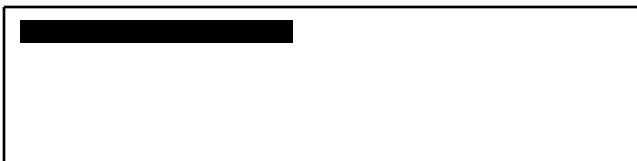
Floor structure maximum loading is 100 lb/sq. ft. (488 kg/m²).

Floor panel maximum loading is 75 lb/sq. ft. (366 kg/m²).

Fuselage structure maximum loading is 500 lb/ft (500 lb/running foot of cabin – fuselage station) (743 kg/running meter of cabin – fuselage station) for the green aircraft.

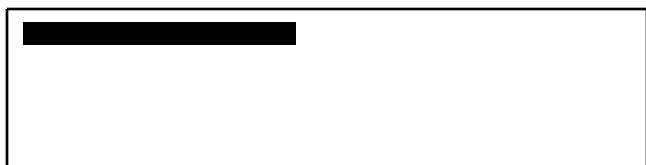
2. General

Refer to Chapter 2, Aircraft Reports provided by the Completion Center for the data related to the cargo compartments.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



PASSENGER COMPARTMENT STORAGE

1. Limitations

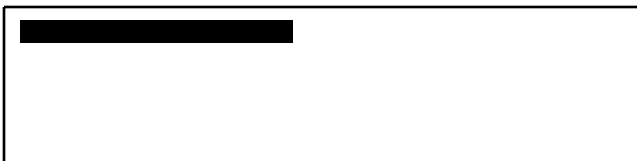
Floor structure maximum loading is 100 lb/sq. ft. (488 kg/m²).

Floor panel maximum loading is 75 lb/sq. ft. (366 kg/m²).

Fuselage structure maximum loading is 500 lb/ft (500 lb/running foot of cabin – fuselage station) (743 kg/running meter of cabin – fuselage station) for the green aircraft.

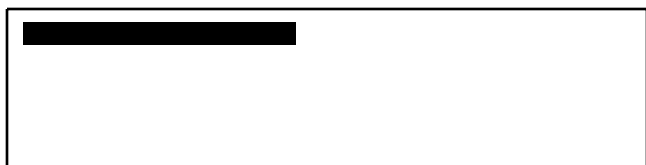
2. General

Refer to Chapter 2, Aircraft Reports provided by the Completion Center for the data related to the passenger compartment storage.



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK

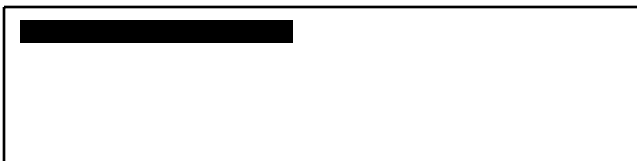


GROUND OPERATIONS

1. General

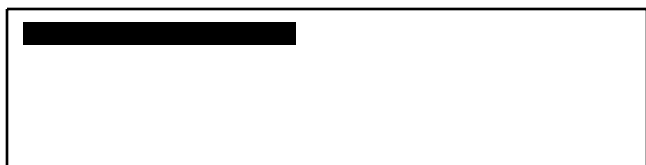
This section contains the subjects related to ground operations that follow:

- Static Loads (Subject 01-80-20)
- Weighing Procedures (Subject 01-80-30)
- Balance Limits (Subject 01-80-60).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



STATIC LOADS

1. Maximum Permitted Jacking Weight

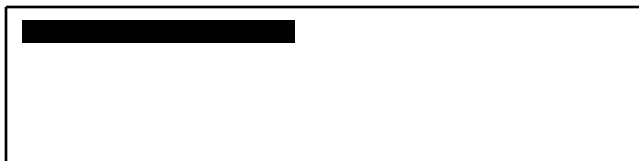
A. Jacking on Structure

The Table 1 that follows gives the maximum permitted aircraft jacking weight and the maximum permitted load for each structural jacking point.

Table 1 – Maximum Jacking Weight		
Description	Maximum Static Load	
	lb	kg
Aircraft (Total Weight)	38 000	17 236.5
Nose Jacking Point	5 200	2 358.6
Wing (each side) Jacking Point	17 400	7 892.5

B. Jacking on Landing Gear

The maximum aircraft weight when jacking on landing gear must not be more than the maximum ramp weight (MRW) (Subject 01-00-50).



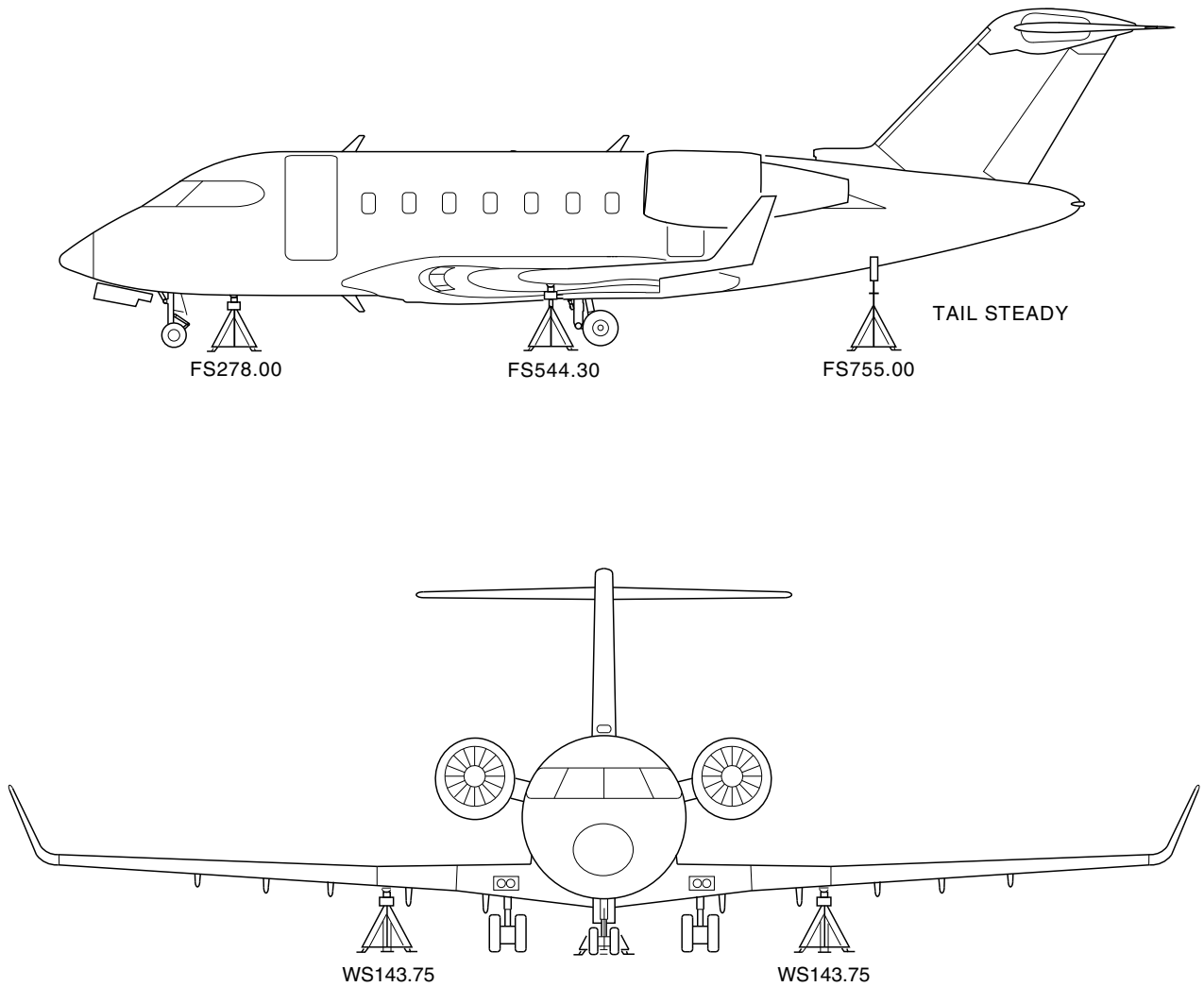
CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

2. Maximum Loading

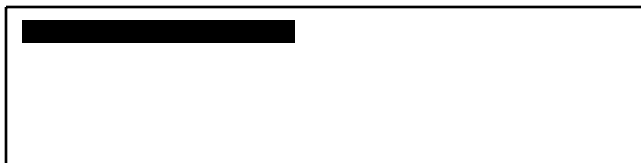
When loading the aircraft, the load must not be more than the maximum load permitted in each area. The Table 2 that follows gives the maximum loading for each aircraft area:

Table 2 – Maximum Loading			
Description	Zone	Loading	Capacity
Floor Structure		100 lb/ft ² (488 kg/m ²)	
Floor Panels		75 lb/ft ² (366 kg/m ²)	
Fuselage Structure		500 lb/Longitudinal ft (743 kg/Longitudinal m)	
Notes: 1. Maximum cargo compartment capacity = 750 lb (340 kg). 2. Maximum cargo compartment capacity = 900 lb (409 kg) if increase baggage STC option selected.			

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

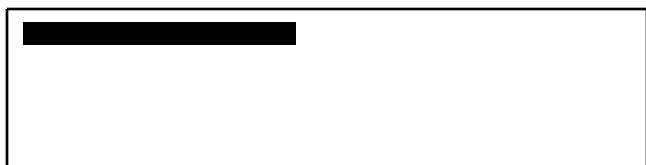


Jacking Locations
Figure 1



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



WEIGHING PROCEDURES

1. General

This section gives the procedures to find the weight and the center of gravity (CG) of the aircraft.

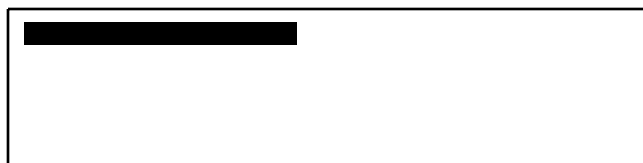
The two different procedures to measure the weight of the aircraft are as follows:

- Weighing on Tripod Jacks Procedure
- Weighing on Platform Scales and Jack Procedure.

2. Job Set-Up Information

A. Tools and Equipment

Reference	Designation
GSE 07-10-01	Jacking Pads
GSE 07-10-02	Wing Jack
GSE 07-10-03	FWD Fuselage Jack
GSE 08-00-01	Leveling Kit
GSE 08-00-02	Aircraft Weighing Kit
GSE 09-10-01	Aircraft Towbar
GSE 10-10-09	Wheel Chocks
Commercially Available	Platform Scale (50 000 lb (22 680 kg))



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

B. Reference Information

Reference	Designation
AMM TASK 12-00-00-867-801	Standard Aircraft Configuration for Maintenance
AMM TASK 12-11-28-650-803	Suction Defueling
AMM TASK 12-12-29-611-801	Servicing of the Hydraulic System Reservoirs
AMM TASK 12-13-79-612-801	Replenishing of the Oil Supply at Engine
AMM TASK 12-20-01-110-804	Cleaning of the Nontextile Floor Coverings
AMM TASK 12-20-01-110-805	Cleaning of the Textile Floor Coverings
AMM TASK 12-20-01-110-801	Cleaning of the External Aircraft Surfaces
AMM TASK 24-00-00-910-801	Electrical/Electronic Safety Precautions
AMM TASK 29-00-00-910-801	Hydraulic Safety Precautions
AMM TASK 29-00-00-910-802	Hydraulic Technical Precautions
AMM TASK 32-11-07-720-801	Functional Test of the Main Landing Gear (MLG) Shock-Strut Assembly
AMM TASK 52-41-00-010-801	Open the Aft Equipment-Compartment Door
AMM TASK 52-41-00-410-801	Close the Aft Equipment-Compartment Door

C. Job Set-Up Procedure

WARNING: WHEN YOU WEIGH THE AIRCRAFT ON JACKS, OBEY THE PRECAUTIONS THAT FOLLOW:

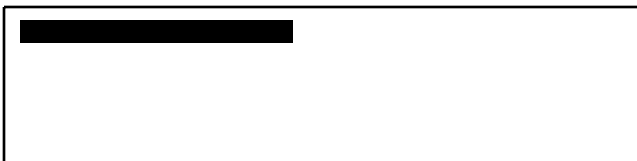
- MAKE SURE THAT NO PERSONS OR TOOLS ARE ON OR IN THE AIRCRAFT.
- INSTALL THE LOCKPINS IN THE NOSE LANDING GEAR AND THE MAIN LANDING GEAR.
- REMOVE THE CHOCKS FROM THE WHEELS.
- RELEASE THE PARKING BRAKE.
- REMOVE THE TAIL STEADY.
- MAKE SURE THAT THE LOAD CELLS ARE INSTALLED CORRECTLY ON THE JACKS.
- MAKE SURE THAT THE JACKS ARE ALIGNED CORRECTLY BELOW THE JACKING PADS.
- MAKE SURE THAT THERE IS SUFFICIENT CLEARANCE AROUND THE AIRCRAFT.
- MAKE SURE THAT THERE IS NO SIDE LOAD ON THE LOAD CELLS.
- MAKE SURE THAT THE PASSENGER DOOR IS CLOSED.

THESE PRECAUTIONS CAN PREVENT INJURY TO PERSONS AND DAMAGE TO THE AIRCRAFT AND EQUIPMENT.

CAUTION: WEIGH THE AIRCRAFT AND MAKE IT LEVEL IN A CLOSED HANGAR. IF YOU DO NOT DO THIS, AIR MOVEMENT CAN CAUSE INCORRECT RESULTS.

Do the steps that follow to prepare the aircraft for weighing:

- (1) Do the suction defueling of the aircraft (AMM TASK 12-11-28-650-803).
- (2) Park the aircraft in a closed hangar.
- (3) Make sure that the hangar floor is hard, level and has no vibration.
- (4) Obey the electrical/electronic safety precautions (AMM TASK 24-00-00-910-801).
- (5) Obey the hydraulic safety precautions (AMM TASK 29-00-00-910-801).
- (6) Obey the hydraulic technical precautions (AMM TASK 29-00-00-910-802).
- (7) Open all fuel drain valves to let residual fuel drain during the weighing procedure.
- (8) Make sure that the hydraulic system reservoirs are full (AMM TASK 12-12-29-611-801).



- (9) Do the servicing of the engine oil tank (AMM TASK 12-13-79-612-801).
- (10) Drain the water system.
- (11) Drain the waste water system or chemical toilet.
- (12) Make sure that the aircraft is in the configuration that follows:
 - (a) The flaps and thrust reversers are retracted.
 - (b) All the flight controls are at the neutral position.
 - (c) All the doors, hatches, and access panels are closed.
 - (d) All the passenger seats are in take-off position.
 - (e) The internal surfaces of the aircraft are dry and clean (AMM TASK 12-20-01-110-804 and AMM TASK 12-20-01-110-805).
 - (f) The external surfaces of the aircraft are dry and clean (AMM TASK 12-20-01-110-801).

3. Weighing on Tripod Jacks Procedure

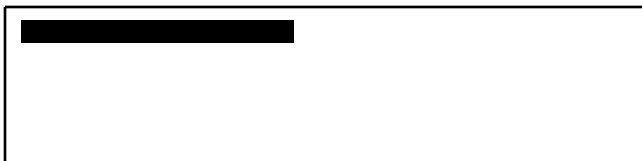
Do the steps that follow to weigh the aircraft on tripod jacks and calculate the center of gravity of the aircraft:

A. Weighing

Refer to Figure 1, Figure 2, Figure 3 and Figure 4.

Do the steps that follow to weigh the aircraft on tripod jacks:

- (1) Do the Job Set-Up Procedure found in the Job Set-Up Information section.
- (2) Remove the nylon plugs from the jacking points. Refer to Figure 1.
- (3) Install the jack pads in the jacking points.
- (4) Put the tripod jacks in position below the jacking points. Refer to Figure 2.
- (5) Follow the manufacturer instructions to install the load cells between the jacks and the jacking points on the aircraft.
- (6) Make sure that the load cell identification agrees with the related jacking point.
- (7) Use a copy of the Aircraft Weighing Records – Tripod Jacks Method form to continue with this procedure. Refer to Figure 3.



CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

- (8) Record the load cell identification into the channel (Chn) column of the Aircraft Weighing Records – Tripod Jacks Method form.

CAUTION: DO NOT OPERATE THE ELECTRONIC WEIGHING KIT UNTIL IT IS SUFFICIENTLY WARM. IF THE ELECTRONIC WEIGHING KIT IS NOT SUFFICIENTLY WARM, YOU CAN GET INCORRECT INDICATIONS ON THE TEST EQUIPMENT.

- (9) Let the load cells get warm for a minimum of 20 minutes.
- (10) Make sure that there is no weight on the load cells.
- (11) Follow the manufacturer instructions to calibrate the weighing kit to zero.
- (12) Open the aft equipment-compartment door 311BB (AMM TASK 52-41-00-010-801).
- (13) Put the plumb line through the hole in the bracket at the top of the frame at FS755.50. Refer to Figure 4.
- (14) Adjust the plumb line until the point of the plumb bob is not more than 0.5 in (1.27 cm) above the target.

NOTE: If the plumb bob is put higher than 0.5 in (1.27 cm) above the target, the result will not be accurate.

- (15) Attach the end of the plumb line to the bracket.

CAUTION: LOWER/EXTEND THE JACKS EQUALLY TO PREVENT SIDE LOADS ON THE LOAD CELLS. THE EQUIPMENT CAN SHOW INCORRECT WEIGHTS, HAVE LOAD CELL FAILURE, OR YOU CAN CAUSE DAMAGE TO THE AIRCRAFT.

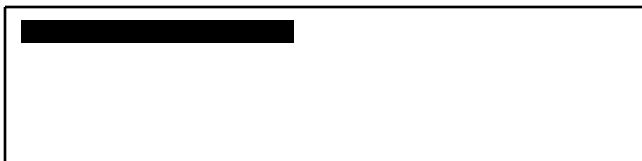
- (16) Extend the jacks to lift the aircraft from 1.0 to 2.0 in (2.54 to 5.08 cm) off the ground.
- (17) Make sure that the landing gears are fully extended.
- (18) Level the aircraft on the jacks.
- (19) Allow any remaining residual fuel to drain.
- (20) After the readings have stabilized, fill the data in the form as follows:
- (a) Find the weight at the nose jacking point as follows:
- 1 Get the indication from the load cell at the NLG tripod jack.
 - 2 Record the value on your copy of the form in the Reading column.
- (b) Find the weight at the left wing jacking point as follows:
- 1 Get the indication from the load cell at the left MLG tripod jack.

--

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

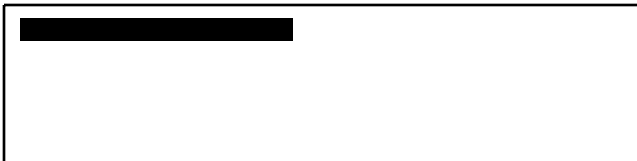
- 2 Record the value on your copy of the form in the Reading column.
 - (c) Find the weight at the right wing jacking point as follows:
 - 1 Get the indication from the load cell at the right MLG tripod jack.
 - 2 Record the value on your copy of the form in the Reading column.
 - (d) Find the weight of the aircraft as follows:
 - 1 Do a sum of the three recorded values.
 - 2 Record the result for Aircraft Total in the Reading Column.
- CAUTION:** LOWER/EXTEND THE JACKS EQUALLY TO PREVENT SIDE LOADS ON THE LOAD CELLS. THE EQUIPMENT CAN SHOW INCORRECT WEIGHTS, HAVE LOAD CELL FAILURE, OR YOU CAN CAUSE DAMAGE TO THE AIRCRAFT.
- (21) Lower the aircraft until there is no load on the jacks and the load cells.
 - (22) Read the residual indication on each of the load cells.
 - (23) Record the residual load cell indications in the related field of the Zero column.
 - (24) Make sure that all of the recorded values are equal to zero:
 - (a) If the values are equal to zero, no change to the weight data is necessary. Copy the value recorded in the related Reading column into the Weight column.
 - (b) If the values are different than zero, change the weight data as follows:
 - 1 If the value is more than zero, subtract it from the weight recorded in the related Reading column. Record the result in the related Weight column.
 - 2 If the value is less than zero, add it to the weight recorded in the related Reading column. Record the result in the related Weight column.
 - 3 If the value is more than the permitted limit of the equipment, do as follows:
 - a Do the Weighing on Tripod Jacks Procedure again with load cells moved clockwise to the next jacking point.
 - b Do this until you get two indications, one after the other, that have the conditions that follow:
 - 1) The residual indications are less than the permitted equipment limit.



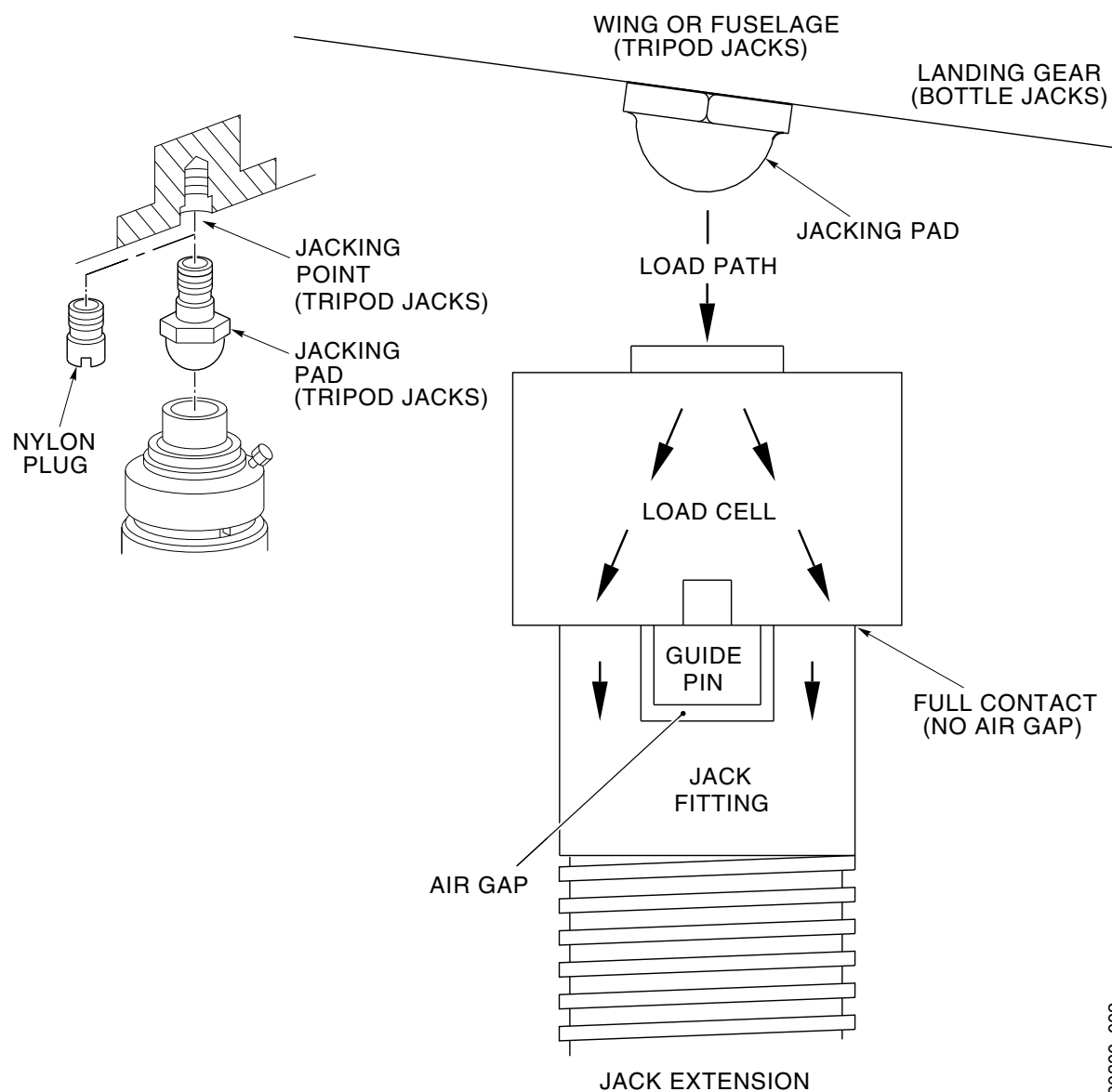
CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

- 2) The two indications that you get one after the other agree with the tolerance of the aircraft weighing kit.



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

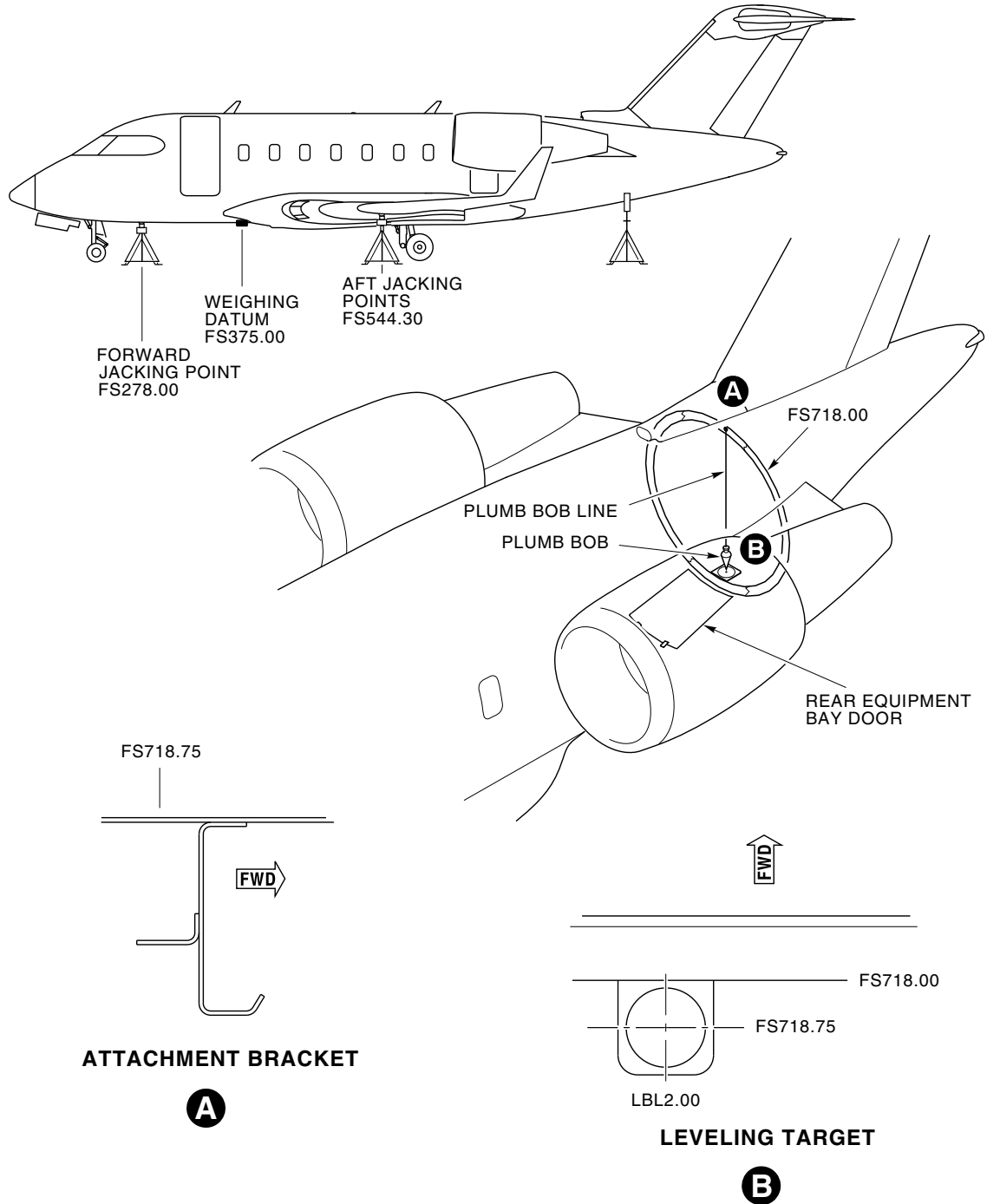


DWB0180300_002

Load Cell and Jacking Pad Arrangement
Figure 1

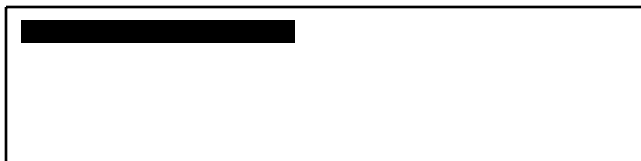
CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL



DWB0180300_001

Weighing Procedure – Leveling
Figure 2



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

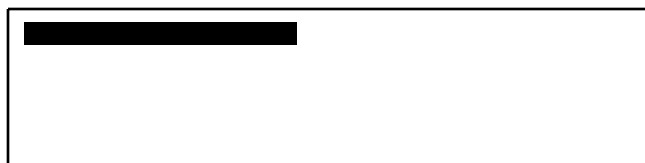
AIRCRAFT WEIGHING RECORDS – TRIPOD JACKS METHOD							
DESCRIPTION		CHN	READING	ZERO	WEIGHT	Xarm	MOMENT
			lb (kg)	lb (kg)	lb (kg)	in. (m)	1 000 lb in. (kg m)
FIRST READING	LEFT MLG TRIPOD JACK						
	RIGHT MLG TRIPOD JACK						
	NLG TRIPOD JACK						
	AIRCRAFT TOTAL						
SECOND READING	LEFT MLG TRIPOD JACK						
	RIGHT MLG TRIPOD JACK						
	NLG TRIPOD JACK						
	AIRCRAFT TOTAL						
AVERAGE OF READINGS	LEFT MLG TRIPOD JACK						
	RIGHT MLG TRIPOD JACK						
	NLG TRIPOD JACK						
	AIRCRAFT TOTAL						

CG position measured in %MAC is calculated as follows:

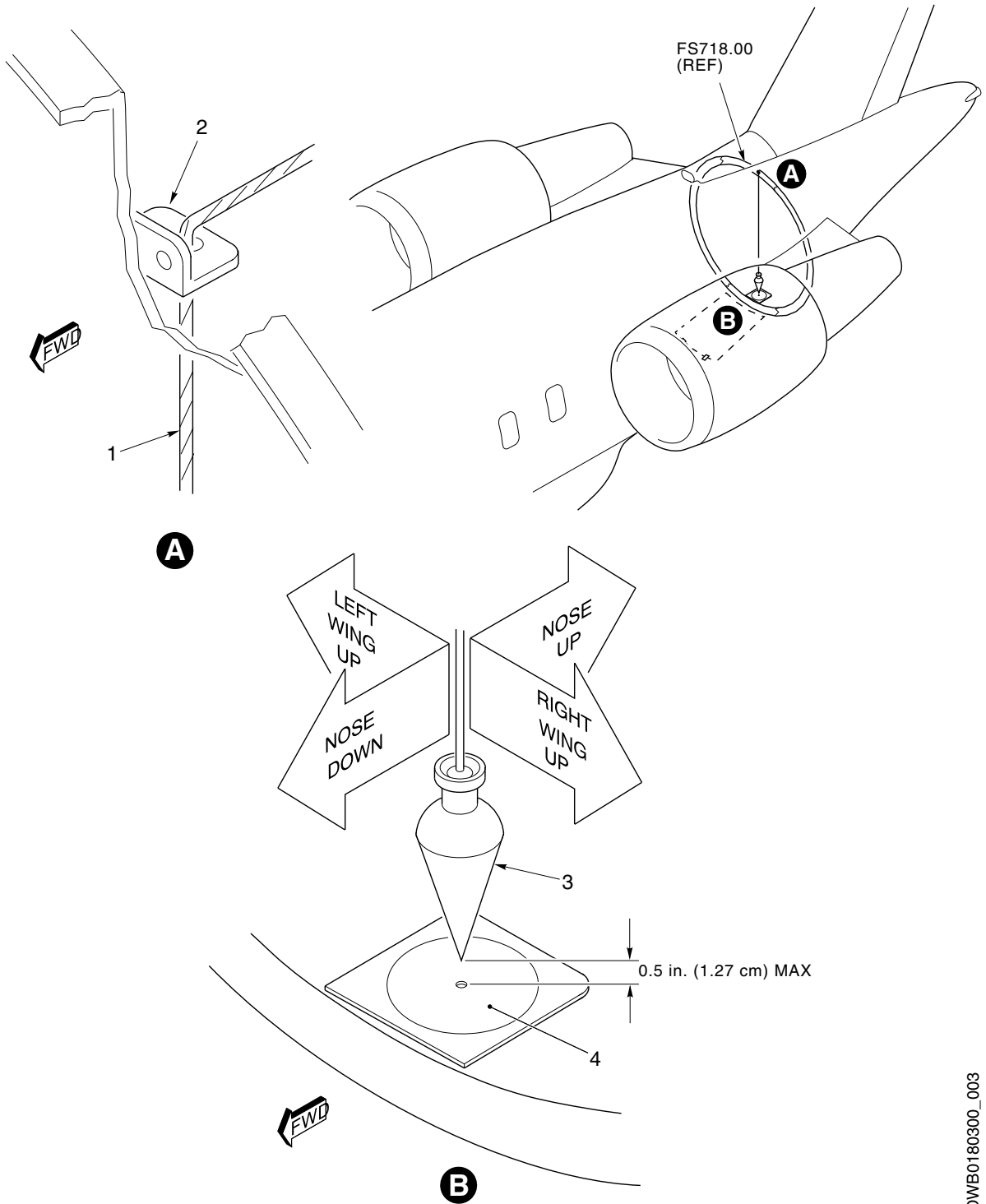
$$\%MAC = [(X_{arm}_{CG} - LEMAC) / MAC] \times 100$$

DWB0180300_012

Weighing Record Form – Tripod Jacks Method
Figure 3



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL



Weighing Procedure – Leveling With the Plumb Bob
Figure 4

DWB0180300_003

B. Center of Gravity

Refer to Figure 1 and Figure 3.

Calculate the CG as follows:

NOTE: Do this procedure with the copy of the Aircraft Weighing Records – Tripod Jacks Method form used during the Weighing procedure. Refer to Figure 3.

- (1) Record the CG of the nose and wing jacking points on the form in the related Xarm fields.

NOTE: The location of the jacking points is always the same unless specified differently. Refer to Figure 1.

- (2) Calculate the moment for each jacking point with the formula:

$$\text{Moment}/1\,000 = (\text{Weight} \times \text{Xarm}) / 1\,000$$

- (3) Record the results on the form in the related Moment/1 000 fields.

- (4) Add the weights from the Weight column and the moments from the Moment/1 000 column.

- (5) Record the results on the form in the related Aircraft Total fields.

- (6) Calculate the CG of the aircraft with the formula:

$$\text{Xarm}_{\text{CG}} = [(\text{Total Moment} \times 1\,000) / \text{Total Weight}]$$

- (7) Calculate the CG as a percentage of mean aerodynamic chord (%MAC) with the value of the CG of the aircraft calculated in the step above.

NOTE: The formula to calculate the CG position as a percentage of the MAC is:

$$\% \text{MAC (imperial)} = [(\text{Xarm}_{\text{CG}} - 488.025) / 92.64] \times 100$$

$$\% \text{MAC (SI)} = [(\text{Xarm}_{\text{CG}} - 12.40) / 2.35] \times 100$$

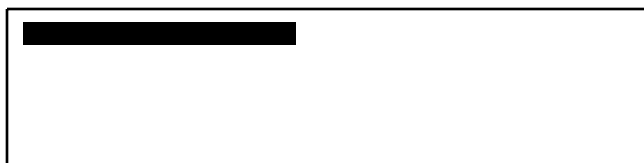
- (8) Fill the Average of Readings section of the form as follows:

- (a) In the Weight column, take the related jacking point values from the First Reading and the Second Reading sections.

- (b) For each jacking point, divide the sum of the two values by two to find the average.

- (c) Record the results in the related Average of Readings fields.

- (d) Do the steps again for the Xarm column and Moment/1 000 column.



C. Example

The Table 1A (imperial units) and Table 1B (SI units) that follow give an example to calculate the aircraft weight and CG of the aircraft when you weigh it on tripod jacks:

Table 1A – Example to Calculate the Aircraft Weight and CG on Tripod Jacks (Imperial Units)							
Description		Chn	Reading	Zero	Weight	Xarm	Moment/ 1 000
			lb	lb	lb	in.	lb in.
First Reading	Left MLG Tripod Jack	1	11 518	0	11 518	544.3	6 269.3
	Right MLG Tripod Jack	2	11 437	0	11 437	544.3	6 225.2
	NLG Tripod Jack	3	2 560	0	2 560	278.0	711.7
	Aircraft Total		25 515	0	25 515	517.6	13 206.1
Second Reading	Left MLG Tripod Jack	2	11 525	4	11 521	544.3	6 270.9
	Right MLG Tripod Jack	3	11 448	0	11 448	544.3	6 231.1
	NLG Tripod Jack	1	2 577	0	2 577	278.0	716.4
	Aircraft Total		25 550	4	27 533	517.4	16 449.1
Average (Avg) of Readings	Left MLG Tripod Jack				11 520	544.3	6 270.1
	Right MLG Tripod Jack				11 443	544.3	6 228.2
	NLG Tripod Jack				2 569	278.0	714.0
	Aircraft Total				25 531	517.5	13 212.3
CG is calculated by equation: $Xarm_{CG} = [(Total\ Moment \times 1\ 000) / Total\ Weight]$							
First reading: $Xarm_{CG} = [(13\ 206.1 \times 1\ 000) / 25\ 515] = 517.6\ in.$							
Average reading: $Xarm_{CG} = [(13\ 212.3 \times 1\ 000) / 25\ 531] = 517.5\ in.$							
CG position measured in %MAC is calculated as follows:							
$\%MAC = [(Xarm_{CG} - 488.025) / 92.64] \times 100$							
Average %MAC = $[(517.5 - 488.025) / 92.64] \times 100 = 31.8\%$							

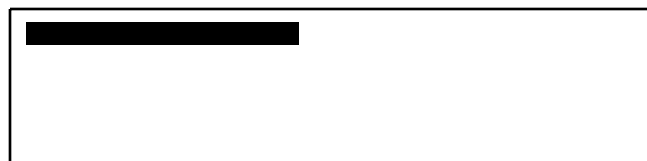
CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Table 1B – Example to Calculate the Aircraft Weight and CG on Tripod Jacks (SI Units)

Description		Chn	Reading	Zero	Weight	Xarm	Moment/ 1 000
			kg	kg	kg	m	kg m
First Reading	Left MLG Tripod Jack	1	5 225	0	5 225	13.825	72.230
	Right MLG Tripod Jack	2	5 188	0	5 188	13.825	71.722
	NLG Tripod Jack	3	1 161	0	1 161	7.061	8.200
	Aircraft Total		11 574	0	11 574	13.146	152.152
Second Reading	Left MLG Tripod Jack	2	5 228	2	5 226	13.825	72.247
	Right MLG Tripod Jack	3	5 193	0	5 193	13.825	71.791
	NLG Tripod Jack	1	1 169	0	1 169	7.061	8.254
	Aircraft Total		11 589	2	11 587	13.141	152.268
Average of Readings	Left MLG Tripod Jack				5 225	13.825	72.236
	Right MLG Tripod Jack				5 190	13.825	71.752
	NLG Tripod Jack				1 165	7.061	8.226
	Aircraft Total				11 580	13.141	152.173
CG is calculated by equation: $Xarm_{CG} = [(Total\ Moment \times 1\ 000) / Total\ Weight]$							
First reading: $Xarm_{CG} = [(152.152 \times 1\ 000) / 11\ 574] = 13.146\ m$							
Average reading: $Xarm_{CG} = [(152.173 \times 1\ 000) / 11\ 580] = 13.141\ m$							
CG position measured in %MAC is calculated as follows:							
$\%MAC = [(Xarm_{CG} - 12.40) / 2.35] \times 100$							
Average %MAC = $[(13.141 - 12.40) / 2.35] \times 100 = 31.5\%$							

4. Weighing on Platform Scales and Jack Procedure

Do the steps that follow to weigh the aircraft on platform scales and jack and calculate the center of gravity of the aircraft:



A. Weighing

Refer to Figure 2, Figure 5 and Figure 6.

Do the steps that follow:

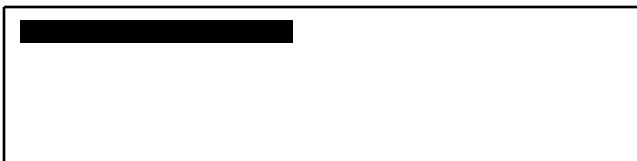
- (1) Do the Job Set-Up Procedure from the Job Set-Up Information section.
- (2) Align the three platform scales and ramps with the left and right main landing gears (MLG) and the nose landing gear (NLG).
- (3) Make sure that the platform scales are leveled and on a level floor.
- (4) Make sure the platform scale identifications agree with the related wheel locations.
- (5) Record the platform scale identifications and the related wheel locations.
- (6) Put the related wheel chocks on their platform scale.
- (7) Set the platform scales to zero.
- (8) Remove the wheel chocks from the platform scales.
- (9) Open the aft equipment-compartment door (311BB) (AMM TASK 52-41-00-010-801).
- (10) Put the plumb line through the hole in the bracket at the top of the frame at FS755.50. Refer to Figure 2.
- (11) Adjust the plumb line until the point of the plumb bob is approximately 0.5 in (1.27 cm) above the target.

NOTE: If the plumb bob is put higher than 0.5 in (1.27 cm) above the target, the result will not be accurate.

- (12) Attach the end of the plumb line to the bracket.
- (13) Remove the rear fuselage support.
- (14) Attach the towing mechanism to the aircraft.
- (15) Move the aircraft slowly up the ramps and on the platform scales.

WARNING: MAKE SURE THAT THE WHEEL CHOCKS ARE PLACED IN FRONT AND BEHIND THE MAIN WHEEL/TIRE ASSEMBLIES. MOVEMENT OF THE AIRCRAFT CAN CAUSE INJURY TO PERSONS AND DAMAGE TO THE EQUIPMENT.

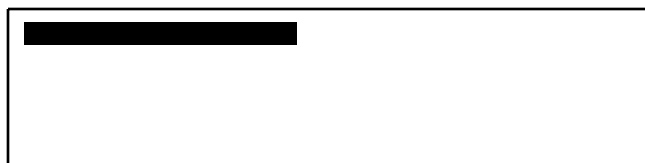
- (16) Put the wheel chocks in front of and behind the MLG and NLG wheels. Use the same wheel chocks used to put the platform scales to zero and in the same locations.



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

- (17) Disconnect the towing mechanism from the aircraft.
- (18) Do not set the parking brake.
- (19) Use a copy of the Aircraft Weighing Records – Platform Scales and Jack Method form to continue with this procedure. Refer to Figure 6.
- (20) Find the weight at the NLG platform scale as follows:
 - (a) Read the NLG weight data from the platform scale.
 - (b) Record the value on your copy of the form in the Scale Reading column.
- (21) Find the weight at the left and right MLG platform scales as follows:
 - (a) Read the weight data from each of the MLG platform scales.
 - (b) Record the related values on your copy of the form in the Scale Reading column.
- (22) Make the aircraft level as follows:
 - (a) Remove the nylon plug from the nose jacking point.
 - (b) Move the ramp away from the NLG platform scale.
 - (c) Install the jack pad in the nose jacking point.
 - (d) Put the jack below the nose jack pad.
 - (e) Extend the jack until it lightly touches the jack pad.
 - (f) Move the NLG chocks away from the NLG wheels. Do not remove the NLG chocks from the platform scales.
 - (g) Extend the jack until the aircraft is level in the longitudinal direction.

NOTE: The NLG wheels may or may not be completely off the platform scale.
 - (h) Lock the jack extension locknut.
 - (i) Adjust the MLG shock struts to laterally make the aircraft level.
- (23) Read the weight data from the left and right MLG platform scales and the NLG platform scale.
- (24) Record the related values on your copy of the form in the MLG/NLG Reading at Level column.



CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL**

- (25) Calculate the Xarm value for the center line position of the MLG wheel axle as follows:

NOTE: The MLG Xarm position changes with a change in aircraft weight.

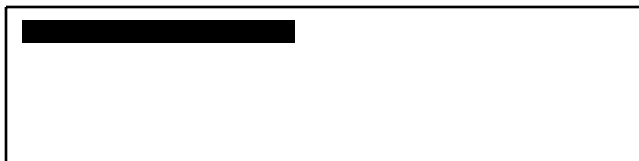
- (a) Measure the exposed chrome length (dimension A) on each MLG shock strut. Refer to Figure 5.
- (b) Take the average chrome length of the two measures to get the A_{avg} .
- (c) Find the MLG center-line position Xarm, refer to Table 2.

- (26) Find the CG of each of the MLG as follows:

- (a) Refer to Table 2 and Refer to Figure 5 to find the MLG Xarm related to the average Chrome length (dimension A).

NOTE: If the measured dimension A is in between the numbers of Table 2, use linear interpolation.

Table 2 – Main Landing Gear Xarm Versus Chrome Length of Main Landing Gear			
Average Chrome Length A_{avg} – (in)	Xarm (in)	Average Chrome Length A_{avg} – (mm)	Xarm (m)
0.50	542.8	12.7	13.788
0.63	542.8	15.9	13.788
0.75	542.8	19.1	13.788
0.88	542.8	22.2	13.788
1.00	542.8	25.4	13.787
1.13	542.8	28.6	13.787
1.25	542.8	31.8	13.787
1.38	542.8	34.9	13.786
1.50	542.8	38.1	13.786
1.63	542.7	41.3	13.785
1.75	542.7	44.5	13.785
1.88	542.7	47.6	13.784
2.00	542.7	50.8	13.784



CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2 – Main Landing Gear Xarm Versus Chrome Length of Main Landing Gear**

Average Chrome Length A_{avg} – (in)	Xarm (in)	Average Chrome Length A_{avg} – (mm)	Xarm (m)
2.13	542.6	54.0	13.783
2.25	542.6	57.2	13.782
2.38	542.6	60.3	13.782
2.50	542.6	63.5	13.781
2.63	542.5	66.7	13.780
2.75	542.5	69.9	13.779
2.88	542.4	73.0	13.778
3.00	542.4	76.2	13.777
3.13	542.4	79.4	13.776
3.25	542.3	82.6	13.775
3.38	542.3	85.7	13.774
3.50	542.2	88.9	13.772
3.63	542.2	92.1	13.771
3.75	542.1	95.3	13.770
3.88	542.1	98.4	13.769
4.00	542.0	101.6	13.767
4.13	542.0	104.8	13.766
4.25	541.9	108.0	13.764
4.38	541.8	111.1	13.762
4.50	541.8	114.3	13.761
4.63	541.7	117.5	13.759
4.75	541.6	120.7	13.757
4.88	541.6	123.8	13.755

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 2 – Main Landing Gear Xarm Versus Chrome Length of Main Landing Gear**

Average Chrome Length A_{avg} – (in)	Xarm (in)	Average Chrome Length A_{avg} – (mm)	Xarm (m)
5.00	541.5	127.0	13.754
5.13	541.4	130.2	13.752
5.25	541.3	133.4	13.750
5.38	541.2	136.5	13.748
5.50	541.2	139.7	13.745
5.63	541.1	142.9	13.743
5.75	541.0	146.1	13.741
5.88	540.9	149.2	13.739
6.00	540.8	152.4	13.736
6.13	540.7	155.6	13.734
6.25	540.6	158.8	13.732
6.38	540.5	161.9	13.729
6.50	540.4	165.1	13.726
6.63	540.3	168.3	13.724
6.75	540.2	171.5	13.721
6.88	540.1	174.6	13.718
7.00	540.0	177.8	13.715
7.13	539.9	181.0	13.712
7.25	539.7	184.2	13.709
7.38	539.6	187.3	13.706
7.50	539.5	190.5	13.703
7.63	539.4	193.7	13.700
7.75	539.2	196.9	13.697

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Table 2 – Main Landing Gear Xarm Versus Chrome Length of Main Landing Gear			
Average Chrome Length A _{avg} – (in)	Xarm (in)	Average Chrome Length A _{avg} – (mm)	Xarm (m)
7.88	539.1	200.0	13.693
8.00	539.0	203.2	13.690
8.13	538.8	206.4	13.687
8.25	538.7	209.6	13.683
8.38	538.6	212.7	13.680
8.50	538.4	215.9	13.676
8.63	538.3	219.1	13.672
8.75	538.1	222.3	13.668
8.88	538.0	225.4	13.664
9.00	537.8	228.6	13.661
9.13	537.7	231.8	13.656
9.25	537.5	235.0	13.652
9.38	537.3	238.1	13.648
9.50	537.2	241.3	13.644
9.63	537.0	244.5	13.640
9.75	536.8	247.7	13.635
9.88	536.6	250.8	13.631
10.00	536.5	254.0	13.626
10.13	536.3	257.2	13.622

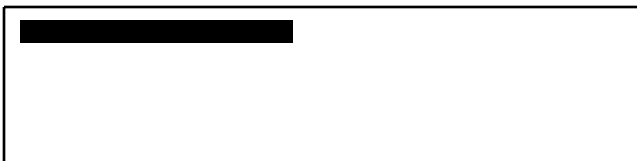
(27) Record the Xarm value in the Total field on your copy of the form.

(28) Lift the aircraft sufficiently to release the jack extension locknuts.

(29) Release the jack extension locknuts.

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

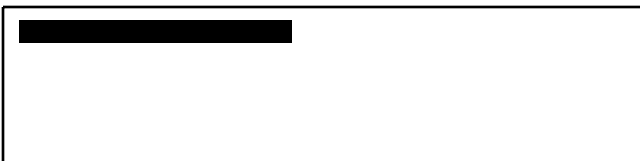
- (30) Lower the aircraft slowly until the NLG wheels are fully supported by the platform scale.
- (31) Remove the jack from the jack pad and the aircraft.
- (32) Remove the jack pad from the jacking point.
- (33) Install the nylon plug in the jacking point.
- (34) Do an extension check and adjustment of the shock strut as follows:
 - (a) Make sure that the shock struts of the MLG are set to the correct pressure (AMM TASK 32-11-07-720-801).
- (35) Remove the chocks from the MLG wheels.
- (36) Put the ramp in position against the platform scale.
- (37) Attach the towing mechanism to the aircraft.
- (38) Tow the aircraft off the platform scales and down the ramps.
- (39) Disconnect the towing mechanism from the aircraft.
- (40) Put the same wheel chocks on the same platform scales.
- (41) Record the indications shown on the three platform scales on your copy of the form. Record the weight indications in the related fields of the Scale Residual column.
- (42) Make sure that all of the recorded values are equal to zero:
 - (a) If the values are equal to zero, no change to the weight data is necessary. Copy the value recorded in the related Scale Reading column into the Weight column.
 - (b) If the values are not equal to zero, change the weight data as follows:
 - 1 If the value is less than zero, add it to the weight recorded in the related Scale Reading column. Record the result in the related Weight column.
 - 2 If the value is more than zero, subtract it from the weight recorded in the related Scale Reading column. Record the result in the related Weight column.
 - 3 If the value is more than the permitted limit of the equipment, do as follows:
 - a Do the Weighing on Platform Scales and Jack Procedure again with platform scales turned clockwise to the next wheel location.



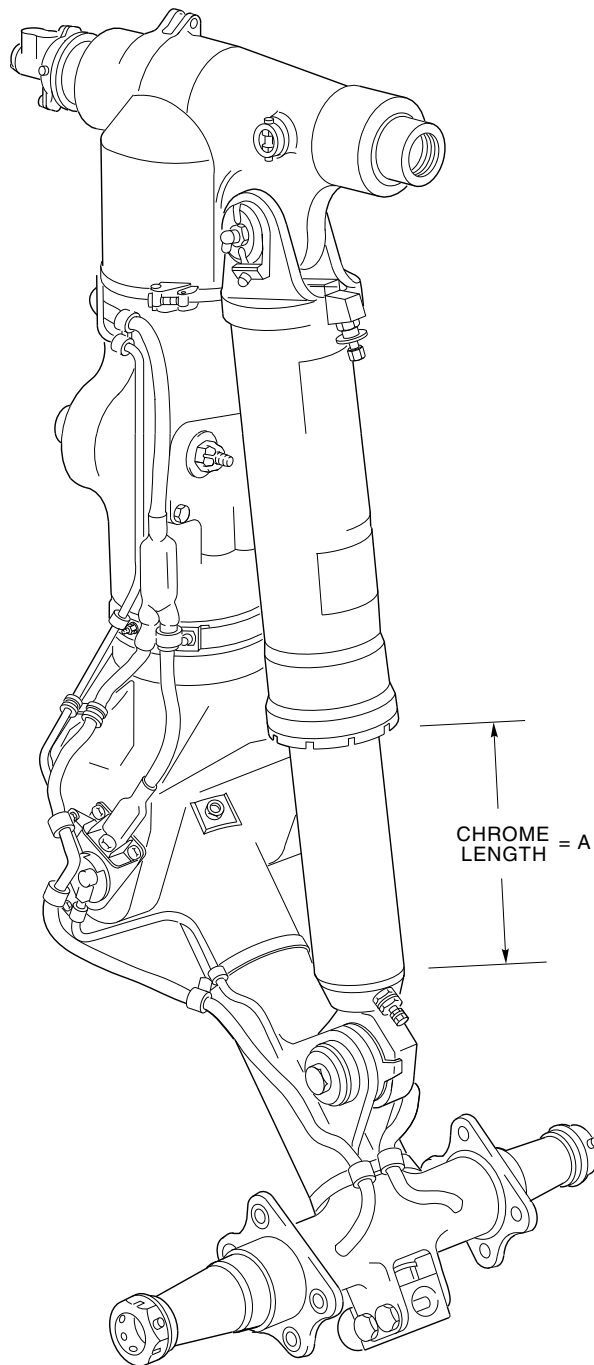
CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

- b Do this until you get two indications, one after the other, that have the conditions that follow:
- 1) The residual indications are less than the permitted equipment limit.
 - 2) The two indications that you get one after the other agree with the tolerance of the platform scales.
- (43) Fill out the Average (Avg) of Readings section on your copy of the form as follows if more than one reading is completed:
- (a) In the Weight column, take the related values from the First Reading and Second Reading fields (if applicable).
 - (b) Make an average of the values.
 - (c) Record the result in the related Average (Avg) of Readings fields. If only one reading is completed, simply copy the results into the related field.
 - (d) Do the steps again for the MLG Weight column and the NLG Reading in the MLG/NLG Reading at the level column.
 - (e) Calculate the average of the Total Xarm values from the First and Second Reading sections.
 - (f) Record the result in the Xarm column in the Total field of the Average (Avg) of Readings section.
 - (g) Copy the weight result from the average Total field of the Weight column to the Aircraft Weight and CG field in the same column.
- (44) Do the Center of Gravity part of the Weighing on Platform Scales and Jack Procedure.

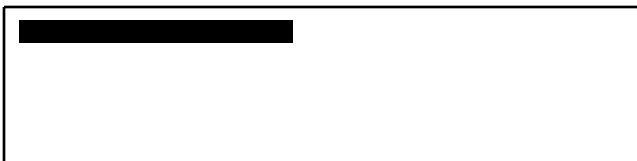


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL



DWB0180300_004

Measurement of Chrome Length on Main Landing Gear
Figure 5



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

AIRCRAFT WEIGHING RECORDS – PLATFORM SCALES AND JACK METHOD								
DESCRIPTION		CHN	SCALE READING	SCALE RESIDUAL	WEIGHT	MLG/NLG READING AT LEVEL	MLG WEIGHT	Xarm
			lb (kg)	lb (kg)	lb (kg)	lb (kg)	lb (kg)	in. (m)
FIRST READING	LEFT MLG PLATFORM SCALE							
	RIGHT MLG PLATFORM SCALE							
	NLG PLATFORM SCALE							
	TOTAL							
SECOND READING	LEFT MLG PLATFORM SCALE							
	RIGHT MLG PLATFORM SCALE							
	NLG PLATFORM SCALE							
	TOTAL							
AVERAGE OF READINGS	LEFT MLG PLATFORM SCALE							
	RIGHT MLG PLATFORM SCALE							
	NLG PLATFORM SCALE							
	TOTAL							
	AIRCRAFT WEIGHT AND CG							

CG is calculated as $Xarm_{CG}$ = Refer to the example.

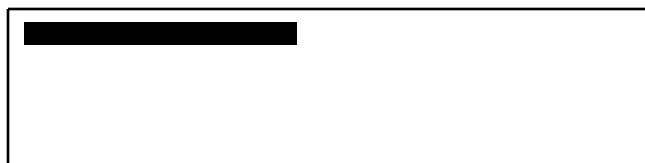
Conversion of CG position measured to %MAC is calculated as follows:

$$\%MAC = [(Xarm_{CG} - LEMAC) / MAC] \times 100$$

MLG Center-Line Position Xarm = Refer to the example.

DWB0180300_011

Weighing Record Form – Platform Scales and Jack Method
Figure 6



B. Center of Gravity

Refer to Figure 6.

Calculate the CG position as follows:

- (1) Use the values from the Total Weight and Total MLG Weight columns of the Average (Avg) of Readings fields as well as the Average NLG reading from the MLG/NLG reading at the level column to continue with this procedure.

- (2) Find the Xarm with the formula:

$$\text{Xarm}_{\text{CG}} (\text{imp}) = 278.00 + [(\text{Total}_{\text{avg}} \text{ MLG Weight} \times (\text{Xarm}_{\text{avg}} - 278.00) - 49.6 (\text{NLG}_{\text{avg}})) / \text{Total}_{\text{avg}} \text{ Weight}]$$

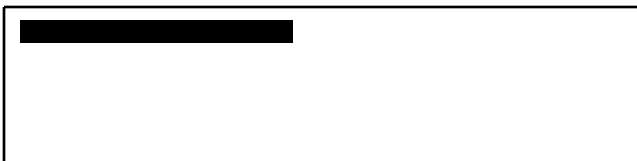
$$\text{Xarm}_{\text{CG}} (\text{SI}) = 7.06 + [(\text{Total}_{\text{avg}} \text{ MLG Weight} \times (\text{Xarm}_{\text{avg}} - 7.06) - 1.26 (\text{NLG}_{\text{avg}})) / \text{Total}_{\text{avg}} \text{ Weight}]$$

- (3) Record the Xarm value in the Aircraft Weight and CG field on your copy of the Aircraft Weighing Records – Platform Scales and Jack Method form. Refer to Figure 6.

C. Example

Refer to Figure 6.

The Table 3A (imperial units) and Table 3B (SI units) that follow give an example to calculate the weight and CG of the aircraft when you weigh it on platform scales and jack:



CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

Table 3A – Example to calculate the Aircraft Weight and CG on Platform Scales and Jack (Imperial units)

Description		Chn	Scale Reading	Scale Residual	Weight	MLG/ NLG Reading at Level	MLG Weight	Xarm
			lb	lb	lb	lb	lb	in.
First Reading	Left MLG Platform Scale	1	14 215	0	14 215	14 215	14 215	n/a
	Right MLG Platform Scale	2	14 790	0	14 790	14 780	14 780	n/a
	NLG Platform Scale	3	1 948	10	1 938	1 802	n/a	n/a
	Total		n/a	10	30 943	n/a	28 995	539.1
Second Reading	Left MLG Platform Scale	2	14 200	0	14 200	14 200	14 200	n/a
	Right MLG Platform Scale	3	14 795	0	14 795	14 805	14 805	n/a
	NLG Platform Scale	1	1 938	0	1 938	1 798	n/a	n/a
	Total		n/a	0	30 933	n/a	29 005	538.9
Average (Avg) of Readings	Left MLG Platform Scale		n/a	n/a	14 208	n/a	14 208	n/a
	Right MLG Platform Scale		n/a	n/a	14 793	n/a	14 793	n/a
	NLG Platform Scale		n/a	n/a	1 938	1 800	n/a	n/a
	Total		n/a	n/a	30 938	n/a	29 000	539.0
	Aircraft Weight and CG		n/a	n/a	30 938	n/a	n/a	519.8

CG is calculated by equation as follows:

$$Xarm_{CG} = 278.00 + [(Total_{avg} \text{ MLG Weight} \times (Xarm_{avg} - 278.00) - 49.6 (NLG_{avg})) / Total_{avg} \text{ Weight}]$$

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 3A – Example to calculate the Aircraft Weight and CG on Platform Scales and Jack
(Imperial units)**

Description	Chn	Scale Reading	Scale Residual	Weight	MLG/ NLG Reading at Level	MLG Weight	Xarm
		lb	lb	lb	lb	lb	in.

Where $X_{arm_{avg}}$ is the average MLG Xarm values between the First and Second Readings (if applicable) with Xarm obtained from Table 2.

$$X_{arm_{CG}} = 278.00 + [(29\,000 \times (539.0 - 278.00) - 49.6 \times 1\,800) / 30\,938] = 519.8 \text{ in.}$$

Conversion of CG position measured to %MAC is calculated as follows:

$$\%MAC = [(X_{arm_{CG}} - 488.025) / 92.64] \times 100$$

$$\%MAC = [(519.8 - 488.025) / 92.64] \times 100 = 34.3\%$$

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

Table 3B – Example to calculate the Aircraft Weight and CG on Platform Scales and Jack (SI units)

Description		Chn	Scale Reading	Scale Residual	Weight	MLG/ NLG Reading at Level	MLG Weight	Xarm
			kg	kg	kg	kg	kg	m
First Reading	Left MLG Platform Scale	1	6 448	0	6 448	6 448	6 448	n/a
	Right MLG Platform Scale	2	6 709	0	6 709	6 704	6 704	n/a
	NLG Platform Scale	3	884	5	879	817	n/a	n/a
	Total		n/a	5	14 036	n/a	13 152	13.69
Second Reading	Left MLG Platform Scale	2	6 441	0	6 441	6 441	6 441	n/a
	Right MLG Platform Scale	3	6 711	0	6 711	6 716	6 716	n/a
	NLG Platform Scale	1	879	0	879	815	n/a	n/a
	Total		n/a	0	14 031	n/a	13 157	13.69
Average (Avg) of Readings	Left MLG Platform Scale		n/a	n/a	6 444	n/a	6 444	n/a
	Right MLG Platform Scale		n/a	n/a	6 710	n/a	6 710	n/a
	NLG Platform Scale		n/a	n/a	879	816	n/a	n/a
	Total		n/a	n/a	14 033	n/a	13 154	13.69
	Aircraft Weight and CG		n/a	n/a	14 033	n/a	n/a	13.20

CG is calculated by equation:

$$Xarm_{CG} = 7.06 + [(Total_{avg} \text{ MLG Weight} \times (Xarm_{avg} - 7.06) - 1.26 (NLG_{avg})) / Total_{avg} \text{ Weight}]$$

Where $Xarm_{avg}$ is the average values between the First and Second Readings (if applicable) with $Xarm$ obtained from Table 2.

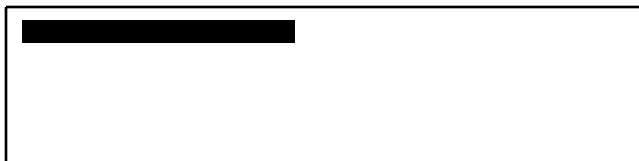
Table 3B – Example to calculate the Aircraft Weight and CG on Platform Scales and Jack (SI units)

Description	Chn	Scale Reading	Scale Residual	Weight	MLG/ NLG Reading at Level	MLG Weight	Xarm
		kg	kg	kg	kg	kg	m
$Xarm_{CG} = 7.06 + [13\ 154 \times (13.69 - 7.06) - 1.26 (816)] / 14\ 033 = 13.20\ m$							
Conversion of CG position measured to %MAC is calculated as follows:							
$\%MAC = [(Xarm_{CG} - 12.40) / 2.35] \times 100$							
$\%MAC = [(13.20 - 12.40) / 2.35] \times 100 = 34.1\%$							

5. Close Out

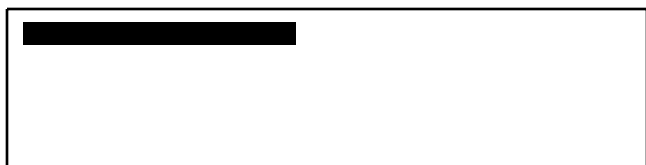
A. Do the close out as follows:

- (1) Remove the plumb bob from the aft equipment compartment. Refer to Figure 2.
- (2) Close the aft equipment-compartment door (311BB) (AMM TASK 52-41-00-410-801).
- (3) Remove all tools, equipment, and unwanted material from the work area.
- (4) Make sure to close all the fuel drain valves.
- (5) Put the aircraft back to the standard configuration for maintenance (AMM TASK 12-00-00-867-801).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



BALANCE LIMITS**1. Static Stability**

The green aircraft is supplied to the Completion Center with ferry ballast and miscellaneous equipment installed. A change in the configuration of the aircraft can cause the aircraft to be in a tip over condition. The changes can include one or more of those that follow:

- Added equipment
- Removal of equipment
- Moved equipment
- Unusual ground equipment
- Wind
- Other conditions.

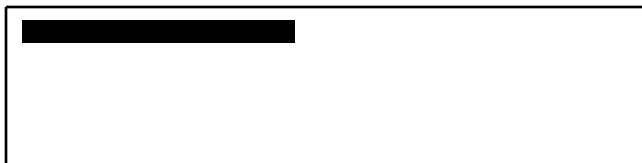
A. Before you complete a configuration change to the aircraft, do the steps that follow to make sure that the aircraft will stay stable:

- (1) Before the removal of the ferry ballast, put a minimum of 800 lb (363 kg) in the passenger compartment, aft of the bulkhead at FS280.00.
- (2) Make an analysis of the aircraft weight and balance moment impact due to the change in configuration.
- (3) Do a check of the new weight and center of gravity of the aircraft as a result of the change in configuration (Subject 01-90-20).
- (4) You must take more precautions when there is fresh fallen snow on the horizontal stabilizer and pylons.

B. If the applicable configuration causes the aircraft to be in the tip over region, do as follows:

CAUTION: MAKE SURE THAT THE REAR FUSELAGE SUPPORT IS PUT DIRECTLY BELOW THE RIVET LINE AT FS755.00. INCORRECTLY LOCATING THE SUPPORT CAN CAUSE DAMAGE TO THE AIRCRAFT AND THE EQUIPMENT.

- (1) Put the rear fuselage support in position for safety reasons in all of these conditions.
- (2) Change the configuration in the flight compartment and passenger compartment until the CG is back to a permitted limit.
- (3) Obey the load limits of the floor structure and floor panels of the flight compartment and passenger compartment, and the load limits of the fuselage. The values of the load limits are given in Table 2 (Subject 01-80-20). If it is not possible, you must use the tie-down point at FS278.00 to tie the nose down.



2. Towing

When you tow the aircraft, you must keep a minimum load on the nose landing gear (NLG). To tow the aircraft when its weight is between MWE and ZFW, a minimum load of 1 650 lb (748 kg) is necessary on the NLG.

A. If the load is not sufficient, you must add ballast on the passenger compartment floor, immediately aft of the bulkhead at FS280.00.

B. Calculate the necessary NLG load as follows:

$$\text{NLG load} = [\text{Aircraft Weight (MLG Xarm - Aircraft CG)}] / (\text{MLG Xarm} - \text{NLG Xarm})$$

NOTE: Where MLG Xarm = Center of gravity (CG) of main landing gear
(Subject 01-00-40) and NLG Xarm = 228.4 in (5.791 m).

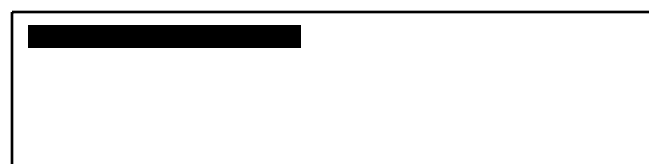
3. Nose Landing Gear Load

Refer to Figure 1.

It is possible to use the monograph to find the approximate NLG load related to the aircraft weight and CG position.

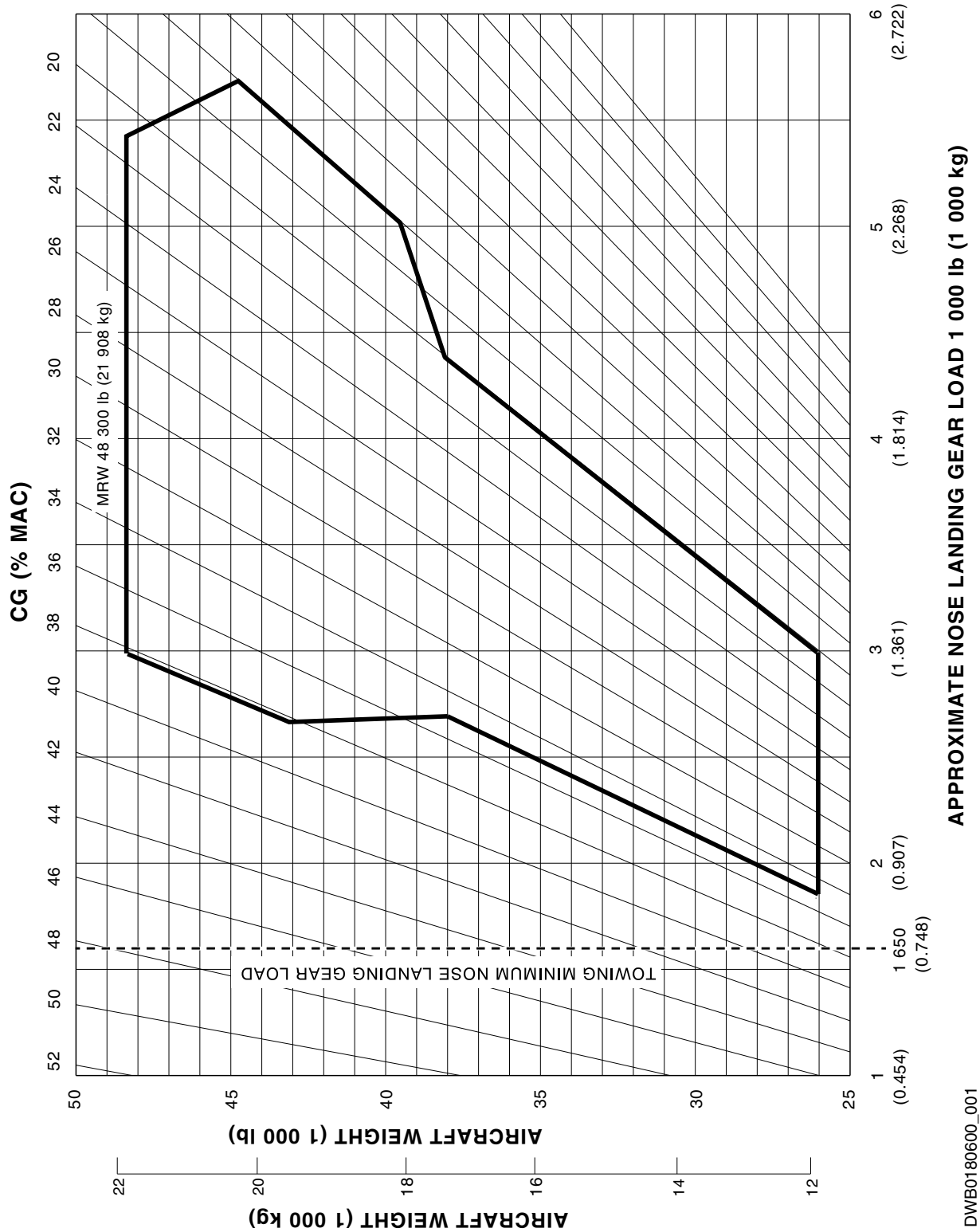
A. Do the steps that follow when the NLG load is less than the minimum load:

- (1) Add ballast in the flight compartment and passenger compartment, as applicable.
- (2) Obey the load limits of the floor structure and floor panels, of the flight compartment and passenger compartment, and the load limits of the fuselage. The values of the load limits are given in Table 2 (Subject 01-80-20).



CHALLENGER 650

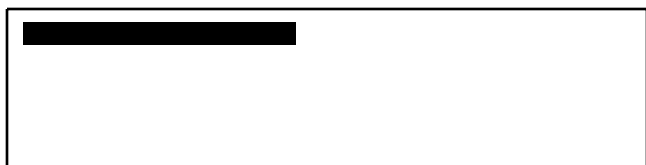
CL-650 WEIGHT AND BALANCE MANUAL



Approximate Nose Landing Gear Load for a Level Aircraft
Figure 1

***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK

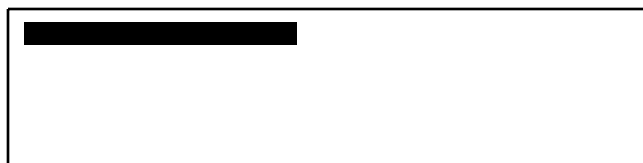


LOADING PROCEDURES AND EXAMPLES

1. General

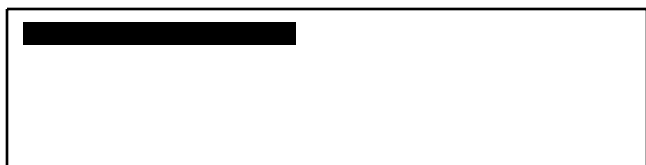
This section contains the subjects that follow related to loading procedures and examples:

- Operating Weight Empty (Subject 01-90-10)
- Weight Build-Up (Subject 01-90-20)
- CG Slide Method (Subject 01-90-50).



***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



OPERATING WEIGHT EMPTY**1. General**

This section shows how to calculate the operating weight empty (OWE) and center of gravity (CG) of an aircraft for a given loading scenario. To find the OWE, it is necessary to add the values of the operating items to the true empty weight (TEW). If the MWE is used, it is also necessary to add the outfitting, unusable fuel, and oil weight to find the OWE.

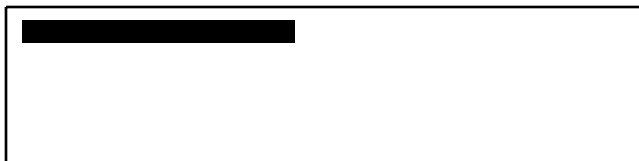
It is necessary to know the value of the OWE to calculate the ramp weight (RW), take-off weight (TOW) and landing weight (LW) with the different procedures given.

2. Procedure

Refer to Figure 1, Figure 2.

A. Do the steps that follow to calculate the OWE:

- (1) Refer to the example in Table 1. Use a copy of the OWE Calculation Form (Refer to Figure 1) to continue with this procedure.
- (2) Find the values for the TEW or MWE, as applicable, in Chapter 2, Aircraft Reports.
- (3) Record these values on the form in the related fields.
- (4) Refer to the example in Table 2. Use a copy of the Multiple Item Form (Refer to Figure 2) to calculate the sums for multiple items.
- (5) Record the sums of the multiple items on your copy of the OWE Calculation Form.
- (6) Do the applicable operations (OP) as shown in Table 1.

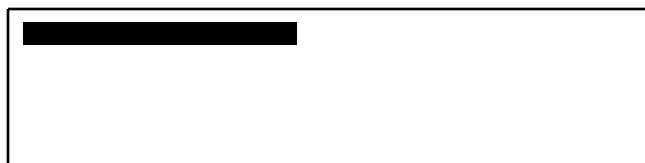


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

OWE Calculation Form							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
True Empty Weight (TEW) or Manufacturer's Weight Empty (MWE) as Applicable						Last Weighing Result	[1] [2]
Items to Add Only if MWE is Used for Calculations							
Outfitting	+					Chapter 2, Aircraft Reports	[4]
Unusable Fuel	+					Subject 01-20-10	[3]
Oil	+					Subject 01-30-10	[3]
Operating Items							
Flight Compartment Crew	+					Subject 01-40-10	
Crew Baggage	+					Chapter 2, Aircraft Reports	[4]
Other Operating Items	+					Chapter 2, Aircraft Reports	[4]
Operating Weight Empty (OWE)	=						[1]
Notes: [1] %MAC (imperial) = [(Xarm - LEMAC) / MAC] x 100 = [(Xarm - 488.025) / 92.64] x 100 %MAC (SI) = [(Xarm - LEMAC) / MAC] x 100 = [(Xarm - 12.40) / 2.35] x 100 [2] Last weighing result for TEW or MWE, as applicable, given in Chapter 2, Aircraft Reports. [3] If you use the true empty weight (TEW) and not the manufacturer's weight empty (MWE), make sure that you do not add this item two times. [4] The operator must use the values related to the applicable flight.							

DWB0190100_011

OWE Calculation Form
Figure 1

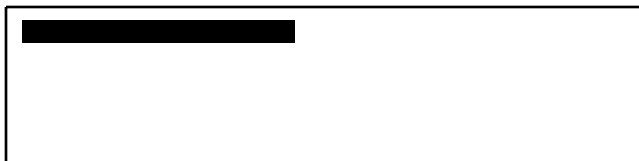


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

MULTIPLE ITEM FORM			
ITEM	WEIGHT	Xarm	$\frac{\text{MOMENT}}{1\ 000}$
	lb (kg)	in (m)	lb in (kg m)

DWB0190100_001

Multiple Item Form
Figure 2



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

3. Example

The example given in the Table 1 and Table 2 that follow shows how to calculate the OWE:

Table 1 – OWE Calculation (Example) [5]							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
True Empty Weight (TEW) or Manufacturer's Weight Empty (MWE) as Applicable		25 480 (11 558)	524.9 (13.33)	13 374 (154.068)	39.8	Last Weighing Result	[1] [2]
Items to Add Only if MWE is Used for Calculations							
Outfitting	+	3 550 (1 610)	430.0 (10.92)	1 526.5 (17.581)		Chapter 2, Aircraft Reports	[4]
Unusable Fuel	+	129 (59)	536.6 (13.6)	69.2 (0.797)		Subject 01-20-10	[3]
Oil	+	47 (21)	673.7 (17.11)	31.7 (0.365)		Subject 01-30-10	[3]
Operating Items							
Flight Compartment Crew (170 lb (77.11 kg) /Crew)	+	340 (154.22)	255.0 (6.48)	86.7 (0.999)		Subject 01-40-10	
Crew Baggage	+	30 (13.61)	290.0 (7.37)	8.7 (0.103)		Chapter 2, Aircraft Reports	[4]
Other Operating Items	+	650 (295)	450.0 (11.43)	292.5 (3.372)		Chapter 2, Aircraft Reports	[4]

CHALLENGER 650**CL-650 WEIGHT AND BALANCE MANUAL****Table 1 – OWE Calculation (Example) [5]**

Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Operating Weight Empty (OWE)	=	26 500 (12 020)	519.3 (13.19)	13 762.5 (158.606)	33.8		[1]

Notes:

[1] %MAC (imperial) = $[(Xarm - LEMAC) / MAC] \times 100 = [(Xarm - 488.025) / 92.64] \times 100$

%MAC (SI) = $[(Xarm - LEMAC) / MAC] \times 100 = [(Xarm - 12.40) / 2.35] \times 100$

[2] Last weighing result for TEW or MWE, as applicable, given in Chapter 2, Aircraft Reports.

[3] If you use the true empty weight (TEW) and not the manufacturer's weight empty (MWE), make sure that you do not add this item two times.

[4] The operator must use the values related to the applicable flight.

[5] Information found here is for example only.

The example given in the Table 1 and Table 2 that follow shows how to calculate the OWE:

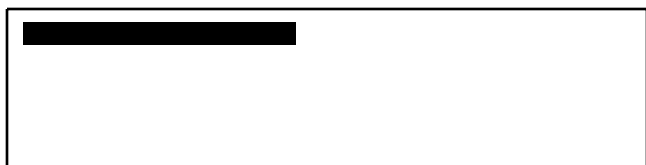
Table 2 – Multiple Items (Example) [1]

Item	Weight	Xarm	Moment/1 000
	lb (kg)	in (m)	lb in (kg m)
Pilot	170 (77.11)	255.0 (6.48)	43.4 (0.500)
Copilot	170 (77.11)	255.0 (6.48)	43.4 (0.500)
Total	340 (154.22)	255.0 (6.48)	86.7 (0.999)
Pilot Baggage	15 (6.80)	290.0 (7.37)	4.4 (0.050)
Copilot Baggage	15 (6.80)	290.0 (7.37)	4.4 (0.050)
Total	30 (13.61)	290.0 (7.37)	8.7 (0.100)

NOTE: [1] information found here is for example only.

***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

THIS PAGE INTENTIONALLY LEFT BLANK



WEIGHT BUILD-UP

1. General

This section shows how to use the weight build-up procedure to calculate the weight and center of gravity (CG) for a given loading scenario. The weight of the passengers and baggage is added to the operating weight empty (OWE) to get the zero fuel weight (ZFW). Then the necessary fuel is added to calculate the ramp weight (RW), take-off weight (TOW), fuel-curve-boundary-points (FCBP) and landing weight (LW).

2. Procedure

A. ZFW Calculation

Refer to Figure 1, Figure 2, Figure 7 and Figure 8.

Do the steps that follow to calculate the ZFW:

- (1) Refer to the example in the Table 1 that follows for the calculation of the ZFW:

Table 1 – Weight Build-Up – ZFW Calculation Example [4]							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Operating Weight Empty (OWE)	=	26 500 (12 020)	519.3 (13.19)	13 762.5 (158.606)	33.8	Subject 01-90-10	[1] [2]
Payload Items							
Check-in Baggage or Cargo	+	150 (68)	605.0 (15.37)	90.8 (1.045)		Chapter 2, Aircraft Reports	[5]
Passengers (5)	+	850 (386)	450.0 (11.43)	382.5 (4.41)		Chapter 2, Aircraft Reports	[5]
Adjustment							
Adjustment	+					Subject 01-40-10	[3]
Zero Fuel Weight – ZFW (Gear Down)							

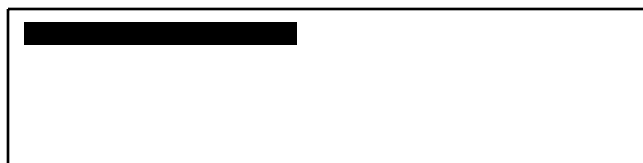


Table 1 – Weight Build-Up – ZFW Calculation Example [4]

Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=	27 500 (12 474)	517.7 (13.15)	14 235.9 (164.061)	32.0	Subject 01–00–50	[2]
Gear Up	+			–3.7 (–0.043)		Subject 01–00–80	[6]
Zero Fuel Weight – ZFW (Gear Up)							
Zero–Fuel–Weight Gear Up (ZFW Gear Up)	=	27 500 (12 474)	517.5 (13.15)	14 232.2 (164.018)	31.9	Subject 01–00–50	[2]

Notes:

[1] Results from previous sum.

[2] %MAC (imperial) = $[(Xarm_{CG} - 488.025) / 92.64] \times 100$

%MAC (SI) = $[(Xarm_{CG} - 12.40) / 2.35] \times 100$

[3] To use when necessary.

[4] Information found here is for example only.

[5] The operator must use the values related to the applicable flight.

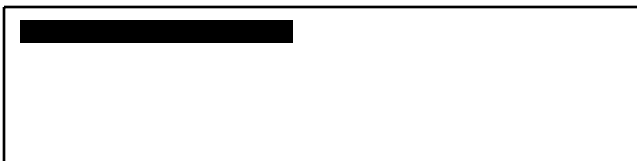
[6] The total aircraft moment decreases by approximately 3 700 lb in (43 kg m) when the landing gears are retracted.

- (2) Use a copy of the Weight Build-Up – ZFW Calculation Form to continue with this procedure. Refer to Figure 2.
- (3) Refer to the example in Table 1.
- (4) Take the values for OWE from your copy of the OWE Calculation Form, refer to (Subject 01–90–10).
- (5) Record these values in the applicable fields of your Weight Build-Up – ZFW Calculation Form.
- (6) Refer to the example in Table 6.
- (7) Use a copy of the Payload Item Form to calculate the sum of payload items. Refer to Figure 7.
- (8) Record the sums of the payload items on your Weight Build-Up – ZFW Calculation Form. Refer to Figure 2.
- (9) Do the applicable operations (Op) as shown in Table 1.

CHALLENGER 650

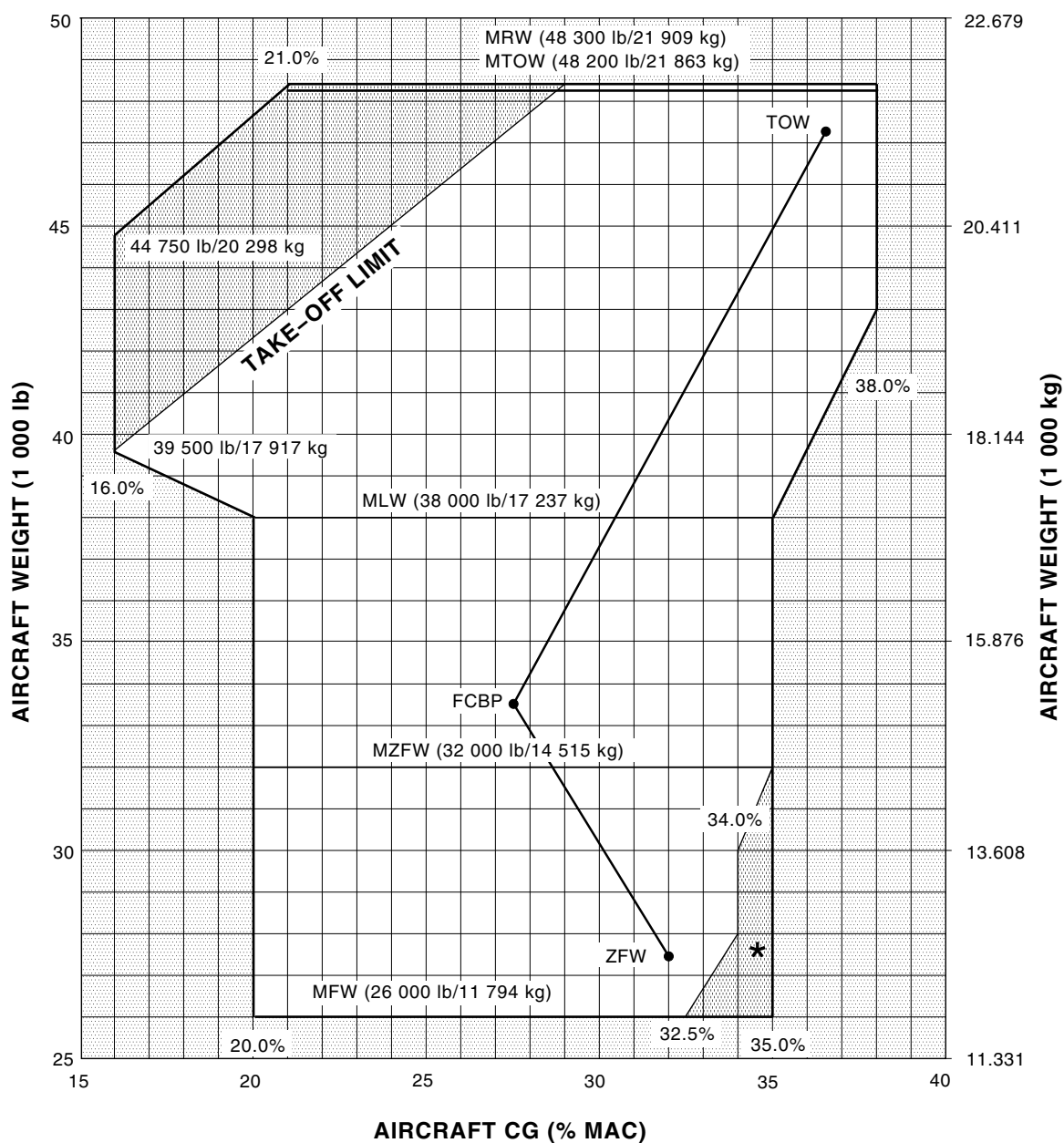
CL-650 WEIGHT AND BALANCE MANUAL

- (10) Make sure that the value of ZFW is not more than the maximum zero fuel weight (MZFW).
- (11) Find if the ZFW point is in or out of the area, identified with an asterisk (*), and refer to the appropriate figure in 01-20-20.
- (12) Refer to Figure 1 for an example.



CHALLENGER 650

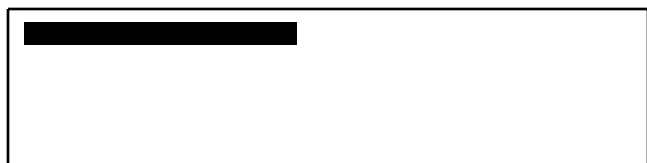
CL-650 WEIGHT AND BALANCE MANUAL



[* For zero fuel weight in that region, refer to Figure 3, 01-20-20, Auxiliary and Tail Tank Admissible Ratio (for ZFW CG inside lower aft zone of CG envelope)]

DWB0190200_081

Weight Build-Up Envelope – Example
Figure 1

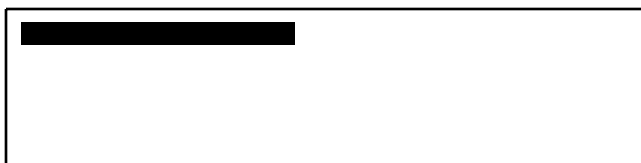


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Weight Build-Up – ZFW Calculation							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Operating Weight Empty (OWE)	=					Subject 01-90-10	[1] [2]
Payload Items							
Check-in Baggage or Cargo	+					Chapter 2, Aircraft Reports	[4]
Passengers	+					Chapter 2, Aircraft Reports	[4]
Adjustment							
Adjustment	+					Subject 01-40-10	[3]
Zero Fuel Weight – ZFW (Gear Down)							
Zero-Fuel-Weight Gear Down (ZFW Gear Down)	=					Subject 01-00-50	[2]
Gear Up	+			-3.7 (-0.043)		Subject 01-00-80	[5]
Zero Fuel Weight– ZFW (Gear Up)							
Zero-Fuel-Weight Gear Up (ZFW Gear UP)	=					Subject 01-00-50	[2]
Notes: [1] Results from previous sum. [2] %MAC (imperial) = [(Xarm CG- 488.025) / 92.64] x 100 %MAC (SI) = [(Xarm CG- 12.40) / 2.35] x 100 [3] To use when necessary. [4] The operator must use the values related to the applicable flight. [5] The total aircraft moment decreases by approximately 3 700 lb in (43 kg m) when the landing gears are retracted.							

DWB0190200_031

Weight Build-Up – ZFW Calculation
Figure 2



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

B. RW Calculation

Refer to Figure 1 and Figure 3.

Do the steps that follow to calculate the RW:

(1) Refer to the example in the Table 2 that follows for the calculation of the RW:

Table 2 – Weight Build-Up – RW Calculation Example [3]							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=	27 500 (12 474)	517.7 (13.15)	14 235.9 (164.061)	32.0	Subject 01–00–50	[1] [2]
Fuel							
Main Tank Fuel	+	9 720 (4 409)	506.6 (12.87)	4 924.2 (56.725)		Subject 01–20–10	[4]
Auxiliary Tank Fuel	+	7 168 (3 252)	450.6 (11.45)	3 230.4 (37.216)		Subject 01–20–10	[4]
Tail Tank Fuel	+	3 112 (1 411)	771.7 (19.60)	2 401.5 (27.655)		Subject 01–20–10	[4]
Ramp Weight – RW (Gear Down)							
Ramp–Weight Gear Down (RW Gear Down)	=	47 500 (21 546)	521.9 (13.26)	24 792.0 (285.704)	36.6	Subject 01–00–50	[2]
Notes: [1] Results from previous sum. [2] %MAC (imperial) = [(XarmCG – 488.025) / 92.64] x 100 %MAC (SI) = [(XarmCG – 12.40) / 2.35] x 100 [3] Information found here is for example only. [4] The operator must use the values related to the applicable flight.							

(2) Use a copy of the Weight Build-Up – RW Calculation Form to continue with this procedure. Refer to Figure 3.

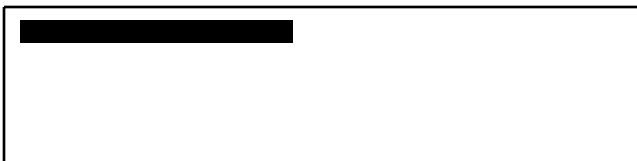
(3) Record the values found for the ZFW gear down.

(4) Record the values for the fuel.

CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

- (5) Do the applicable operations (Op) as shown in Table 2.
- (6) Make sure that the value of RW is not more than the maximum ramp weight (MRW) shown on your copy of the Weight Build-Up Envelope.
- (7) Refer to Figure 1 for an example.

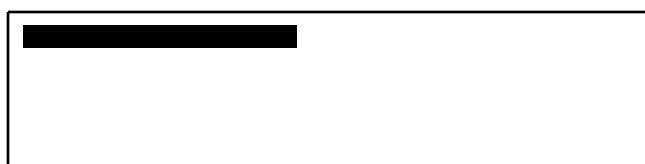


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Weight Build-Up – RW Calculation							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero-Fuel-Weight Gear Down (ZFW Gear Down)	=					Subject 01-00-50	[1] [2]
Fuel							
Main Tank Fuel	+					Subject 01-20-10	[3]
Auxiliary Tank Fuel	+					Subject 01-20-10	[3]
Tail Tank Fuel	+					Subject 01-20-10	[3]
Ramp Weight – RW (Gear Down)							
Ramp-Weight Gear Down (RW Gear Down)	=					Subject 01-00-50	[2]
Notes: [1] Results from previous sum. [2] %MAC (imperial) = $[(Xarm_{CG} - 488.025) / 92.64] \times 100$ %MAC (SI) = $[(Xarm_{CG} - 12.40) / 2.35] \times 100$ [3] The operator must use the values related to the applicable flight.							

LWB0190200_021

Weight Build-Up – RW Calculation
Figure 3



C. TOW Calculation

Refer to Figure 1 and Figure 4.

Do the steps that follow to calculate the TOW:

(1) Refer to the example in the Table 3 that follows for the calculation of the TOW:

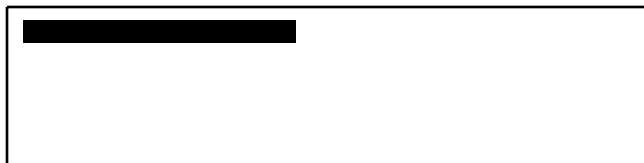
Table 3 – Weight Build-Up – TOW Calculation Example [3]							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=	27 500 (12 474)	517.7 (13.15)	14 235.9 (164.061)	32.0	Subject 01–00–50	[1] [2]
Fuel							
Main Tank Fuel	+	9 620 (4 364)	506.1 (12.85)	4868.7 (56.077)		Subject 01–20–10	[4] [5]
Auxiliary Tank Fuel	+	7 168 (3 252)	450.6 (11.45)	3 230.4 (37.216)		Subject 01–20–10	[4] [5]
Tail Tank Fuel	+	3 112 (1 411)	771.7 (19.60)	2 401.5 (27.655)		Subject 01–20–10	[4] [5]
Take–Off Weight – TOW (Gear Down)							
Take–Off–Weight Gear Down (TOW Gear Down)	=	47 400 (21 500)	521.8 (13.25)	24 736.5 (285.012)	36.5	Subject 01–00–50	[2]
Notes: [1] Results from previous sum. [2] %MAC (imperial) = [(Xarm _{CG} – 488.025) / 92.64] x 100 %MAC (SI) = [(Xarm _{CG} – 12.40) / 2.35] x 100 [3] Information found here is for example only. [4] The operator must use the values related to the applicable flight. [5] Fuel quantity remaining prior take–off. Equivalent to main tank fuel weight at RW reduced by quantity burned during taxi.							

(2) Use a copy of the Weight Build-Up – TOW Calculation Form to continue with this procedure. Refer to Figure 4.

(3) Record the values found for the ZFW gear down.

***CHALLENGER* 650**
CL-650 WEIGHT AND BALANCE MANUAL

- (4) Record the values for the fuel.
- (5) Do the applicable operations (Op) as shown in Table 3.
- (6) Make sure that the value of TOW is not more than the maximum take-off weight (MTOW) shown on your copy of the Weight Build-Up Envelope.
- (7) Refer to Figure 1 for an example.

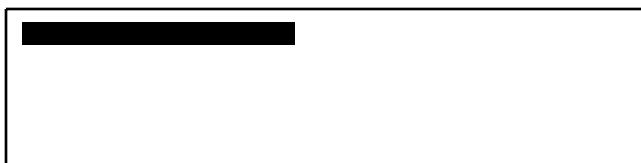


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Weight Build-Up – TOW Calculation							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=					Subject 01–00–50	[1] [2]
Fuel							
Main Tank Fuel	+					Subject 01–20–10	[3] [4]
Auxiliary Tank Fuel	+					Subject 01–20–10	[3] [4]
Tail Tank Fuel	+					Subject 01–20–10	[3] [4]
Take–Off Weight – TOW (Gear Down)							
Take–Off–Weight Gear Down (TOW Gear Down)	=					Subject 01–00–50	[2]
Notes: [1] Results from previous sum. [2] %MAC (imperial) = [(Xarm CG– 488.025) / 92.64] x 100 %MAC (SI) = [(Xarm CG– 12.40) / 2.35] x 100 [3] The operator must use the values related to the applicable flight. [4] Fuel quantity remaining prior take–off. Equivalent to main tank fuel weight at RW reduced by quantity burned during taxi.							

DWB0190200_071

Weight Build-Up – TOW Calculation
Figure 4



CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

D. FCBP Calculation

Refer to Figure 1 and Figure 5.

Do the steps that follow to calculate the FCBP:

(1) Refer to the example in the Table 4 that follows for the calculation of the FCBP:

Table 4 – Weight Build-Up – FCBP Calculation Example [3]							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=	27 500 (12 474)	517.7 (13.15)	14 235.9 (164.061)	32.0	Subject 01–00–50	[1] [2]
Fuel							
Fuel Quantity for Fuel–Curve Boundary–Point	+	6 075 (2 756)	494.9 (12.06)	3 006.5 (34.635)		Subject 01–20–10	[5]
Fuel–Curve Boundary–Point – FCBP (Gear Down)							
Fuel–Curve Boundary–Point – FCBP (Gear Down)	=	33 575 (15 229)	513.5 (13.05)	17 242.4 (198.696)	27.6		[2]
Add Gear Retraction Moment	+			–3.7 (–0.043)		Subject 01–00–80	[4]
Fuel–Curve Boundary–Point – FCBP (Gear Up)							

Table 4 – Weight Build-Up – FCBP Calculation Example [3]

Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Fuel–Curve Boundary–Point – FCBP (Gear Up)	=	33 575 (15 229)	513.4 (13.04)	17 238.7 (198.653)	27.4	Subject 01–00–50	[2]

Notes:

[1] Results from previous sum.

[2] %MAC (imperial) = $[(Xarm_{CG} - 488.025) / 92.64] \times 100$ %MAC (SI) = $[(Xarm_{CG} - 12.40) / 2.35] \times 100$

[3] Information found here is for example only.

[4] The total aircraft moment decreases by approximately 3 700 lb in (43 kg m) when the landing gears are retracted.

[5] The quantity of fuel in the main tank gives the most forward center of gravity variation due to fuel in main tank. In the case where the quantity of fuel necessary in the main tank for the applicable flight is less than 6 075 lb (2 755 kg), then the values found here must be related to the smaller quantity of fuel.

- (2) Use a copy of the Weight Build-Up – FCBP Calculation Form to continue with this procedure. Refer to Figure 5.
- (3) Record the values found for the ZFW gear down.
- (4) Do the applicable operations (Op) as shown in Table 4.

NOTE: The values found for the FCBP will be used to complete your weight build-up envelope.

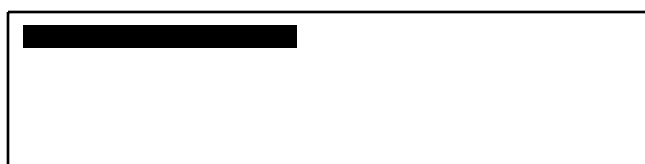
- (5) Refer to Figure 1 for an example.

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Weight Build-Up – FCBP Calculation							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=					Subject 01–00–50	[1] [2]
Fuel							
Fuel Quantity for Fuel–Curve Boundary–Point	+					Subject 01–20–10	[4]
Fuel–Curve Boundary–Point – FCBP (Gear Down)							
Fuel–Curve Boundary–Point – FCBP (Gear Down)	=						[2]
Add Gear Retraction Moment	+			–3.7 (–0.043)		Subject 01–00–80	[3]
Fuel–Curve Boundary–Point – FCBP (Gear Up)							
Fuel–Curve Boundary–Point – FCBP (Gear Up)	=					Subject 01–00–50	[2]
Notes:							
[1] Results from previous sum.							
[2] %MAC (imperial) = [(Xarm CG– 488.025) / 92.64] x 100							
%MAC (SI) = [(Xarm CG– 12.40) / 2.35] x 100							
[3] The total aircraft moment decreases by approximately 3 700 lb in (43 kg m) when the landing gears are retracted.							
[4] The quantity of fuel in the main tank gives the most forward center of gravity variation due to fuel in main tank. In the case where the quantity of fuel necessary in the main tank for the applicable flight is less tha 6 075 lb (2 755 kg), then the values found here must be related to the smaller quantity of fuel.							

DWB0190200_061

Weight Build-Up – FCBP Calculation
Figure 5



E. LW Calculation

Refer to Figure 1 and Figure 6.

Do the steps that follow to calculate the LW:

- (1) Refer to the example in the Table 5 that follows for the calculation of the LW:

Table 5 – Weight Build-Up – LW Calculation Example [3]							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=	27 500 (12 474)	517.7 (13.15)	14 235.9 (164.061)	32.0	Subject 01–00–50	[1] [2]
Fuel							
Remaining Fuel at Landing	+	2 025 (919)	481.5 (12.23)	975.0 (11.239)		Subject 01–20–10	
Landing Weight – LW (Gear Down)							
Landing–Weight Gear Down (LW Gear Down)	=	29 525 (13 393)	515.2 (13.09)	15 210.9 (175.300)	29.3	Subject 01–00–50	[2]
Notes: [1] Results from previous sum. [2] %MAC (imperial) = [(Xarm _{CG} – 488.025) / 92.64] x 100 %MAC (SI) = [(Xarm _{CG} – 12.40) / 2.35] x 100 [3] Information found here is for example only.							

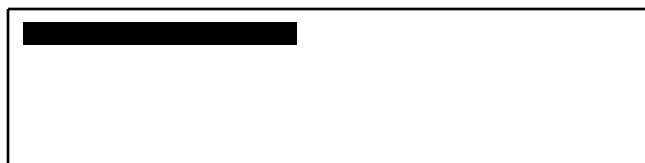
- (2) Use a copy of the Weight Build-Up – LW Calculation Form to continue with this procedure. Refer to Figure 6.
- (3) Record the values found for the ZFW gear down.
- (4) Record the values for the fuel.
- (5) Do the applicable operations (Op) as shown in Table 5.
- (6) Make sure that the value of LW is not more than the maximum landing weight (MLW) shown on your copy of the Weight Build-Up Envelope.
- (7) Refer to Figure 1 for an example.

CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

Weight Build-Up – LW Calculation							
Item	Op	Weight	Xarm	Moment/ 1 000	%MAC	Reference	Note
		lb (kg)	in (m)	lb in (kg m)			
Zero Fuel Weight – ZFW (Gear Down)							
Zero–Fuel–Weight Gear Down (ZFW Gear Down)	=					Subject 01–00–50	[1] [2]
Fuel							
Remain Fuel at Landing	+					Subject 01–20–10	
Landing Weight – LW (Gear Down)							
Landing–Weight Gear Down (LW Gear Down)	=					Subject 01–00–50	[2]
Notes: [1] Results from previous sum. [2] %MAC (imperial) = [(Xarm CG– 488.025) / 92.64] x 100 %MAC (SI) = [(Xarm CG– 12.40) / 2.35] x 100							

DWB0190200_051

Weight Build-Up – LW Calculation
Figure 6



F. Payload Item Calculation

Refer to Figure 7.

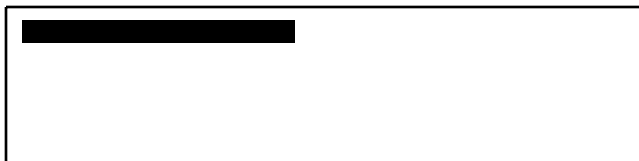
Do the steps that follow to calculate the payload items:

- (1) Refer to the example in the Table 6 that follows for the calculation of the payload items:

Table 6 – Payload Item Calculation Example [1]			
Item	Weight	Xarm	Moment/1 000
	lb (kg)	in (m)	lb in (kg m)
Baggage	80 (36)	605.0 (15.37)	48.4 (0.553)
Baggage	70 (32)	605.0 (15.37)	42.4 (0.492)
Total	150 (68)	605.0 (15.37)	90.8 (1.045)
NOTE: [1] Information found here is for example only.			

- (2) Use a copy of the Payload Item Form to continue with this procedure. Refer to Figure 7.
- (3) Record the applicable values for each of the payload items on your copy of the Payload Item Form.
- (4) Calculate the sum of payload items as shown in Table 6.

NOTE: The sum of payload items is used for the calculation of the ZFW.



CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL

DWB0190200_041

Payload Item Calculation

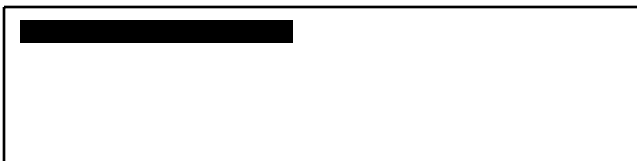
Figure 7

G. Weight Build-Up Envelope

Refer to Figure 1 and Figure 8.

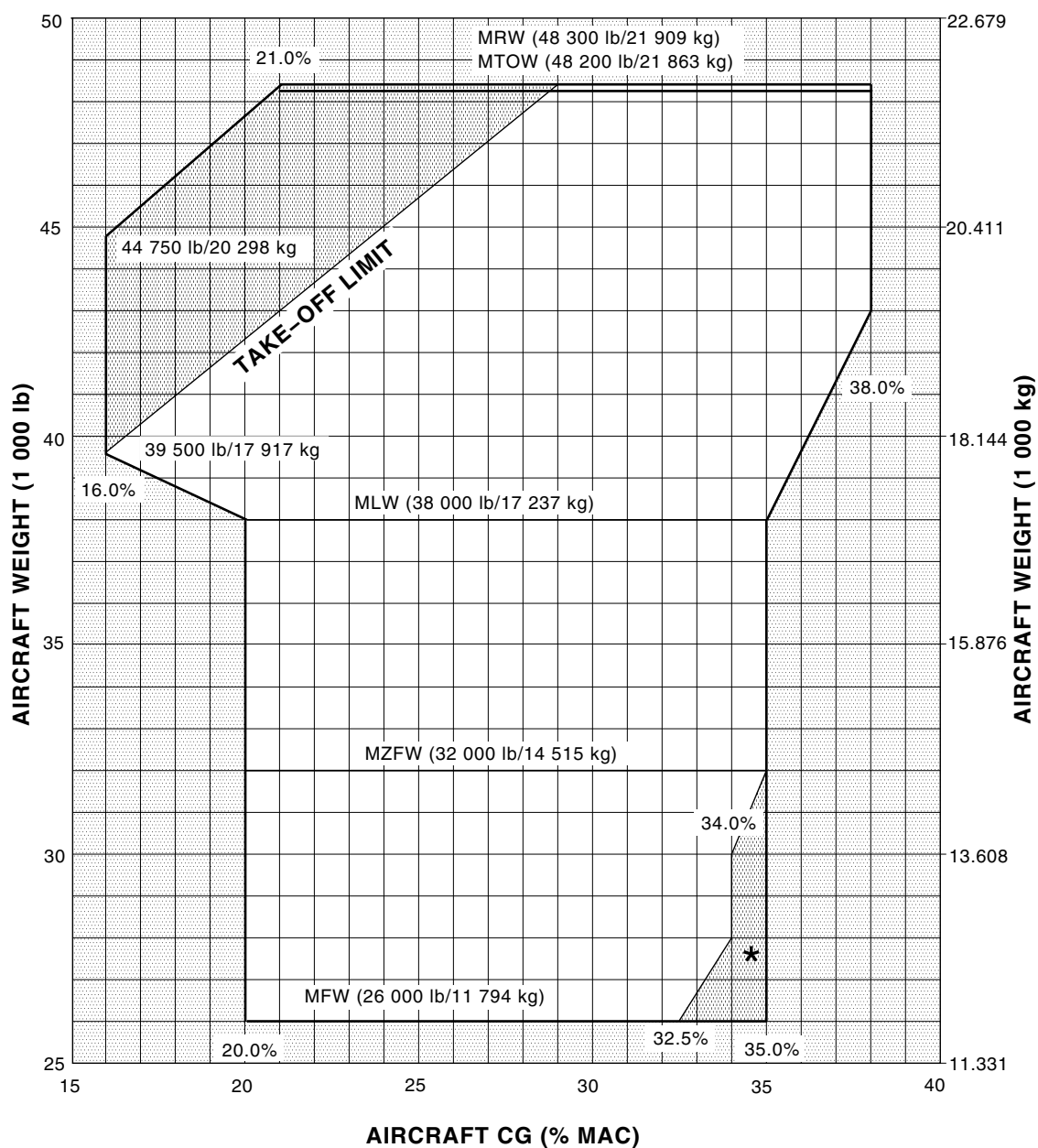
Do the steps that follow to complete the weight build-up envelope:

- (1) Refer to Figure 1 for an example.
- (2) Use a copy of the Weight Build-Up Envelope to continue with this procedure. Refer to Figure 8.
- (3) Make points on your copy of the Weight Build-Up Envelope to show the coordinates for the ZFW, TOW and FCBP.
- (4) Make lines to attach the points in sequence TOW, FCBP and ZFW.
 - (a) Do not include the fuel-curve boundary-points (FCBP) if they are heavier than the TOW:
 - 1 If FCBP is heavier than the TOW make only lines to attach the TOW and ZFW points, in sequence.
- (5) Make sure that the curve obtained is in the inner area of the envelope limits:
 - (a) If the curve is in the inner area of the envelope limits, the aircraft is safe for take-off.
 - (b) If a part of the curve is out of the envelope limits, the aircraft is not safe for take-off. You must move, add or remove payload or ballast until the weight and CG of the aircraft are into the envelope limits.



CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL



[* For zero fuel weight in that region, refer to Figure 3, 01-20-20, Auxiliary and Tail Tank Admissible Ratio (for ZFW CG inside lower aft zone of CG envelope)]

Weight Build-Up Envelope
Figure 8

DWB0190200_011

CG SLIDE METHOD**1. Procedure**

The slides are used as an alternative procedure to graphically get the approximate weight and longitudinal CG position during flight.

NOTE: The slides used for this procedure are at the end of this section.

A. Do the steps that follow to correctly use the CG Slide Method:

- (1) Calculate the ZFW and longitudinal CG position (%MAC) (Refer to the ZFW Calculation procedure (Subject 01-90-20)).
- (2) Make a copy of the CG SLIDE ENVELOPE. On your copy, make a point to record the coordinates for ZFW and %MAC.
- (3) Put the transparent CG SLIDE CHART squarely on top of the copy of the CG SLIDE ENVELOPE.
- (4) Align the bottom point of the fuel curve (ZFW) with the point that you made on the CG SLIDE ENVELOPE.

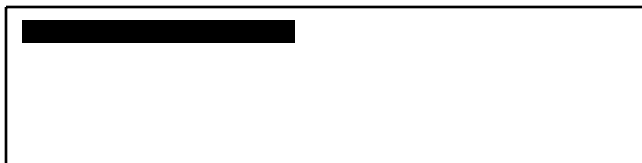
NOTE: The fuel curve that you get with this procedure shows the set of weight and longitudinal CG points along the fuel burn path. In other words, the highest point of the fuel curve shows an aircraft with full fuel tanks. During the flight, the point moves down the fuel curve as the fuel burns.

If the fuel tanks do not contain the maximum of fuel, it is necessary to use only the applicable part of the fuel curve. The applicable part goes from the ZFW point to the point equivalent to the RW of the aircraft for the flight.

- (5) Find the weight and longitudinal CG position as a result of the fuel load for the flight.

WARNING: THERE MUST BE NO PART OF THE FUEL CURVE OUT OF THE LONGITUDINAL CG ENVELOPE LIMITS. IT IS NOT SAFE TO OPERATE THE AIRCRAFT IF PART OF THE FUEL CURVE IS OUT OF THE LONGITUDINAL CG ENVELOPE LIMITS. THIS COULD CAUSE INJURY TO PERSONS AND DAMAGE TO THE AIRCRAFT.

- (6) Make sure that all of the applicable part of the fuel curve is in the inner area of the longitudinal CG envelope limits.

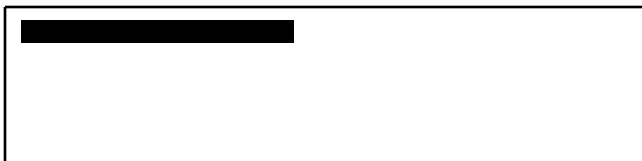


CHALLENGER 650
CL-650 WEIGHT AND BALANCE MANUAL

B. Example

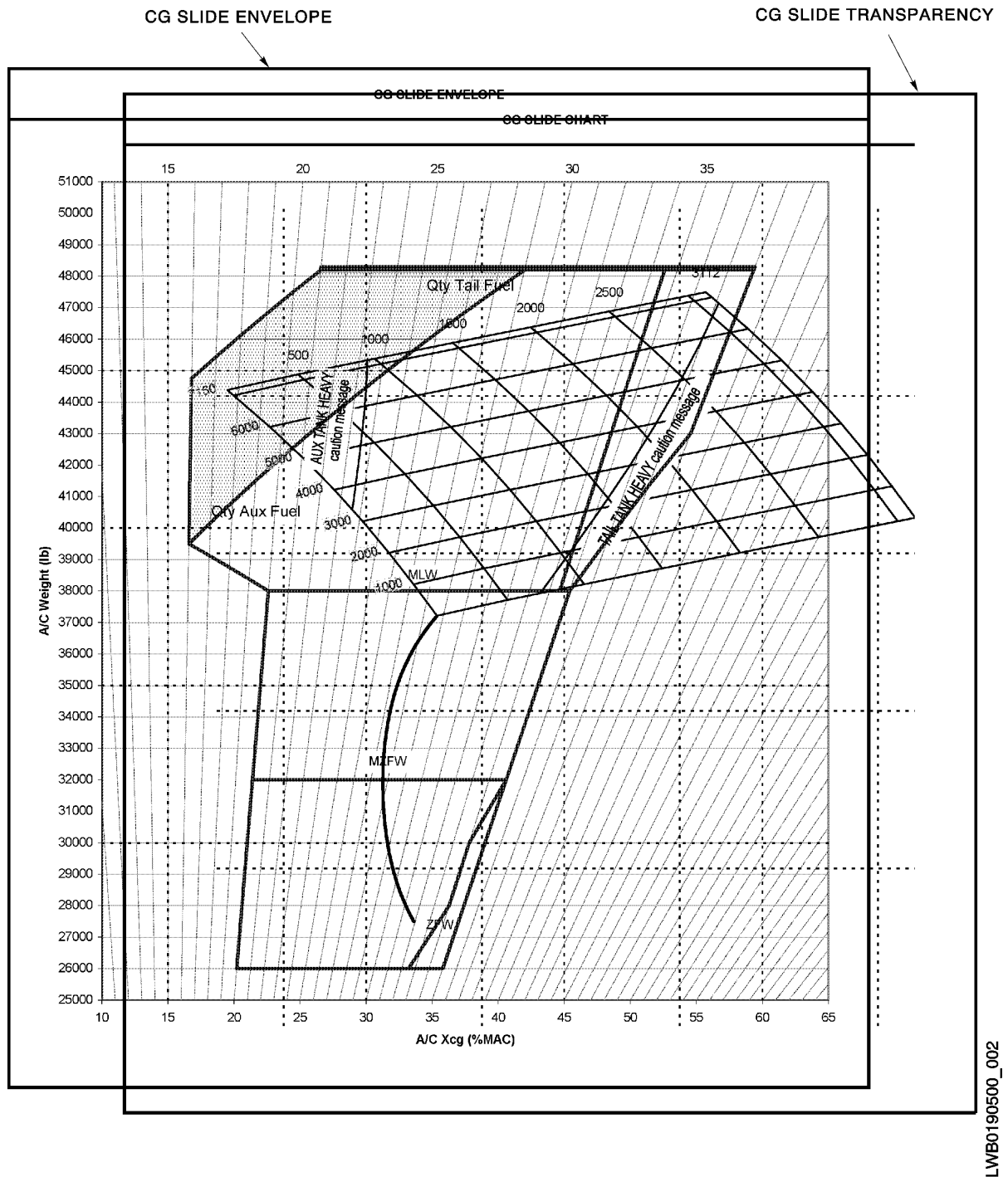
This procedure shows an example of how to use the CG Slide Method. This example uses the values found in the ZFW Calculation procedure (Refer to Table 1 (Subject 01-90-20)) and Refer to Figure 1. Do the steps that follow:

- (1) The values for the example are ZFW=27 500 lb (12 474 kg) and %MAC=32.0.
- (2) The aircraft fuel tanks are full. The weight of the fuel load is 20 000 lb (9 072 kg). This fuel load added to the aircraft weight gives an approximate RW of 47 500 lb (21 546 kg).
- (3) From this RW of 47 500 lb (21 546 kg) the CG Slide Method procedure is used to find that %MAC=36.6 approximately.
- (4) It is safe to operate the aircraft if all of the applicable part of the fuel curve is in the inner area of the longitudinal CG envelope limits.



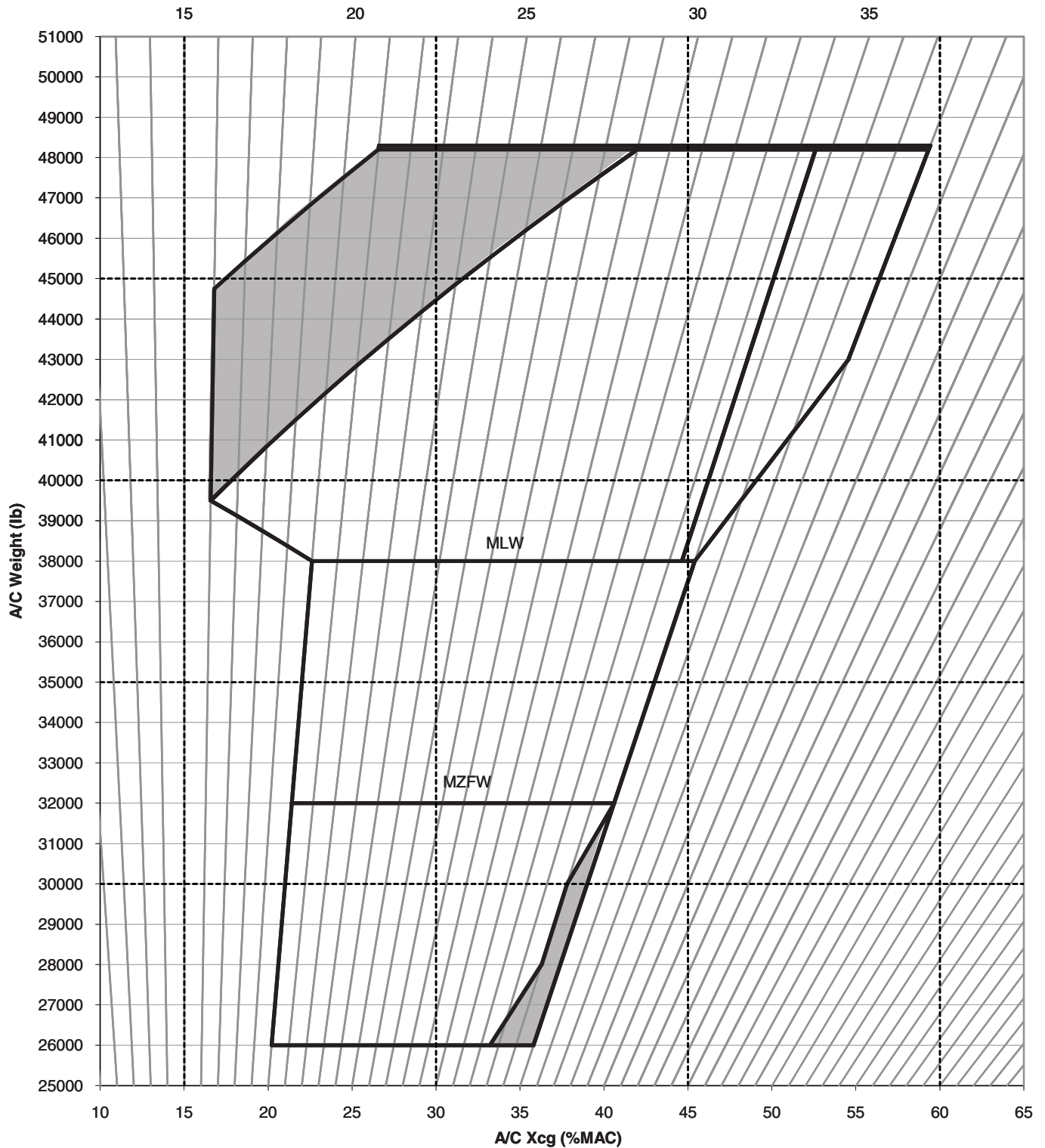
CHALLENGER 650

CL-650 WEIGHT AND BALANCE MANUAL



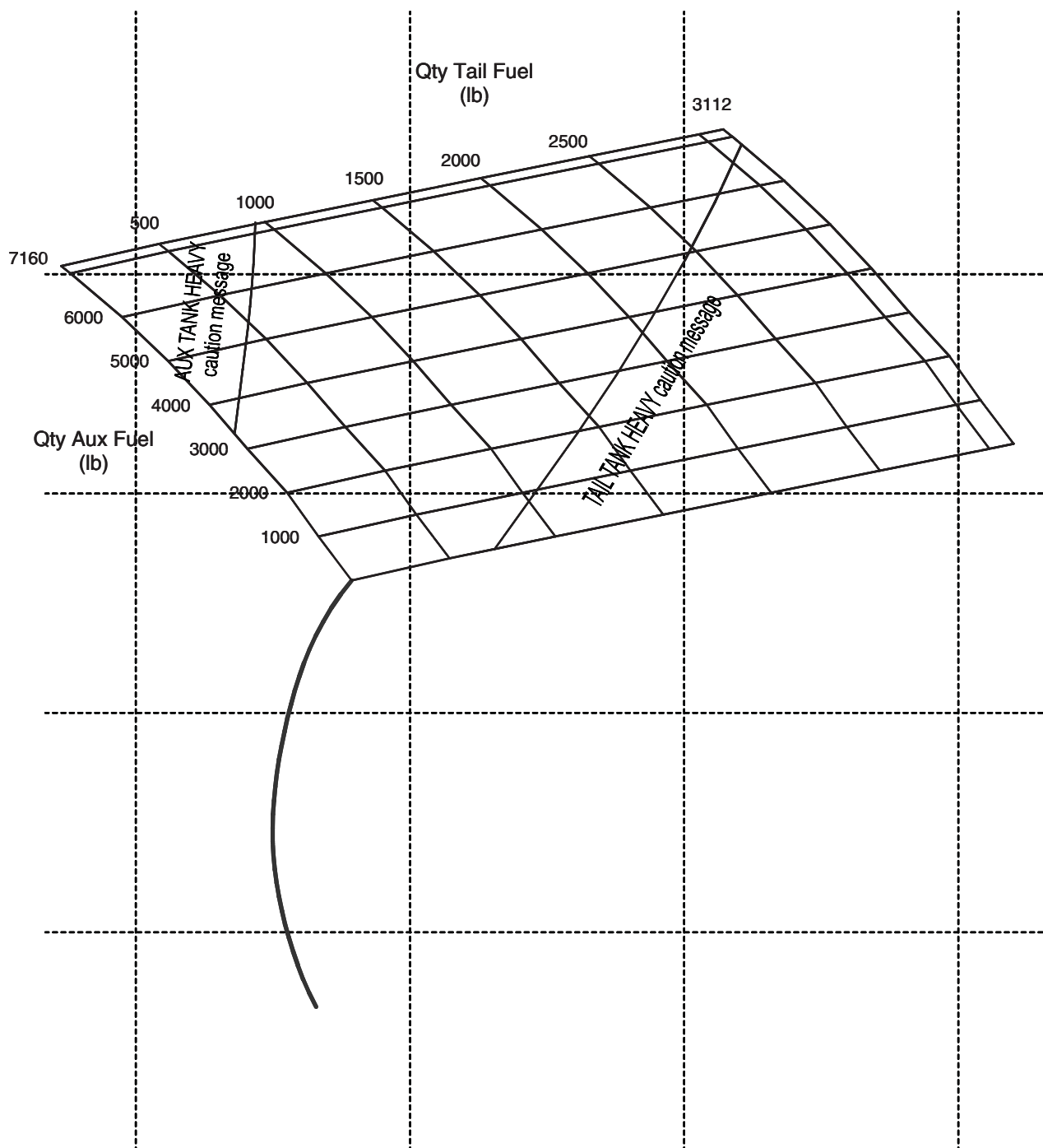
CG Slide Method – Example
Figure 1

CG SLIDE ENVELOPE - IMPERIAL UNITS



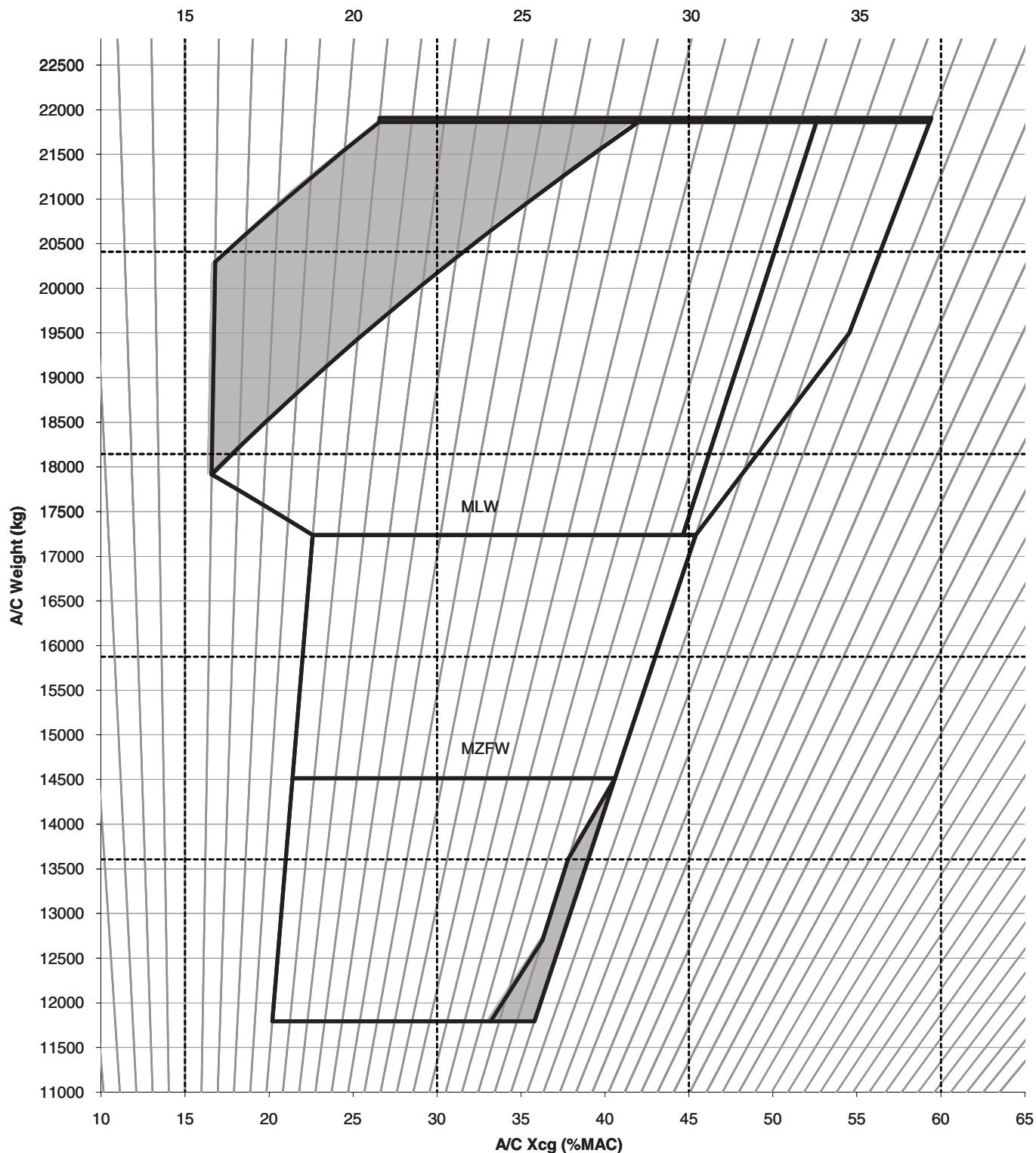
EFFECTIVITY: ALL

CG SLIDE CHART - IMPERIAL UNITS



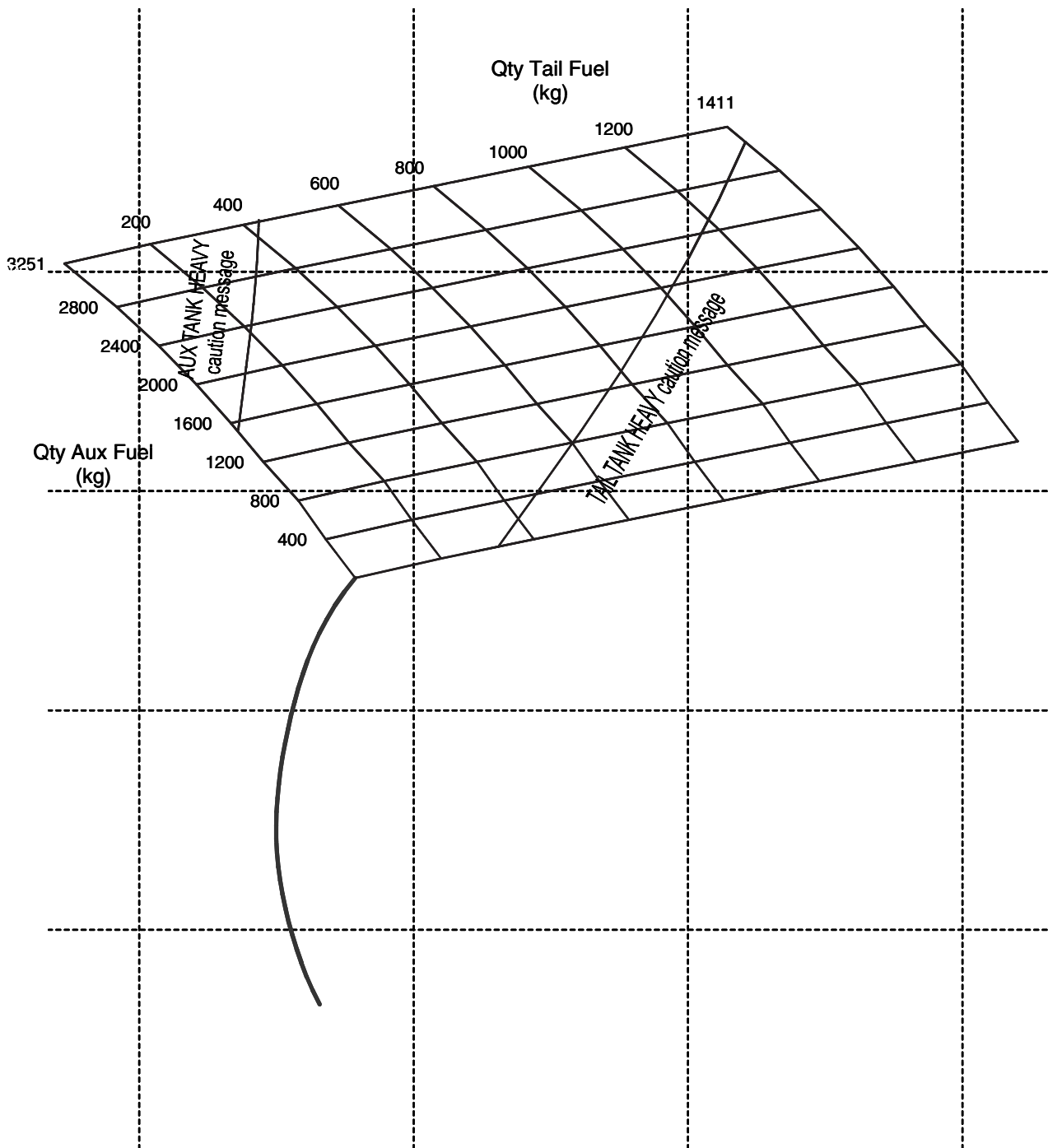
EFFECTIVITY: ALL

CG SLIDE ENVELOPE - SI UNITS



EFFECTIVITY: ALL

CG SLIDE CHART - SI UNITS



EFFECTIVITY: ALL